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PERÚ | Ministerio de Transportes y Comunicaciones

# FICHA TÉCNICA ESTÁNDAR PARA LA FORMULACIÓN Y EVALUACIÓN DE PROYECTOS DE INVERSIÓN EN CARRETERAS INTERURBANAS - V 2.0

(La información registrada tiene carácter de Declaración Jurada - DS. N° 284-2018-EF) - ACTUALIZADA A JULIO 2023

## I. ASPECTOS GENERALES

**1. NOMBRE DEL PROYECTO**

**1.1 Naturaleza de intervención**: MEJORAMIENTO

**1.2 Enunciado del nombre del proyecto**

MEJORAMIENTO DEL SERVICIO DE TRANSITABILIDAD VIAL DEL/LA DE LA CARRETERA SM-107, TRAMO: TINGO DE PONASA - SHAMBOYACU DEL DISTRITO DE TINGO DE PONASA , DE LA PROVINCIA DE PICOTA, DEL DEPARTAMENTO DE SAN MARTIN.

**1.3 ¿El proyecto pertenece a un Programa de Inversión?**: ○ SI ● NO

**En caso sea afirmativo, poner nombre y codigo del Programa:**

**Cod. Prog.**

**2. ALINEAMIENTO Y CONTRIBUCIÓN AL CIERRE DE BRECHAS PRIORITARIAS**

**2.1 Responsabilidad Funcional (Según Anexo N° 02 – Decreto Supremo N° 284-2018-EF )**

| | |
|---|---|
| FUNCIÓN | 015 TRANSPORTE |
| DIVISION FUNCIONAL | 033 TRANSPORTE TERRESTRE |
| GRUPO FUNCIONAL | 066 VIAS VECINALES |
| SECTOR RESPONSABLE | TRANSPORTES Y COMUNICACIONES |

**2.2 Servicios públicos con brecha identificada y priorizada:** SERVICIO DE TRANSITABILIDAD VIAL INTERURBANO

**2.3 Indicador de producto asociado a la brecha de servicios:** "PORCENTAJE DE LA RED VIAL VECINAL CON INADECUADAS CONDICIONES DE SERVICIO"

**Valor de contribución al cierre de brechas**: 24.7 Km

**2.4 Definición del servicio público**

Es la provisión de la infraestructura vial que asegura las condiciones adecuadas que permiten un flujo vehicular regular a lo lardo de la via de la Red Vial Vecinal del Sistema Nacional de Carreteras (SINAC)

**3. INSTITUCIONALIDAD**

**OFICINA DE PROGRAMACIÓN MULTIANUAL DE INVERSIONES (OPMI)**

| | |
|---|---|
| Entidad : | GOBIERNO REGIONAL SAN MARTIN |
| Nombre de la OPMI: (Nombre de la Unidad Orgánica a la que pertenece la OPMI) | OPMI DEL GOBIERNO REGIONAL SAN MARTIN |
| Responsable de la OPMI: | CLEBER PERALTA FERNANDEZ |

**UNIDAD FORMULADORA DEL PROYECTO DE INVERSIÓN (UF)**

| | |
|---|---|
| Entidad | GOBIERNO REGIONAL SAN MARTIN |
| Nombre de la UF: (Nombre de la Unidad Orgánica a la que pertenece la UF) | UF DIRECCIÓN REGIONAL DE TRANSPORTES Y COMUNICACIONES |
| Responsable de la UF: | DAVID GRANDEZ SILVA |

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Dirección Regional de Transportes y Comunicaciones
Ing. David Grandez Silva
CIP. N° 284250
RESPONSABLE DE LA UF DRTC SM

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**UNIDAD EJECUTORA DE INVERSIONES (UEI)**

| | |
|---|---|
| Entidad : | GOBIERNO REGIONAL SAN MARTIN |
| Nombre de la UEI: (Nombre de la Unidad Orgánica a la que pertenece la UEI) | UEI DIRECCION REGIONAL DE TRANSPORTES Y COMUNICACIONES |
| Responsable de la UEI: | LAEL TASILLA MONTALVAN |

**UNIDAD EJECUTORA PRESUPUESTAL (UEP)**

| | |
|---|---|
| Entidad | DIRECCION REGIONAL DE TRANSPORTES Y COMUNICACIONES |
| Nombre de la UEP: | 925 - REGION SAN MARTIN-TRANSPORTES |

**4. UBICACIÓN GEOGRÁFICA**

| Tramo | Departamento | Provincia | Distrito | Región Geográfica (1) | Coordenadas geográficas de la intervención (2) | Centro Poblado (3) | Ubigeo (4) |
|---|---|---|---|---|---|---|---|
| 1 - Origen | SAN MARTIN. | PICOTA | TINGO DE PONASA | SELVA | 361584.86 m E / 9233261.73 m S | TINGO DE PONASA | 220708 |
| 2 -Destino | SAN MARTIN. | PICOTA | SHAMBOYACU | SELVA | 374970.65 m E / 9223434.58 m S | SHAMBOYACU | 220708 |

(1)Seleccionar en caso que el proyecto abarque más de una región geográfica
(2)Consignar la latitud y longitud del punto de origen y destino del proyecto (WGS84)
(3) Llenar la información en caso corresponda
(4) Especificar si el Ubigeo es a nivel de Departamento, Provincia y Distrito → https://n9.cl/ub1xr

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## II. IDENTIFICACIÓN

**5. DIAGNÓSTICO DE LA UNIDAD PRODUCTORA (UP)**

Tramo - Denominacón de la vía (*)

**DE LA CARRETERA SM-107, TRAMO: TINGO DE PONASA - SHAMBOYACU**

**(*) Según Regitro Nacional de Carreteras del SINAC:**
https://portal.mtc.gob.pe/transportes/caminos/renac.html

| | |
|---|---|
| **Longitud:** | **24.7 Km** |
| | **SELVA** |
| **Progresiva (inicio de tramo):** | **19+343** |
| | **SELVA** |
| **Progresiva (fin de tramo):** | **44+090** |
| **Código de la UP** | **SM-107** |

**https://portal.mtc.gob.pe/transportes/caminos/renac.html**

Insertar mapa o esquema del trazo según fuentes oficiales



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## 6. DIAGNOSTICO DE LA POBLACIÓN BENEFICIADA Y ÁREA DE INFLUENCIA

| Población beneficiaria directa | |
|---|---|
| Cantidad | 11960 |

Link de consulta
https://www.inei.gob.pe/media/MenuRecursivo/publicaciones_digitales/Est/Lib1541/index.htm

## 7. DIAGNOSTICO DE OTROS AGENTES INVOLUCRADOS

| Involucrado | Ámbito del participante | Entidad a la que pertenece | Posición (Cooperante, Beneficiario, Oponente, Perjudicado) | Intereses | Contribución |
|---|---|---|---|---|---|
| Direccion Regional de Transportes y Comunicaciones | Regional | Gobierno Regional San Martín | Cooperante | Generar desarrollo integral entre localidades rurales y su área de influencia del proyecto, traducido en su mejoras de su ingreso y el nivel de vida del poblador | Apoyar el desarrollo de los pueblos, mejorando el acceso a los mercados y el crecimiento económico, sostenible y equitativo, apoyo logístico y tecnico en el momento de la ejecución del PIP y apoyo en gestionar el financiamiento |
| Distrito Tingo de Ponasa | Local | Municipalidad Distrital de Ponasa | Beneficiario | Integrar al distrito y la Provincia mediante infraestructura vial, salvaguardar las actividades socioeconómicas local mejorando los ingresos de los pobladores | Gestionar los recursos financieros para su ejecución ante el Gobierno central u otra entidad que estará a cargo la fase de ejecución y/o post inversión |
| Distrito de Shamboyacu | Local | Municipalidad Distrital de Shamboyacu | Beneficiario | Integrar al distrito y la Provincia mediante infraestructura vial, salvaguardar las actividades socioeconómicas local mejorando los ingresos de los pobladores | Gestionar los recursos financieros para su ejecución ante el Gobierno central u otra entidad que estará a cargo la fase de ejecución y/o post inversión |
| | | | | | |
| | | | | | |

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## 8. PROBLEMA CENTRAL, CAUSAS Y EFECTOS

**Descripción del problema central**

Inadecuadas Condiciones de Transitabilidad Vial de la Carretera del DE LA CARRETERA SM-107, TRAMO: TINGO DE PONASA - SHAMBOYACU del Distrito de TINGO DE PONASA , de la Provincia de PICOTA, del Departamento de SAN MARTIN.

| Causas directas | Causas Indirectas | Marcar |
|---|---|---|
| Deficiente infraestructura vial | Inadecuadas condiciones del pavimento | x |
| | Inadecuadas condiciones del puente | x |
| | Inadecuadas obras de artes y drenaje | x |
| | Limitadas actividades de conservación Vial | x |

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| | | |
|---|---|---|
| Deficiente seguridad vial | Inadecuada geometría vial | |
| | Inadecuada señalización | x |

| Efectos directos | Efectos Indirecto | Marcar |
|---|---|---|
| Dificultad en la articulación de mercados locales y regionales | Bajos ingresos y pérdidas económicas de la población | x |
| | Deficiente acceso de servicios básicos | x |
| Limitada circulación vehicular | Aumento de los costos asociados al transporte | x |
| | Incremento de los tiempos de viaje | x |
| | Incremento de los accidentes de tránsito | x |

## 9. DEFINICION DE LOS OBJETIVOS DEL PROYECTO

### 9.1. Objetivo

| Descripción del Objetivo central | | | | |
|---|---|---|---|---|
| Adecuadas condiciones de transitabilidad vial del DE LA CARRETERA SM-107, TRAMO: TINGO DE PONASA - SHAMBOYACU del Distrito de TINGO DE PONASA , de la Provincia de PICOTA, del Departamento de SAN MARTIN. | | | | |
| **Principales indicadores del objetivo** | **Unidad de medida** | **Magnitud sin Proyecto** | **Magnitud con Proyecto** | **Fuente de verificación** |
| IMDA | Veh/día | 254 | 298 | Estudio de Tráfico |
| Tiempo de viaje | Min | | | Estudio de Tráfico |
| Costos de Operación Vehicular acumumlado anual | Soles | | | Estudio de Tráfico y Evaluación Económica |

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### 9.2. Medios fundamentales

| N° | Medios fundamentales | Marcar |
|---|---|---|
| 1 | Mejoramiento o recuperación del pavimento | x |
| 2 | Adecuación y/o dotación de obras de arte y drenaje | x |
| 3 | Adecuación y/o dotación de la señalización | x |

## 10. DESCRIPCIÓN DE LA(S) ALTERNATIVA(S) DE SOLUCIÓN

☑ 1 Alternativa ☐ 2 Alternativas

| Alternativa (*) | Descripción |
|---|---|
| Alternativa de Solución 1 | MEJORAMIENTO DE LA INFRAESTRUCTURA VIAL<br>•CONSTRUCCIÓN DE PLATAFORMA (6.00M DE CALZADA Y 0.50M DE BERMAS A CADA LADO) CON CARPETA ASFÁLTICA EN UNA LONGITUD DE 24.70 KM.<br>•CONSTRUCCIÓN DE MURO DE CONTENCION Y MUROS DE GAVION TIPO CAJA EN UNA LONGITUD TOTAL DE 540 M.<br>•CONSTRUCCIÓN DE 774.35M DE ALCANTARILLAS CON TMC; 20M DE ALCANTARILLA CON MCA.<br>•CONSTRUCCIÓN DE 9,410.00 M DE CUNETAS TRIANGULARES; 1,070.49 M DE CUNETAS RECTANGULARES.<br>•05 PUENTES TANTO DE LOSA, COMO VIGA-LOSA.<br>•116 SEÑALES PREVENTIVAS, 23 SEÑALES REGLAMENTARIAS, 34 SEÑALES INFORMATIVAS, 25 POSTES DE KILOMETRAJE. |

**NOTA: De evidenciarse solo una alternativa de solución, el formulador deberá sustentar las razones.**

## 11. REQUERIMIENTOS TÉCNICOS, REGULATORIOS Y/O NORMATIVOS

| DOCUMENTOS A GESTIONAR EN LA FASE DE EJECUCIÓN | MARQUE CON (x) |
|---|---|
| Acta de Libre disponibilidad del Terreno (Acuerdo Institucional) | x |
| Plan de Compensación y Reasentamiento Involuntario - (PACRI) | |
| Estudio de Impacto Ambiental o lo requerido según nivel de estudio. | x |

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| | | |
|---|---|---|
| Estudio de Factibilidad de los Servicios (Interferencias) | | |
| Certificado de Inexistencia de Restos Arqueológicos (CIRA) | | |
| Otros (Señale) | | |

**NORMAS TÉCNICAS RELACIONADAS**

**RD N° 22-2013-MTC/14 "(Manual de Carreteras: Especificaciones Técnicas Generales para Construcción"**
**RD N° 03-2018-MTC 14 "Manual de Carreteras: Diseño Geométrico"**
**RD N° 19-2018-MTC14 "Manual de Carreteras: Puentes"**
**RD N° 10-2014-MTC14 "Manual de Carreteras: Suelos, Geología, Geotecnia y Pavimentos"**

**(*): Revisar ejemplo detallado en el instructivo**

## III. FORMULACIÓN Y EVALUACIÓN

### 12. HORIZONTE DE EVALUACIÓN

**Número de años del horizonte de evaluación** **20 Años**

### 13. ESTUDIO DE MERCADO DEL SERVICIO PÚBLICO

**(De ser el Proyecto en Tramos, clasificados por demanda)** **Nro de Tramos** 1

**13.1 Análisis de la demanda (*)**

| Descripción | Unidad de Medida | Año 1 | Año 2 | Año 3 | Año 4 | Año 5 | Año 6 | Año 7 | Año 8 | Año 9 | Año 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| DE LA CARRETERA SM-107, TRAMO: TINGO DE PONASA - | Veh/día | 304 | 311 | 318 | 321 | 330 | 337 | 345 | 351 | 359 | 364 |
| | | Año 11 | Año 12 | Año 13 | Año 14 | Año 15 | Año 16 | Año 17 | Año 18 | Año 19 | Año 20 |
| | | 371 | 381 | 388 | 398 | 405 | 415 | 423 | 431 | 438 | 452 |

**13.2 Análisis de la oferta (*)**

| Descripción | Unidad de Medida | Año 1 | Año 2 | Año 3 | Año 4 | Año 5 | Año 6 | Año 7 | Año 8 | Año 9 | Año 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| DE LA CARRETERA SM-107, TRAMO: TINGO DE PONASA - | Veh/día | 259 | 265 | 271 | 274 | 281 | 287 | 294 | 299 | 306 | 311 |
| | | Año 11 | Año 12 | Año 13 | Año 14 | Año 15 | Año 16 | Año 17 | Año 18 | Año 19 | Año 20 |
| | | 318 | 326 | 332 | 340 | 347 | 354 | 361 | 369 | 376 | 388 |

**(*) La Oferta y Demanda puede contabilizarse en tramos de ser el caso.**

**13.3 Brecha Oferta - Demanda**

| Descripción | Unidad de Medida | Año 1 | Año 2 | Año 3 | Año 4 | Año 5 | Año 6 | Año 7 | Año 8 | Año 9 | Año 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| DE LA CARRETERA SM-107, TRAMO: TINGO DE PONASA - | Veh/día | -45 | -46 | -47 | -47 | -49 | -50 | -51 | -52 | -53 | -53 |
| | | Año 11 | Año 12 | Año 13 | Año 14 | Año 15 | Año 16 | Año 17 | Año 18 | Año 19 | Año 20 |
| | | -53 | -55 | -56 | -58 | -58 | -61 | -62 | -62 | -62 | -64 |

### 14. ANÁLISIS TÉCNICO DE LAS ALTERNATIVAS:

**El primer medio fundamental: Mejoramiento del Pavimento, se consideran las siguientes acciones:**
Acción 1.1: Construcción de 24.7 km de la infraestructura vial con carpeta asfáltica en un ancho total de plataforma de 7.00m (6.00m de calzada y 0.50m de bermas).
Acción 1.2: Construcción de muro de contención y muros de gavión tipo caja en total 540 M.

**El segundo medio fundamental: Adecuación y/o dotación de obras de arte y drenaje, se consideran las siguientes acciones:**
Acción 2.1: Construcción de 774.35m de alcantarillas con TMC y 20m de alcantarilla con MCA.
Acción 2.2: Construcción de 9,410.00 m de cunetas triangulares y 1,070.49 m de cunetas rectangulares.

**El tercer medio fundamental: Adecuación y/o dotación de la señalización, se consideran las siguientes acciones:**
Acción 3.1: Instalación de 116 señales preventivas, 23 señales reglamentarias, 34 señales informativas, 25 postes de kilometraje.

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## 15. COSTO DEL PROYECTO

### 15.1 Costos de inversión

| PRODUCTO / PROYECTO | | Unidad de medida representativa | Cantidad | Costo/km | Inversión total |
|---|---|---|---|---|---|
| MEJORAMIENTO DEL SERVICIO DE TRANSITABILIDAD VIAL DEL/LA DE LA CARRETERA SM-107, TRAMO: TINGO DE PONASA - SHAMBOYACU DEL DISTRITO DE TINGO DE PONASA , DE LA PROVINCIA DE PICOTA, DEL DEPARTAMENTO DE SAN MARTIN. | | Km | 24.7 | 5,346,121.44 | 132,049,199.60 |
| **ACCIONES** | | **Unidad de medida** | **Cantidad** | **Costo/km** | **Costo por Componente (soles)** |
| Región Geográfica | SELVA | | | | |
| Tramo 01 | Carretera | Km | 24.7 | 1,687,212.24 | 41674142.22 |
| | Puente | m | 127.00 | 197,961.88 | 25141159.14 |
| Tramo 02 | Carretera | Km | 0 | | 0 |
| | Puente | Km | 0 | | 0 |
| Tramo 03 | Carretera | Km | 0 | | 0 |
| | Puente | Km | 0 | | 0 |

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### 15.2 Cronograma de ejecución financiera

○ MENSUAL ○ BIMENSUAL ● TRIMESTRAL

| Región Geográfica | SELVA | | Cronograma | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Proyecto/ Tramo | Actividad | Costo por Componente (soles) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| Tramo 01 | Carretera | 41,674,142.2 | - | - | 5,209,268 | 5,209,268 | 5,209,268 | 5,209,268 | 5,209,268 | 5,209,268 | 5,209,268 | 5,209,268 | - | - |
| | Puente | 25,141,159.1 | - | - | 3,142,645 | 3,142,645 | 3,142,645 | 3,142,645 | 3,142,645 | 3,142,645 | 3,142,645 | 3,142,645 | - | - |
| Tramo 02 | Carretera | - | - | - | - | - | - | - | - | - | - | - | - | - |
| | Puente | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Tramo 03 | Carretera | - | - | - | - | - | - | - | - | - | - | - | - | - |
| | Puente | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Costo total | | 66,815,301 | - | - | 8,351,913 | 8,351,913 | 8,351,913 | 8,351,913 | 8,351,913 | 8,351,913 | 8,351,913 | 8,351,913 | - | - |

### 15.3 Cronograma de ejecución física

○ MENSUAL ○ BIMENSUAL ● TRIMESTRAL

| Región Geográfica | SELVA | | | Cronograma | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Proyecto/ Tramo | Actividad | Unidad de medida | Meta | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| Tramo 01 | Carretera | Km | 100% | | | 12.50% | 12.50% | 12.50% | 12.50% | 12.50% | 12.50% | 12.50% | 12.50% | | |
| | Puente | m | 100% | | | 12.50% | 12.50% | 12.50% | 12.50% | 12.50% | 12.50% | 12.50% | 12.50% | | |
| Tramo 02 | Carretera | Km | 0% | | | | | | | | | | | | |
| | Puente | Km | 0% | | | | | | | | | | | | |

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| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Tramo 03 | Carretera | Km | 0% | | | | | | | | | | | | |
| | Puente | Km | 0% | | | | | | | | | | | | |

**15.4 Costos de operación y mantenimiento con y sin proyecto**

| COSTOS | | AÑOS (Soles) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Año 1 | Año 2 | Año 3 | Año 4 | Año 5 | Año 6 | Año 7 | Año 8 | Año 9 | Año 10 |
| SIN PROYECTO | OPERACIÓN | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - |
| | MANTENIMIENTO | S/ 1,500,218.90 | S/ 1,500,218.90 | S/ 1,500,218.90 | S/ 1,500,218.90 | S/ 1,500,218.90 | S/ 1,500,218.90 | S/ 1,500,218.90 | S/ 1,500,218.90 | S/ 1,500,218.90 | S/ 1,500,218.90 |
| CON PROYECTO | OPERACIÓN | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - |
| | MANTENIMIENTO | S/ 432,250.00 | S/ 432,250.00 | S/ 453,900.00 | S/ 432,250.00 | S/ 432,250.00 | S/ 605,200.00 | S/ 432,250.00 | S/ 432,250.00 | S/ 605,200.00 | S/ 432,250.00 |
| INCREMENTAL | OPERACIÓN | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - |
| | MANTENIMIENTO | S/ 1,067,968.90 | S/ 1,067,968.90 | S/ 1,046,318.90 | S/ 1,067,968.90 | S/ 1,067,968.90 | S/ 895,018.90 | S/ 1,067,968.90 | S/ 1,067,968.90 | S/ 895,018.90 | S/ 1,067,968.90 |

| COSTOS | | AÑOS (Soles) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Año 11 | Año 12 | Año 13 | Año 14 | Año 15 | Año 16 | Año 17 | Año 18 | Año 19 | Año 20 |
| SIN PROYECTO | OPERACIÓN | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - |
| | MANTENIMIENTO | S/ 1,500,218.90 | S/ 1,500,218.90 | S/ 1,500,218.90 | S/ 1,500,218.90 | S/ 1,500,218.90 | S/ 1,500,218.90 | S/ 1,500,218.90 | S/ 1,500,218.90 | S/ 1,500,218.90 | S/ 1,500,218.90 |
| CON PROYECTO | OPERACIÓN | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - |
| | MANTENIMIENTO | S/ 432,250.00 | S/ 605,200.00 | S/ 432,250.00 | S/ 432,250.00 | S/ 605,200.00 | S/ 432,250.00 | S/ 432,250.00 | S/ 605,200.00 | S/ 432,250.00 | S/ 432,250.00 |
| INCREMENTAL | OPERACIÓN | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - | S/ - |
| | MANTENIMIENTO | S/ 1,067,968.90 | S/ 895,018.90 | S/ 1,067,968.90 | S/ 1,067,968.90 | S/ 895,018.90 | S/ 1,067,968.90 | S/ 1,067,968.90 | S/ 895,018.90 | S/ 1,067,968.90 | S/ 1,067,968.90 |

**16. CRITERIO DE DECISIÓN DE INVERSIÓN**

**Beneficio/Costo**

**17. EVALUACION SOCIAL**

| Tipo* | | Alternativa 1 | Alternativa 2 |
|---|---|---|---|
| Beneficio/Costo | VAN | S/. 30,682,072 | S/. 19,605,335 |
| | B/C | S/. 2.24 | S/. - |
| | TIR | S/. 8.08 | S/. - |
| Costo/Eficiencia | Valor Actual de los Costos (VACS) | S/. 96,829,222 | S/. - |
| | Costo Anual Equivalente (CAE) | S/. 14,430,409 | S/. - |
| | VACS/Km | S/. 3,920,211 | S/. - |
| | Línea de Corte | S/. - | S/. - |
| | Cumple Linea de Corte (S/ / Km) | SI CUMPLE | SI CUMPLE |

* A precios sociales

GOBIERNO REGIONAL SAN MARTIN
Dirección Regional de Transportes y Comunicaciones
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PERÚ Ministerio de Transportes y Comunicaciones

# FICHA TÉCNICA ESTÁNDAR PARA LA FORMULACIÓN Y EVALUACIÓN DE PROYECTOS DE INVERSIÓN EN CARRETERAS INTERURBANAS - V 2.0

(La información registrada tiene carácter de Declaración Jurada - DS. N° 284-2018-EF) - ACTUALIZADA A JULIO 2023

## 18. SOSTENIBILIDAD

### 18.1 Responsable de la operación y mantenimiento del proyecto

LA VÍA SM-107 PERTENECE A LA RED VIAL DEPARTAMENTAL, POR TANTO, ES RESPONSABILIDAD FUNCIONAL DEL GOBIERNO REGIONAL SAN MARTIN REALIZAR EL FINANCIAMIENTO PARA LA FASE DE EJECUCIÓN Y OPERACIÓN Y MANTENIMIENTO

### 18.2 ¿Es la Unidad Ejecutora de Inversiones la responsable de la Operación y Mantenimiento del proyecto de inversión con cargo a su Presupuesto Institucional?

(●) SÍ ( ) NO ( ) PARCIALMENTE

| Documento | Entidad / Organización | Compromiso |
|---|---|---|
| ACTA DE OPERACIÓN Y MANTENIMIENTO | DIRECCIÓN REGIONAL DE TRANSPORTES Y COMUNICACIONES - SM | MANTENER LA VÍA EN OPTIMAS CONDICIONES CON LA INTERVENCIÓN OPORTUNA PARA EL MANTENIMIENTO PERIODICO Y RUTINARIO DE LA VÍA |
| | | |
| | | |
| | | |

### 18.3 ¿El área donde se ubica el proyecto ha sido afectada por algún desastre?

( ) SÍ (●) NO ( ) PARCIALMENTE

| Medidas consideradas en el Proyecto para Mitigar el Riesgo de Desastre | | | | |
|---|---|---|---|---|
| Peligros | Ocurrencia | Nivel | Probabilidad de Ocurrencia | Medida de Reducción de Riesgo en el Contexto de Cambio Climático |
| Sismos | SI | Bajo | Bajo | DISEÑO DE LA INFRAESTRUCTURA DE ACUERDO CON LA NORMA E.030 DEL RNE, Y ADECUADOS ESTUDIOS DE INGENIERÍA |
| Tsunamis | NO | | | |
| Heladas | NO | | | |
| Friajes | NO | | | |
| Erupciones Volcánicas | NO | | | |
| Sequías | NO | | | |
| Granizadas | NO | | | |
| Lluvias intensas. | SI | Medio | Medio | DISEÑO DEL SISTEMA DE DRENAJE DE ACORDE AL MARCO NORMATIVO Y LOS ESTUDIOS BÁSICOS DE INGENIERÍA |
| Avalanchas | NO | | | |
| Flujos de lodo (Huaicos) | NO | | | |
| Deslizamientos | NO | | | |
| Inundaciones | NO | | | |
| Vientos fuertes | NO | | | |

GOBIERNO REGIONAL SAN MARTIN
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CIP. N° 284250
RESPONSABLE DE LA UF DRTC SM

FECHA DE REGISTRO:
**01/04/2025 15:31**

**Soporte: opmi@mtc.gob.pe**

PERÚ Ministerio de Transportes y Comunicaciones

# FICHA TÉCNICA ESTÁNDAR PARA LA FORMULACIÓN Y EVALUACIÓN DE PROYECTOS DE INVERSIÓN EN CARRETERAS INTERURBANAS - V 2.0

(La información registrada tiene carácter de Declaración Jurada - DS. N° 284-2018-EF) - ACTUALIZADA A JULIO 2023

| | | |
|---|---|---|
| **Otros (Especificar):** | **NO** | |

**19. MODALIDAD DE EJECUCIÓN**

| N° | TIPO DE EJECUCIÓN | Elegir Modalidad de Ejecución (X) |
|---|---|---|
| 1 | ADMINISTRACIÓN DIRECTA | |
| 2 | ADMINISTRACIÓN INDIRECTA - POR CONTRATA | x |
| 3 | ADMINISTRACIÓN INDIRECTA - ASOCIACIÓN PÚBLICA PRIVADA (APP) | |
| 4 | ADMINISTRACIÓN INDIRECTA - OBRAS POR IMPUESTOS | |
| 5 | ADMINISTRACIÓN INDIRECTA - GOBIERNO A GOBIERNO | |

**20. TIPO DE FUENTE DE FINANCIAMIENTO**

| | | |
|---|---|---|
| 1 | RECURSOS ORDINARIOS | |
| 2 | RECURSOS DIRECTAMENTE RECAUDADOS | |
| 3 | RECURSOS POR OPERACIONES OFICIALES DE CREDITO | |
| 4 | DONACIONES Y TRANSFERENCIAS | |
| 5 | RECURSOS DETERMINADOS | x |

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**21. IMPACTO AMBIENTAL** **El Proyecto cuenta con Clasificación Ambiental**

| ETAPAS DEL PROYECTO | IMPACTOS NEGATIVOS | MEDIDAS DE PREVENCIÓN, CONTROL Y/O MITIGACIÓN | MEDIOS DE VERIFICACIÓN DEL CUMPLIMIENTO | FRECUENCIA DE EJECUCIÓN | COSTO (S/) |
|---|---|---|---|---|---|
| EJECUCIÓN | | | | | |
| | | | | | |
| | | | | | |
| FUNCIONAMIENTO | | | | | |
| | | | | | |
| | | | | | |

(*): El Monto de Prevención, Control y/o Mitigación deberá estar incluido en el Costo de Inversión (16.1) en la etapa de ejecución y los costos durante el funcionamiento en los costos de operación y mantenimiento (16.4)

**22. COMPETENCIA EN LAS QUE SE ENMARCA EL PROYECTO DE INVERSIÓN**

La Unidad Formuladora declara que el presente proyecto de inversión es competencia de su nivel de Gobierno. SI [x] NO [ ]

Caso contrario y sólo de ser competencia local, el GL involucrado autoriza su Formulación y Evaluación mediante (Convenio): _____ de fecha: _____

FECHA DE REGISTRO:
**01/04/2025 15:31**

**Soporte: opmi@mtc.gob.pe**

PERÚ | Ministerio de Transportes y Comunicaciones

# FICHA TÉCNICA ESTÁNDAR PARA LA FORMULACIÓN Y EVALUACIÓN DE PROYECTOS DE INVERSIÓN EN CARRETERAS INTERURBANAS - V 2.0

(La información registrada tiene carácter de Declaración Jurada - DS. N° 284-2018-EF) - ACTUALIZADA A JULIO 2023

## 23. CONCLUSIONES Y RECOMENDACIONES

CONCLUSIONES

Con el proyecto se solucionará el problema de "Inadecuadas condiciones de Transitabilidad Vial en la Carretera SM-107, Tramo: "Tingo de Ponasa - Shamboyacu distrito de Shamboyacu de la provincia de Picota del Departamento de San Martin".

La inversión como alternativa única a precios de mercado asciende a S/. 130,844,691.59 Soles y a Precio Social S/. 103,367,306.36 Soles, sin incluir control de costo concurrente.

La evaluación económica a precios sociales muestra un VANS social ascendente a S/. 30682071.67 soles y la TIR 8.1%; siendo un indicador social y económico aceptable.

La sostenibilidad del presente proyecto está plenamente asegurada debido a que el financiamiento del proyecto en su etapa de inversión y operación y mantenimiento está garantizado por el Gobierno Regional de San Martín en coordinación con la Dirección Regional de Trasportes y Comunicaciones.

RECOMENDACIONES

Ceñirse a los parámetros, la concepción técnica y dimensionamiento del presente estudio

## 24 RESULTADO DE LA FORMULACION Y EVALUACIÓN

**VIABLE :** x

**NO VIABLE :**

## 25. ANEXOS

**Anexo N° 01** Estudio de Tráfico
**Anexo N° 02** Formatos de Reconocimiento de campo
Formato N° 1: Datos Generales.
Formato N° 2: Topografía.
Formato N° 3A: Registro de excavación de calicatas
Formato N° 3B: Daños en la superficie de rodadura
Formato N° 4: Canteras y Fuentes de Agua.
Formato N° 5A: Obras de Arte.
Formato N° 5B: Obras de Drenaje.
Formato N° 6: Señalización.
Formato N° 7: Puntos Críticos.
**Anexo N° 03** Sustentos del dimensionamiento, ubicación y tecnología del proyecto de inversión
Planos de Ubicación en *KMZ
Planos en Planta
Planos de Sección Transversal Típica y perfil
Diseño de Pavimentos
**Anexo N° 04** Presupuesto del proyecto de inversión
**Anexo N° 05** Actas y/o Acuerdos del proyecto de inversión.

GOBIERNO REGIONAL SAN MARTÍN
Dirección Regional de Transportes y Comunicaciones
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RESPONSABLE DE LA UF DRTC SM

**NOTA:**
**Los Anexos y Formatos indicados se encuentran en el Instructivo**

## 26. FIRMAS

**Responsable de la formulación y evaluación del proyecto**

**Responsable de la Unidad Formuladora**

PERÚ | Ministerio de Transportes y Comunicaciones

# FORMATO DE CLASIFICACION VEHICULAR

## ESTUDIO DE TRAFICO - ESTACIÓN PRINCIPAL 1 - LUNES

| | | | |
|---|---|---|---|
| ESTACIÓN PRINCIPAL DE LA CARRETERA | 1 | TINGO DE PONAZA - SHAMBOYACU | |
| SENTIDO | | E ← | S → |
| UBICACIÓN | | TINGO DE PONASA - PROVINCIA DE PICOTA | |
| DIA | Lunes | | |

| | | | | |
|---|---|---|---|---|
| ESTACION | | DV A PASE PROVISIONAL | | |
| CODIGO DE LA ESTACION | | E-01 | | |
| DIA Y FECHA | LUNES | 14 | 10 | 2024 |

| HORA | SENTIDO | AUTO | STATION WAGON | CAMIONETAS | | RURAL | MICRO | BUS | | | CAMION | | | SEMI TRAYLER | | | | TRAYLER | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | PICK UP | PANEL | Combi | | 2 E | 3 E | 4 E | 2 E | 3 E | 4 E | 2S1/2S2 | 2S3 | 3S1/3S2 | >= 3S3 | 2T2 | 2T3 | 3T2 | >=3T3 |
| 00-01 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 01-02 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 02-03 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 03-04 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 04-05 | E | 2 | | 1 | | | | | | | | | | | | | | | | | |
| | S | 4 | | 3 | | 1 | | | | | 1 | 1 | | | | | | | | | |
| 05-06 | E | 3 | | 2 | | | | | | | 1 | 2 | | | | | | | | | |
| | S | 2 | 1 | 2 | | 1 | | | | | 2 | 1 | | | | | | | | | |
| 06-07 | E | 2 | | 2 | | 1 | | | | | 3 | 1 | | | | | | | | | |
| | S | 1 | 1 | 2 | | 1 | | | | | 3 | | | | | | | | | | |
| 07-08 | E | 1 | | 3 | | 2 | | | | | 1 | | | | | | | | | | |
| | S | 4 | 1 | 1 | | 2 | | | | | 1 | | | | | | | | | | |
| 08-09 | E | 3 | | | | 2 | | | | | 1 | 3 | | | | | | | | | |
| | S | 2 | | 1 | 1 | 3 | | | | | 2 | 4 | | | | | | | | | |
| 09-10 | E | 1 | | 2 | | 1 | | | | | 4 | 2 | | | | | | | | | |
| | S | 3 | | 1 | 2 | 1 | | | | | 3 | 2 | | | | | | | | | |
| 10-11 | E | 1 | | 4 | | 3 | | | | | 4 | 2 | | | | | | | | | |
| | S | 1 | | 5 | 1 | 2 | | | | | 5 | 4 | | | | | | | | | |
| 11-12 | E | 2 | | 1 | | 4 | | | | | 2 | | | | | | | | | | |
| | S | 2 | | 1 | | 3 | | | | | 3 | 3 | | | | | | | | | |
| 12-13 | E | 3 | | 2 | | | | | | | 1 | 2 | | | | | | | | | |
| | S | 2 | | 1 | 2 | | | | | | 2 | 5 | | | | | | | | | |
| 13-14 | E | 2 | | 2 | | 1 | | | | | 1 | | | | | | | | | | |
| | S | 1 | 1 | 1 | | 1 | | | | | 1 | | | | | | | | | | |
| 14-15 | E | 1 | 1 | 2 | | | | | | | | | | | | | | | | | |
| | S | 2 | | 2 | | | | | | | | | | | | | | | | | |
| 15-16 | E | 2 | | 3 | | | | | | | | | | | | | | | | | |
| | S | 2 | | 1 | | | | | | | | 1 | | | | | | | | | |
| 16-17 | E | 0 | | 3 | | | | | | | 5 | 2 | | | | | | | | | |
| | S | 1 | 1 | 2 | 1 | | | | | | 2 | 3 | | | | | | | | | |
| 17-18 | E | 2 | 1 | | | | | | | | 2 | 4 | | | | | | | | | |
| | S | 2 | 1 | 1 | | 1 | | | | | 2 | 5 | | | | | | | | | |
| 18-19 | E | 2 | 1 | 3 | | | | | | | 2 | 4 | | | | | | | | | |
| | S | 2 | | 3 | 1 | 1 | | | | | 2 | 6 | | | | | | | | | |
| 19-20 | E | 3 | | 3 | | | | | | | 3 | 2 | | | | | | | | | |
| | S | 2 | | 3 | | | | | | | 2 | 2 | | | | | | | | | |
| 20-21 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | 1 | | | | | | | | | | | | | | | | |
| 21-22 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 22-23 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 23-24 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| PARCIAL: | | 63 | 9 | 63 | 9 | 31 | 0 | 0 | 0 | 0 | 61 | 61 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

ENCUESTADOR : _____ JEFE DE BRIGADA : _____ ING.RESPONS: _____ SUPERV.MTC : _____

GOBIERNO REGIONAL SAN MARTÍN
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PERÚ | Ministerio de Transportes y Comunicaciones

# FORMATO DE CLASIFICACION VEHICULAR

## ESTUDIO DE TRAFICO - ESTACIÓN PRINCIPAL 1 - MARTES

| | | | | | |
|---|---|---|---|---|---|
| ESTACIÓN PRINCIPAL DE LA CARRETERA | 1 | TINGO DE PONAZA - SHAMBOYACU | | | |
| SENTIDO | | E | | S | |
| UBICACIÓN | TINGO DE PONASA - PROVINCIA DE PICOTA | | | | |
| DIA | Martes | | | | |

| | | | | |
|---|---|---|---|---|
| ESTACION | DV A PASE PROVISIONAL | | | |
| CODIGO DE LA ESTACION | E-01 | | | |
| DIA Y FECHA | MARTES | 15 | 10 | 2024 |

| HORA | SENTIDO | AUTO | STATION WAGON | CAMIONETAS | | RURAL Combi | MICRO | BUS | | | CAMION | | | SEMI TRAYLER | | | | TRAYLER | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | PICK UP | PANEL | | | 2 E | 3 E | 4 E | 2 E | 3 E | 4 E | 2S1/2S2 | 2S3 | 3S1/3S2 | >= 3S3 | 2T2 | 2T3 | 3T2 | >=3T3 |
| 00-01 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 01-02 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 02-03 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 03-04 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 04-05 | E | 3 | 1 | | | | | | | | | | | | | | | | | | |
| | S | 2 | | 1 | | | | | | | | | | | | | | | | | |
| 05-06 | E | 2 | 1 | 1 | | | | | | | 2 | 1 | | | | | | | | | |
| | S | 1 | | 1 | | | | | | | 2 | 1 | | | | | | | | | |
| 06-07 | E | 3 | 2 | 2 | | 3 | | | | | 2 | 1 | | | | | | | | | |
| | S | 2 | | 3 | | | | | | | 2 | 1 | | | | | | | | | |
| 07-08 | E | 4 | 3 | 3 | | 2 | | | | | 1 | | | | | | | | | | |
| | S | 2 | | 2 | | 3 | | | | | 1 | | | | | | | | | | |
| 08-09 | E | 6 | | 3 | | 1 | | | | | 2 | | | | | | | | | | |
| | S | 1 | | 2 | | 1 | | | | | 1 | | | | | | | | | | |
| 09-10 | E | 3 | | 1 | | 2 | | | | | 3 | | | | | | | | | | |
| | S | | | 2 | | 2 | | | | | 2 | | | | | | | | | | |
| 10-11 | E | 2 | | 1 | 1 | 3 | | | | | 3 | | | | | | | | | | |
| | S | | | | | 2 | | | | | 3 | | | | | | | | | | |
| 11-12 | E | 1 | | 2 | | | | | | | 2 | | | | | | | | | | |
| | S | | | 1 | | | | | | | 2 | | | | | | | | | | |
| 12-13 | E | 4 | | 3 | | 3 | | | | | 1 | 1 | | | | | | | | | |
| | S | | | 1 | 1 | 3 | | | | | 2 | | | | | | | | | | |
| 13-14 | E | 2 | 2 | 2 | | | | | | | 2 | | | | | | | | | | |
| | S | | | 1 | 1 | 1 | | | | | 1 | | | | | | | | | | |
| 14-15 | E | 5 | 1 | 2 | | 1 | | | | | | | | | | | | | | | |
| | S | | | 2 | | | | | | | | | | | | | | | | | |
| 15-16 | E | 3 | 1 | 5 | 1 | | | | | | 1 | 1 | | | | | | | | | |
| | S | 2 | | 2 | | | | | | | | | | | | | | | | | |
| 16-17 | E | | 2 | 3 | | 1 | | | | | 1 | 1 | | | | | | | | | |
| | S | | | 1 | | 1 | | | | | 1 | 1 | | | | | | | | | |
| 17-18 | E | 3 | 3 | 3 | 1 | 2 | | | | | 1 | 1 | | 2 | | | | | | | |
| | S | 3 | | | 1 | | | | | | 3 | 1 | | | | | | | | | |
| 18-19 | E | 2 | | 1 | | | | | | | 1 | | | | | | | | | | |
| | S | 1 | | 2 | | | | | | | 1 | | | | | | | | | | |
| 19-20 | E | 2 | | 2 | 1 | 1 | | | | | 2 | | | | | | | | | | |
| | S | 3 | | 2 | 1 | | | | | | 1 | | | | | | | | | | |
| 20-21 | E | 2 | 1 | 3 | | 1 | | | | | | | | | | | | | | | |
| | S | | | 2 | | | | | | | | | | | | | | | | | |
| 21-22 | E | | | | 1 | | | | | | | | | | | | | | | | |
| | S | | | | 1 | | | | | | | | | | | | | | | | |
| 22-23 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 23-24 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| PARCIAL: | | 64 | 17 | 62 | 10 | 33 | 0 | 0 | 0 | 0 | 46 | 10 | 0 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

ENCUESTADOR : _____ JEFE DE BRIGADA : _____ ING.RESPONS: _____ SUPERV.MTC : _____

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PERÚ | Ministerio de Transportes y Comunicaciones

# FORMATO DE CLASIFICACION VEHICULAR

## ESTUDIO DE TRAFICO - ESTACIÓN PRINCIPAL 1 - MIERCOLES

| | | | |
|---|---|---|---|
| ESTACIÓN PRINCIPAL DE LA CARRETERA | 1 | TINGO DE PONAZA - SHAMBOYACU | |
| SENTIDO | E ← | | S → |
| UBICACIÓN | TINGO DE PONASA - PROVINCIA DE PICOTA | | |
| DIA | Miércoles | | |

| | | | | |
|---|---|---|---|---|
| ESTACION | DV A PASE PROVISIONAL | | | |
| CODIGO DE LA ESTACION | E-01 | | | |
| DIA Y FECHA | MIERCOLES | 16 | 10 | 2024 |

| HORA | SENTIDO | AUTO | STATION WAGON | CAMIONETAS | | RURAL Combi | MICRO | BUS | | | CAMION | | | SEMI TRAYLER | | | | TRAYLER | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | PICK UP | PANEL | | | 2 E | 3 E | 4 E | 2 E | 3 E | 4 E | 2S1/2S2 | 2S3 | 3S1/3S2 | >= 3S3 | 2T2 | 2T3 | 3T2 | >=3T3 |
| 00-01 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 01-02 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 02-03 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 03-04 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 04-05 | E | 3 | 1 | | | 1 | | | | | | | | | | | | | | | |
| | S | 3 | 1 | 2 | | 2 | | | | | 1 | | | | | | | | | | |
| 05-06 | E | 4 | 4 | 2 | 1 | | | | | | 2 | 2 | | | | | | | | | |
| | S | 4 | 1 | 2 | | | | | | | 2 | 1 | | | 2 | | | | | | |
| 06-07 | E | 2 | | 2 | | 2 | | | | | 1 | 1 | | | | | | | | | |
| | S | 2 | 1 | 2 | | 1 | | | | | 3 | 1 | | | | | | | | | |
| 07-08 | E | 2 | 3 | 2 | | 2 | | | | | 1 | 1 | | | | | | | | | |
| | S | 2 | | | | 1 | | | | | 1 | | | | | | | | | | |
| 08-09 | E | 2 | | 1 | | 2 | | | | | 2 | | | | | | | | | | |
| | S | 1 | 2 | 2 | | 2 | | | | | 2 | | | | | | | | | | |
| 09-10 | E | 2 | | 1 | | 3 | | | | | 1 | | | 1 | | | | | | | |
| | S | | | | 1 | | | | | | 2 | | | | | | | | | | |
| 10-11 | E | | | 2 | | | | | | | 3 | | | | | | | | | | |
| | S | | | 2 | 1 | | | | | | 3 | | | | | | | | | | |
| 11-12 | E | 1 | | 2 | | | | | | | 2 | | | | | | | | | | |
| | S | 2 | | 1 | | | | | | | 2 | | | | | | | | | | |
| 12-13 | E | 3 | 1 | 1 | | 1 | | | | | 1 | | | | | | | | | | |
| | S | 3 | 2 | | 1 | | | | | | 1 | | | | | | | | | | |
| 13-14 | E | 2 | 1 | 2 | | 1 | | | | | 1 | | | | | | | | | | |
| | S | 2 | | 3 | | | | | | | 2 | | | | | | | | | | |
| 14-15 | E | 1 | 1 | 1 | | | | | | | | | | | | | | | | | |
| | S | 2 | 1 | 2 | | 1 | | | | | | | | | | | | | | | |
| 15-16 | E | 2 | 1 | 1 | | | | | | | 2 | | | | | | | | | | |
| | S | 1 | 3 | 3 | | 1 | | | | | 3 | | | | | | | | | | |
| 16-17 | E | | 1 | 1 | 1 | | | | | | | | | | | | | | | | |
| | S | 1 | 1 | 1 | 1 | | | | | | 1 | | | | | | | | | | |
| 17-18 | E | 1 | 1 | 1 | | 3 | | | | | 1 | 1 | | | | | | | | | |
| | S | 3 | | | | 4 | | | | | 2 | 1 | | | | | | | | | |
| 18-19 | E | 1 | | 1 | | 3 | | | | | 2 | 1 | | | | | | | | | |
| | S | 2 | 1 | 1 | | 3 | | | | | 1 | 1 | | | | | | | | | |
| 19-20 | E | 2 | 2 | 1 | | 2 | | | | | 1 | 1 | | | | | | | | | |
| | S | 2 | | | 1 | 2 | | | | | 1 | | | | | | | | | | |
| 20-21 | E | | | | 1 | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 21-22 | E | | | | 1 | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 22-23 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 23-24 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| PARCIAL: | | 58 | 29 | 42 | 9 | 37 | 0 | 0 | 0 | 0 | 47 | 11 | 0 | 1 | 2 | 0 | 0 | 0 | 0 | 0 | 0 |

ENCUESTADOR : _____ JEFE DE BRIGADA : _____ ING.RESPONS: _____ SUPERV.MTC : _____

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Dirección Regional de Transportes y Comunicaciones
Ing. David Grandez Silva
CIP. N° 284250
RESPONSABLE DE LA UF DRTC SM

PERÚ | Ministerio de Transportes y Comunicaciones

# FORMATO DE CLASIFICACION VEHICULAR

## ESTUDIO DE TRAFICO - ESTACIÓN PRINCIPAL 1 - JUEVES

| | | | | | |
|---|---|---|---|---|---|
| ESTACIÓN PRINCIPAL DE LA CARRETERA | 1 | TINGO DE PONAZA - SHAMBOYACU | | | |
| SENTIDO | | E ← | | S → | |
| UBICACIÓN | TINGO DE PONASA - PROVINCIA DE PICOTA | | | | |
| DIA | Jueves | | | | |

| | | | | |
|---|---|---|---|---|
| ESTACION | DV A PASE PROVISIONAL | | | |
| CODIGO DE LA ESTACION | E-01 | | | |
| DIA Y FECHA | JUEVES | 17 | 10 | 2024 |

| HORA | SENTIDO | AUTO | STATION WAGON | CAMIONETAS | | RURAL Combi | MICRO | BUS | | | CAMION | | | SEMI TRAYLER | | | | TRAYLER | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | PICK UP | PANEL | | | 2 E | 3 E | 4 E | 2 E | 3 E | 4 E | 2S1/2S2 | 2S3 | 3S1/3S2 | >= 3S3 | 2T2 | 2T3 | 3T2 | >=3T3 |
| 00-01 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 01-02 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 02-03 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 03-04 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 04-05 | E | 2 | | 1 | | | | 4 | | | | | | | | | | | | | |
| | S | 4 | | 3 | | 1 | | 2 | | | 1 | 1 | | | | | | | | | |
| 05-06 | E | 3 | | 2 | | | | 4 | | | 1 | 2 | | | | | | | | | |
| | S | 3 | 1 | 2 | | 1 | | 2 | | | 2 | 1 | | | | | | | | | |
| 06-07 | E | 2 | | 2 | | 1 | | 2 | | | 3 | 1 | | | | | | | | | |
| | S | 1 | 1 | | | 1 | | 2 | | | 3 | | | | | | | | | | |
| 07-08 | E | 1 | | 2 | | 2 | | 1 | | | 1 | | | | | | | | | | |
| | S | 1 | 1 | 1 | | 2 | | 2 | | | 1 | | | | | | | | | | |
| 08-09 | E | 3 | | | | 2 | | 1 | | | 1 | 3 | | | | | | | | | |
| | S | 2 | | 1 | | 3 | | 1 | | | 2 | 4 | | | | | | | | | |
| 09-10 | E | 1 | | 2 | | 1 | | 3 | | | 4 | 2 | | | | | | | | | |
| | S | 3 | | 1 | | 1 | | 1 | | | 3 | 2 | | | | | | | | | |
| 10-11 | E | 1 | | 4 | | 3 | | 4 | | | 1 | 2 | | | | | | | | | |
| | S | 1 | | 5 | 1 | 2 | | 1 | | | 3 | 4 | | | | | | | | | |
| 11-12 | E | 2 | | 1 | | 4 | | 1 | | | 2 | 0 | | | | | | | | | |
| | S | 2 | | 1 | | 3 | | 1 | | | 3 | 3 | | | | | | | | | |
| 12-13 | E | 3 | | 1 | | | | | | | 1 | 2 | | | | | | | | | |
| | S | 2 | | | | | | | | | 2 | 5 | | | | | | | | | |
| 13-14 | E | 1 | | 2 | | 1 | | | | | 1 | | | | | | | | | | |
| | S | | 1 | 1 | | 1 | | | | | 1 | | | | | | | | | | |
| 14-15 | E | | 1 | 2 | | | | | | | | | | | | | | | | | |
| | S | 2 | | 1 | | | | | | | | | | | | | | | | | |
| 15-16 | E | 1 | | 3 | | | | | | | 1 | | | | | | | | | | |
| | S | 2 | 1 | 1 | | | | | | | | 1 | | | | | | | | | |
| 16-17 | E | 1 | | 3 | | | | | | | 5 | 2 | | | | | | | | | |
| | S | | 1 | 2 | | | | | | | 2 | 3 | | | | | | | | | |
| 17-18 | E | 3 | 1 | | | | | | | | 2 | 4 | | | | | | | | | |
| | S | 1 | 1 | 1 | | 1 | | | | | 2 | 5 | | | | | | | | | |
| 18-19 | E | 1 | 1 | 1 | | | | | | | 1 | 4 | | | | | | | | | |
| | S | 2 | | 2 | 1 | 1 | | | | | 2 | 6 | | | | | | | | | |
| 19-20 | E | 1 | | 2 | | | | | | | 3 | 2 | | | | | | | | | |
| | S | 2 | | 3 | | | | | | | 2 | 2 | | | | | | | | | |
| 20-21 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | 1 | | | | | | | | | | | | | | | | |
| 21-22 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 22-23 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 23-24 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| PARCIAL: | | 54 | 10 | 53 | 3 | 31 | 0 | 32 | 0 | 0 | 56 | 61 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

ENCUESTADOR : _____ JEFE DE BRIGADA : _____ ING.RESPONS: _____ SUPERV.MTC : _____

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PERÚ | Ministerio de Transportes y Comunicaciones

# FORMATO DE CLASIFICACION VEHICULAR

## ESTUDIO DE TRAFICO - ESTACIÓN PRINCIPAL 1 - VIERNES

| | | | | | |
|---|---|---|---|---|---|
| ESTACIÓN PRINCIPAL DE LA CARRETERA | 1 | TINGO DE PONAZA - SHAMBOYACU | | | |
| SENTIDO | | E | | S | |
| UBICACIÓN | TINGO DE PONASA - PROVINCIA DE PICOTA | | | | |
| DIA | Viernes | | | | |

| | | | | |
|---|---|---|---|---|
| ESTACION | DV A PASE PROVISIONAL | | | |
| CODIGO DE LA ESTACION | E-01 | | | |
| DIA Y FECHA | VIERNES | 18 | 10 | 2024 |

| HORA | SENTIDO | AUTO | STATION WAGON | CAMIONETAS | | RURAL Combi | MICRO | BUS | | | CAMION | | | SEMI TRAYLER | | | | TRAYLER | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | PICK UP | PANEL | | | 2 E | 3 E | 4 E | 2 E | 3 E | 4 E | 2S1/2S2 | 2S3 | 3S1/3S2 | >= 3S3 | 2T2 | 2T3 | 3T2 | >=3T3 |
| 00-01 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 01-02 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 02-03 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 03-04 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 04-05 | E | 3 | | 2 | | | | | | | 2 | | | | | | | | | | |
| | S | 2 | | 3 | 1 | 1 | | | | | 1 | | | | | | | | | | |
| 05-06 | E | 1 | | 2 | | | | | | | 2 | 1 | | | | | | | | | |
| | S | 3 | | 1 | | | | | | | 2 | 2 | | | | | | | | | |
| 06-07 | E | 1 | 1 | 1 | | 1 | | | | | 2 | 1 | | | | | | | | | |
| | S | 3 | 2 | 3 | | 1 | | | | | 2 | 1 | | | | | | | | | |
| 07-08 | E | 2 | | 3 | | 2 | | | | | 2 | | | | | | | | | | |
| | S | 2 | | 1 | | | | | | | 1 | | | | | | | | | | |
| 08-09 | E | 2 | | 2 | | 4 | | | | | 2 | | | 1 | | | | | | | |
| | S | 1 | | 0 | | 4 | | | | | 2 | | | | | | | | | | |
| 09-10 | E | 2 | | 5 | | 3 | | | | | 2 | 1 | | | | | | | | | |
| | S | 1 | | 2 | | 3 | | | | | 2 | | | | | | | | | | |
| 10-11 | E | 1 | | 2 | | 2 | | | | | 2 | | | | | | | | | | |
| | S | 2 | | 1 | | 2 | | | | | 2 | | | | | | | | | | |
| 11-12 | E | 3 | | 3 | | 1 | | | | | | | | | | | | | | | |
| | S | 2 | | 1 | | 1 | | | | | | | | | | | | | | | |
| 12-13 | E | 2 | | 3 | | | | | | | 1 | 1 | | 1 | | | | | | | |
| | S | 3 | 1 | 2 | | 1 | | | | | 2 | | | | | | | | | | |
| 13-14 | E | 1 | 1 | 2 | | 1 | | | | | 2 | | | | | | | | | | |
| | S | 2 | | 1 | | | | | | | 2 | | | | | | | | | | |
| 14-15 | E | 2 | | 2 | | | | | | | | | | | | | | | | | |
| | S | 3 | | 1 | | | | | | | | | | | | | | | | | |
| 15-16 | E | 2 | | 2 | | | | | | | 1 | | | | | | | | | | |
| | S | | | 5 | | | | | | | | | | | | | | | | | |
| 16-17 | E | 1 | | 1 | | | | | | | 1 | | | | | | | | | | |
| | S | 3 | | 3 | | | | | | | 2 | 1 | | | | | | | | | |
| 17-18 | E | 2 | | 1 | | 1 | | | | | 1 | 1 | | | | | | | | | |
| | S | 1 | 1 | | | 1 | | | | | 2 | 1 | | | | | | | | | |
| 18-19 | E | 3 | 1 | 3 | | | | | | | 1 | | | | | | | | | | |
| | S | 2 | | 1 | | | | | | | 2 | 1 | | | | | | | | | |
| 19-20 | E | 2 | | 3 | | | | | | | 1 | | | | | | | | | | |
| | S | 1 | | 1 | | | | | | | | | | | | | | | | | |
| 20-21 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | 1 | | | | | | | | | | | | | | | | |
| 21-22 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 22-23 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 23-24 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| PARCIAL: | | 61 | 7 | 63 | 2 | 29 | 0 | 0 | 0 | 0 | 44 | 11 | 0 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

ENCUESTADOR : _____ JEFE DE BRIGADA : _____ ING.RESPONS: _____ SUPERV.MTC : _____

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PERÚ | Ministerio de Transportes y Comunicaciones

# FORMATO DE CLASIFICACION VEHICULAR

## ESTUDIO DE TRAFICO - ESTACIÓN PRINCIPAL 1 - SABADO

| | | | | | |
|---|---|---|---|---|---|
| ESTACIÓN PRINCIPAL DE LA CARRETERA | 1 | TINGO DE PONAZA - SHAMBOYACU | | | |
| SENTIDO | | E ← | | S → | |
| UBICACIÓN | TINGO DE PONASA - PROVINCIA DE PICOTA | | | | |
| DIA | Sábado | | | | |

| | | | | |
|---|---|---|---|---|
| ESTACION | DV A PASE PROVISIONAL | | | |
| CODIGO DE LA ESTACION | E-01 | | | |
| DIA Y FECHA | SABADO | 19 | 10 | 2024 |

| HORA | SENTIDO | AUTO | STATION WAGON | CAMIONETAS | | RURAL Combi | MICRO | BUS | | | CAMION | | | SEMI TRAYLER | | | | TRAYLER | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | PICK UP | PANEL | | | 2 E | 3 E | 4 E | 2 E | 3 E | 4 E | 2S1/2S2 | 2S3 | 3S1/3S2 | >= 3S3 | 2T2 | 2T3 | 3T2 | >=3T3 |
| 00-01 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 01-02 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 02-03 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 03-04 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 04-05 | E | 4 | | 2 | | | | | | | | | | | | | | | | | |
| | S | 3 | | 3 | | | | | | | | | | | | | | | | | |
| 05-06 | E | 1 | | 1 | | | | | | | 1 | 1 | | | | | | | | | |
| | S | 3 | 2 | 2 | | | | | | | 3 | 1 | | | | | | | | | |
| 06-07 | E | 1 | 1 | | | 1 | | | | | 3 | 2 | | | | | | | | | |
| | S | 2 | 1 | 1 | | 2 | | | | | 2 | | | | | | | | | | |
| 07-08 | E | 3 | 1 | 2 | | | | | | | 3 | | | | | | | | | | |
| | S | 1 | | 2 | | 1 | | | | | 3 | | | | | | | | | | |
| 08-09 | E | 1 | 1 | | | 2 | | | | | 1 | | | | | | | | | | 1 |
| | S | 1 | | | | 2 | | | | | 2 | 1 | | | | | | | | | |
| 09-10 | E | 1 | | | | 3 | | | | | | | | | | | | | | | |
| | S | 2 | | | | 1 | | | | | | | | | | | | | | | |
| 10-11 | E | 2 | | 1 | | 3 | | | | | 1 | | | | | | | | | | 1 |
| | S | 2 | | | | 5 | | | | | | | | | | | | | | | |
| 11-12 | E | 1 | | | | 3 | | | | | | | | | | | | | | | |
| | S | | | 1 | | 6 | | | | | 2 | | | | | | | | | | |
| 12-13 | E | | | 3 | | 1 | | | | | 1 | 1 | 1 | | | | | | | | |
| | S | 1 | | 1 | | 3 | | | | | 2 | | | | | | | | | | |
| 13-14 | E | 3 | | 2 | | 5 | | | | | 1 | | | 1 | | | | | | | |
| | S | 8 | 1 | 4 | | 5 | | | | | 3 | | | | | | | | | | |
| 14-15 | E | 1 | | 1 | | | | | | | 1 | | | | | | | | | | |
| | S | | | 1 | | | | | | | 2 | | | | | | | | | | |
| 15-16 | E | | 1 | 2 | | | | | | | | | | | | | | | | | |
| | S | 3 | | | | | | | | | | | | | | | | | | | |
| 16-17 | E | 2 | | | | | | | | | 1 | | | 1 | | | | | | | |
| | S | 2 | | | | 1 | | | | | 1 | 1 | | | | | | | | | |
| 17-18 | E | 1 | | 2 | | | | | | | 3 | | 1 | | | | | | | | |
| | S | 3 | 1 | 1 | | | | | | | 1 | 1 | | | | | | | | | |
| 18-19 | E | 2 | | 2 | | | | | | | 1 | 1 | | | | | | | | | |
| | S | 2 | | 1 | | 1 | | | | | | | | | | | | | | | |
| 19-20 | E | 2 | | 2 | | | | | | | 1 | | | | | | | | | | |
| | S | 3 | | 1 | | | | | | | 2 | | | | | | | | | | |
| 20-21 | E | 2 | | 3 | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 21-22 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 22-23 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 23-24 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| PARCIAL: | | 63 | 9 | 41 | 0 | 45 | 0 | 0 | 0 | 0 | 41 | 9 | 2 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 2 |

ENCUESTADOR : _____ JEFE DE BRIGADA : _____ ING.RESPONS: _____ SUPERV.MTC : _____

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PERÚ | Ministerio de Transportes y Comunicaciones

# FORMATO DE CLASIFICACION VEHICULAR

## ESTUDIO DE TRAFICO - ESTACIÓN PRINCIPAL 1 - DOMINGO

| | | | |
|---|---|---|---|
| ESTACIÓN PRINCIPAL DE LA CARRETERA | 1 | TINGO DE PONAZA - SHAMBOYACU | |
| SENTIDO | E ← | | S → |
| UBICACIÓN | TINGO DE PONASA - PROVINCIA DE PICOTA | | |
| DIA | Domingo | | |

| | | | | |
|---|---|---|---|---|
| ESTACION | DV A PASE PROVISIONAL | | | |
| CODIGO DE LA ESTACION | E-01 | | | |
| DIA Y FECHA | DOMINGO | 20 | 10 | 2024 |

| HORA | SENTIDO | AUTO | STATION WAGON | CAMIONETAS | | RURAL Combi | MICRO | BUS | | | CAMION | | | SEMI TRAYLER | | | | TRAYLER | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | PICK UP | PANEL | | | 2 E | 3 E | 4 E | 2 E | 3 E | 4 E | 2S1/2S2 | 2S3 | 3S1/3S2 | >= 3S3 | 2T2 | 2T3 | 3T2 | >=3T3 |
| 00-01 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 01-02 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 02-03 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 03-04 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 04-05 | E | 3 | | 1 | | | | | | | | | | | | | | | | | |
| | S | 1 | | 3 | | | | | | | 1 | | | | | | | | | | |
| 05-06 | E | 4 | 1 | 1 | | | | | | | 2 | 1 | | | | | | | | | |
| | S | 1 | 1 | 3 | | | | | | | 1 | 1 | | | | | | | | | |
| 06-07 | E | 6 | 1 | | | 4 | | | | | 1 | | | | | | | | | | |
| | S | 1 | 1 | 2 | | 5 | | | | | 2 | | | | | | | | | | |
| 07-08 | E | 2 | 1 | | | 6 | | | | | 3 | | | | | | | | | | |
| | S | 1 | 1 | 3 | 1 | 5 | | | | | 3 | | | | | | | | | | |
| 08-09 | E | 1 | | 4 | | 5 | | | | | 1 | | | | | | | | | | |
| | S | 4 | | 3 | | 5 | | | | | 1 | | | | | | | | | | |
| 09-10 | E | 4 | | 1 | | 4 | | | | | 1 | | | | | | | | | | |
| | S | 1 | | | | 4 | | | | | 2 | | | | | | | | | | |
| 10-11 | E | | | 3 | | 3 | | | | | 1 | | | | | | | | | | |
| | S | | | 1 | | 5 | | | | | 1 | | | | | | | | | | |
| 11-12 | E | 1 | | 3 | | 4 | | | | | | | | | | | | | | | |
| | S | | | 1 | | 5 | | | | | | | | | | | | | | | |
| 12-13 | E | 2 | | 2 | | 4 | | | | | 2 | | | 1 | | | | | | | |
| | S | | 1 | 1 | 1 | 7 | | | | | 2 | | | | | | | | | | |
| 13-14 | E | 1 | | 1 | 1 | | | | | | | | | | | | | | | | |
| | S | 3 | | | | 1 | | | | | | | | | | | | | | | |
| 14-15 | E | 2 | | | | | | | | | | 1 | | | | | | | | | |
| | S | 3 | | | | | | | | | | | | | | | | | | | |
| 15-16 | E | 1 | | 1 | | | | | | | 1 | | | | | | | | | | |
| | S | 2 | | 2 | | | | | | | 1 | | | | | | | | | | |
| 16-17 | E | 2 | | 2 | 1 | | | | | | | | | | | | | | | | |
| | S | 3 | | 2 | 1 | | | | | | 3 | | | | | | | | | | |
| 17-18 | E | 4 | 1 | 1 | | 1 | | | | | 1 | 1 | | | | | | | | | |
| | S | 2 | 1 | 2 | | 1 | | | | | 2 | | | | | | | | | | |
| 18-19 | E | 2 | 1 | | | | | | | | 2 | | | | | | | | | | |
| | S | 2 | | | | | | | | | 2 | | | | | | | | | | |
| 19-20 | E | 3 | | 1 | | | | | | | 1 | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 20-21 | E | | | | | | | | | | 1 | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 21-22 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 22-23 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| 23-24 | E | | | | | | | | | | | | | | | | | | | | |
| | S | | | | | | | | | | | | | | | | | | | | |
| PARCIAL: | | 62 | 10 | 44 | 5 | 69 | 0 | 0 | 0 | 0 | 38 | 4 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

ENCUESTADOR : _____ JEFE DE BRIGADA : _____ ING.RESPONS: _____ SUPERV.MTC : _____

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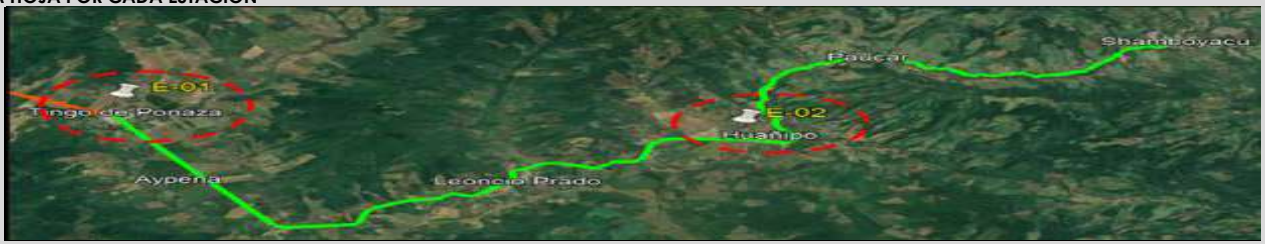
# ASPECTOS DE LA DEMANDA



## 1.GENERALIDADES

**NOTA: USAR ESTA HOJA POR CADA ESTACIÓN**

**LEYENDA:**



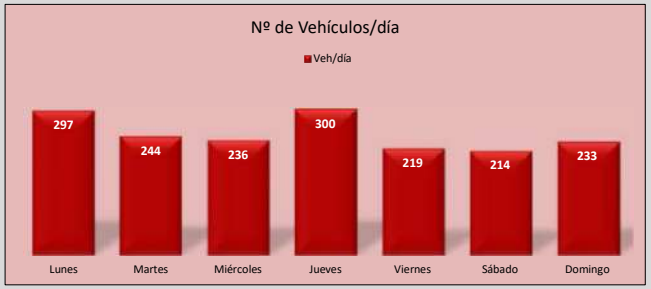

| | Datos a ingresar |
|---|---|
| Departamento: | SAN MARTIN. |
| Provincia: | PICOTA |
| Distrito: | TINGO DE PONASA |
| Horizonte del Proyecto (en años): | 20 Años |

### 1.1 Determinación del tráfico actual

**i) Resumir los conteos de tránsito a nivel del día y tipo de vehículo**

Resultados de los conteo de tráfico: Mes: 10 Año: 2024

| Tipo de Vehículo | Lunes | Martes | Miércoles | Jueves | Viernes | Sábado | Domingo | |
|---|---|---|---|---|---|---|---|---|
| Automovil | 63 | 64 | 58 | 54 | 61 | 63 | 62 | 425 |
| Station Wagon | 9 | 17 | 29 | 10 | 7 | 9 | 10 | 91 |
| Camioneta (Pickup/Panel) | 72 | 72 | 51 | 56 | 65 | 41 | 49 | 406 |
| C.Rural (Combi) | 31 | 33 | 37 | 31 | 29 | 45 | 69 | 275 |
| Micro | | | | | | | | |
| Bus 2E | | | | 32 | | | | 32 |
| Bus 3E | | | | | | | | |
| Bus 4E | | | | | | | | |
| Camión 2E | 61 | 46 | 47 | 56 | 44 | 41 | 38 | 333 |
| Camión 3E | 61 | 10 | 11 | 61 | 11 | 9 | 4 | 167 |
| Camión 4E | | | | | | 2 | | 2 |
| Semitrayler (T2S1 / T2S2) | | 2 | 1 | | 2 | 2 | 1 | 8 |
| Semitrayler (T2S3) | | | 2 | | | | | 2 |
| Semitrayler (T3S1 / T3S2) | | | | | | | | |
| Semitrayler (T3S3) | | | | | | | | |
| Trayler (C2R2) | | | | | | | | |
| Trayler (C2R3) | | | | | | | | |
| Trayler (C3R2) | | | | | | | | |
| Trayler (C3R3) | | | | | | 2 | | 2 |
| TOTAL | 297 | 244 | 236 | 300 | 219 | 214 | 233 | |




**ii) Determinar los factores de corrección estacional de una estación de peaje cercano al camino**

| | | |
|---|---|---|
| F.C.E. Vehículos ligeros: | 1.000000 | Ver hoja A3. FC (Llenar campo manuamente) |
| F.C.E. Vehículos pesados: | 1.000000 | Ver hoja A3. FC (Llenar campo manuamente) |

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**iii) Aplicar la siguiente fórmula, para un conteo de 7 días**

$$IMD_A = IMD_S * FC \quad IMD_S = \frac{(\Sigma Vi)}{7}$$

**Donde:**

$IMD_S$ = Índice Medio Diario Semanal de la Muestra Vehícular Tomada
$IMD_A$ = Índice Medio Anual
Vi = Volumen Vehícular diario de cada uno de los días de conteo
FCE= Factores de Corrección Estacional

| Tipo de Vehículo | Tráfico Vehícular en dos Sentidos por Día | | | | | | | TOTAL | $IMD_S$ | FC | $IMD_a$ | Distribución (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Lunes | Martes | Miércoles | Jueves | Viernes | Sábado | Domingo | | | | | |
| Automovil | 63 | 64 | 58 | 54 | 61 | 63 | 62 | 425 | 61 | 1.000 | 61 | 24.0 |
| Station Wagon | 9 | 17 | 29 | 10 | 7 | 9 | 10 | 91 | 13 | 1.000 | 13 | 5.1 |
| Camioneta (Pickup/Panel) | 72 | 72 | 51 | 56 | 65 | 41 | 49 | 406 | 58 | 1.000 | 58 | 22.8 |
| C.Rural (Combi) | 31 | 33 | 37 | 31 | 29 | 45 | 69 | 275 | 39 | 1.000 | 40 | 15.7 |
| Micro | | | | | | | | | | 1.000 | | |
| Bus 2E | | | | 32 | | | | 32 | 5 | 1.000 | 5 | 2.0 |
| Bus 3E | | | | | | | | | | 1.000 | | |
| Bus 4E | | | | | | | | | | 1.000 | | |
| Camión 2E | 61 | 46 | 47 | 56 | 44 | 41 | 38 | 333 | 48 | 1.000 | 48 | 18.9 |
| Camión 3E | 61 | 10 | 11 | 61 | 11 | 9 | 4 | 167 | 24 | 1.000 | 24 | 9.4 |
| Camión 4E | | | | | | 2 | | 2 | 0 | 1.000 | 1 | 0.4 |
| Semitrayler (T2S1 / T2S2) | | 2 | 1 | | 2 | 2 | 1 | 8 | 1 | 1.000 | 2 | 0.8 |
| Semitrayler (T2S3) | | | 2 | | | | | 2 | 0 | 1.000 | 1 | 0.4 |
| Semitrayler (T3S1 / T3S2) | | | | | | | | | | 1.000 | | |
| Semitrayler (T3S3) | | | | | | | | | | 1.000 | | |
| Trayler (C2R2) | | | | | | | | | | 1.000 | | |
| Trayler (C2R3) | | | | | | | | | | 1.000 | | |
| Trayler (C3R2) | | | | | | | | | | 1.000 | | |
| Trayler (C3R3) | | | | | | 2 | | 2 | 0 | 1.000 | 1 | 0.4 |
| TOTAL | 297 | 244 | 236 | 300 | 219 | 214 | 233 | 1743 | 249 | | 254 | 100.0 |

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## 2. ANALISIS DE LA DEMANDA

### 2.1 Demanda Actual

**Tráfico Actual por Tipo de Vehículo**

| Tipo de Vehículo | IMD | Distribución (%) |
|---|---|---|
| Automovil | 61 | 24.0 |
| Station Wagon | 13 | 5.1 |
| Camioneta (Pickup/Panel) | 58 | 22.8 |
| C.Rural (Combi) | 40 | 15.7 |
| Micro | | |
| Bus 2E | 5 | 2.0 |
| Bus 3E | | |
| Bus 4E | | |
| Camión 2E | 48 | 18.9 |
| Camión 3E | 24 | 9.4 |
| Camión 4E | 1 | 0.4 |
| Semitrayler (T2S1 / T2S2) | 2 | 0.8 |
| Semitrayler (T2S3) | 1 | 0.4 |
| Semitrayler (T3S1 / T3S2) | | |
| Semitrayler (T3S3) | | |
| Trayler (C2R2) | | |
| Trayler (C2R3) | | |
| Trayler (C3R2) | | |
| Trayler (C3R3) | 1 | 0.4 |
| IMD | 254.0 | 100.0 |

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### 2.2 Demanda Proyectada

$$T_n = T_0 * (1 + r)^n$$

**Donde:**

$T_n$ = Tránsito proyectado al año "n" en veh/día
$T_0$ = Tránsito actual (año base) en veh/día
n = año futuro de proyeccción
r = tasa anual de crecimiento de tránsito

**Tasa de Crecimiento x Región en %**

$r_{vp}$ = 2.11% (Ver A4.TC - Tasa de Crecimiento Anual de la Población **(para vehículos de pasajeros)**
$r_{vc}$ = 2.88% (Ver A4.TC - Tasa de Crecimiento Anual del PBI Region **(para vehículos de carga)**

**Proyección de Tráfico - Situación Sin Proyecto**

| Tipo de Vehículo | Año 0 | Año 1 | Año 2 | Año 3 | Año 4 | Año 5 | Año 6 | Año 7 | Año 8 | Año 9 | Año 10 | Año 11 | Año 12 | Año 13 | Año 14 | Año 15 | Año 16 | Año 17 | Año 18 | Año 19 | Año 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Tráfico Normal | 254 | 259 | 265 | 271 | 274 | 281 | 287 | 294 | 299 | 306 | 311 | 318 | 326 | 332 | 340 | 347 | 354 | 361 | 369 | 376 | 388 |
| Automovil | 61 | 62 | 64 | 65 | 66 | 68 | 69 | 71 | 72 | 74 | 75 | 77 | 78 | 80 | 82 | 83 | 85 | 87 | 89 | 91 | 93 |
| Station Wagon | 13 | 13 | 14 | 14 | 14 | 14 | 15 | 15 | 15 | 16 | 16 | 16 | 17 | 17 | 17 | 18 | 18 | 19 | 19 | 19 | 20 |
| Camioneta (Pickup/Panel) | 58 | 59 | 60 | 62 | 63 | 64 | 66 | 67 | 69 | 70 | 71 | 73 | 75 | 76 | 78 | 79 | 81 | 83 | 85 | 86 | 88 |
| C.Rural (Combi) | 40 | 41 | 42 | 43 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 54 | 55 | 56 | 57 | 58 | 60 | 61 |
| Micro | | | | | | | | | | | | | | | | | | | | | |
| Bus 2E | 5 | 5 | 5 | 5 | 5 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 8 |
| Bus 3E | | | | | | | | | | | | | | | | | | | | | |
| Bus 4E | | | | | | | | | | | | | | | | | | | | | |
| Camión 2E | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 56 | 57 | 58 | 59 | 60 | 62 | 63 | 64 | 66 | 67 | 68 | 70 | 71 | 73 |
| Camión 3E | 24 | 25 | 25 | 26 | 26 | 27 | 27 | 28 | 28 | 29 | 30 | 30 | 31 | 31 | 32 | 33 | 34 | 34 | 35 | 36 | 36 |
| Camión 4E | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2 |
| Semitrayler (T2S1 / T2S2) | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| Semitrayler (T2S3) | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2 |
| Semitrayler (T3S1 / T3S2) | | | | | | | | | | | | | | | | | | | | | |
| Semitrayler (T3S3) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C2R2) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C2R3) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C3R2) | | | | | | | | | | | | | | | | | | | | | |

| | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Trayler (C3R3) | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2 |

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## 2.3 Demanda Proyectada "Con Proyecto"

**Tráfico Generado por Tipo de Proyecto**

| Tipo de Intervención | % de Tráfico Normal |
|---|---|
| MEJORAMIENTO | 15 |

| ¿Existe vía alterna? | NO |
|---|---|

**Tráfico Proyectado - Con Proyecto**

| Tipo de Vehículo | Año 0 | Año 1 | Año 2 | Año 3 | Año 4 | Año 5 | Año 6 | Año 7 | Año 8 | Año 9 | Año 10 | Año 11 | Año 12 | Año 13 | Año 14 | Año 15 | Año 16 | Año 17 | Año 18 | Año 19 | Año 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Tráfico Normal** | **254** | **259** | **265** | **271** | **274** | **281** | **287** | **294** | **299** | **306** | **311** | **318** | **326** | **332** | **340** | **347** | **354** | **361** | **369** | **376** | **388** |
| Automovil | 61 | 62 | 64 | 65 | 66 | 68 | 69 | 71 | 72 | 74 | 75 | 77 | 78 | 80 | 82 | 83 | 85 | 87 | 89 | 91 | 93 |
| Station Wagon | 13 | 13 | 14 | 14 | 14 | 14 | 15 | 15 | 15 | 16 | 16 | 16 | 17 | 17 | 17 | 18 | 18 | 19 | 19 | 19 | 20 |
| Camioneta (Pickup/Panel) | 58 | 59 | 60 | 62 | 63 | 64 | 66 | 67 | 69 | 70 | 71 | 73 | 75 | 76 | 78 | 79 | 81 | 83 | 85 | 86 | 88 |
| C.Rural (Combi) | 40 | 41 | 42 | 43 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 54 | 55 | 56 | 57 | 58 | 60 | 61 |
| Micro | | | | | | | | | | | | | | | | | | | | | |
| Bus 2E | 5 | 5 | 5 | 5 | 5 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 8 |
| Bus 3E | | | | | | | | | | | | | | | | | | | | | |
| Bus 4E | | | | | | | | | | | | | | | | | | | | | |
| Camión 2E | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 56 | 57 | 58 | 59 | 60 | 62 | 63 | 64 | 66 | 67 | 68 | 70 | 71 | 73 |
| Camión 3E | 24 | 25 | 25 | 26 | 26 | 27 | 27 | 28 | 28 | 29 | 30 | 30 | 31 | 31 | 32 | 33 | 34 | 34 | 35 | 36 | 36 |
| Camión 4E | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2 |
| Semitrayler (T2S1 / T2S2) | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| Semitrayler (T2S3) | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2 |
| Semitrayler (T3S1 / T3S2) | | | | | | | | | | | | | | | | | | | | | |
| Semitrayler (T3S3) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C2R2) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C2R3) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C3R2) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C3R3) | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2 |
| **Tráfico Generado** | **44** | **45** | **46** | **47** | **47** | **49** | **50** | **51** | **52** | **53** | **53** | **53** | **55** | **56** | **58** | **58** | **61** | **62** | **62** | **62** | **64** |
| Automovil | 10 | 10 | 10 | 10 | 10 | 11 | 11 | 11 | 11 | 12 | 12 | 12 | 12 | 12 | 13 | 13 | 13 | 14 | 14 | 14 | 14 |
| Station Wagon | 2 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| Camioneta (Pickup/Panel) | 9 | 9 | 9 | 10 | 10 | 10 | 10 | 11 | 11 | 11 | 11 | 11 | 12 | 12 | 12 | 12 | 13 | 13 | 13 | 13 | 14 |
| C.Rural (Combi) | 6 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 8 | 8 | 8 | 8 | 8 | 8 | 9 | 9 | 9 | 9 | 9 | 9 | 10 |
| Micro | | | | | | | | | | | | | | | | | | | | | |
| Bus 2E | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| Bus 3E | | | | | | | | | | | | | | | | | | | | | |
| Bus 4E | | | | | | | | | | | | | | | | | | | | | |
| Camión 2E | 8 | 8 | 8 | 8 | 8 | 8 | 9 | 9 | 9 | 9 | 9 | 9 | 10 | 10 | 10 | 10 | 11 | 11 | 11 | 11 | 11 |
| Camión 3E | 4 | 4 | 4 | 4 | 4 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 6 | 6 | 6 | 6 | 6 |
| Camión 4E | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| Semitrayler (T2S1 / T2S2) | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| Semitrayler (T2S3) | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| Semitrayler (T3S1 / T3S2) | | | | | | | | | | | | | | | | | | | | | |
| Semitrayler (T3S3) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C2R2) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C2R3) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C3R2) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C3R3) | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| **Tráfico Desviado** | | | | | | | | | | | | | | | | | | | | | |
| Automovil | | | | | | | | | | | | | | | | | | | | | |
| Station Wagon | | | | | | | | | | | | | | | | | | | | | |
| Camioneta (Pickup/Panel) | | | | | | | | | | | | | | | | | | | | | |
| C.Rural (Combi) | | | | | | | | | | | | | | | | | | | | | |
| Micro | | | | | | | | | | | | | | | | | | | | | |
| Bus 2E | | | | | | | | | | | | | | | | | | | | | |
| Bus 3E | | | | | | | | | | | | | | | | | | | | | |
| Bus 4E | | | | | | | | | | | | | | | | | | | | | |
| Camión 2E | | | | | | | | | | | | | | | | | | | | | |
| Camión 3E | | | | | | | | | | | | | | | | | | | | | |
| Camión 4E | | | | | | | | | | | | | | | | | | | | | |
| Semitrayler (T2S1 / T2S2) | | | | | | | | | | | | | | | | | | | | | |
| Semitrayler (T2S3) | | | | | | | | | | | | | | | | | | | | | |
| Semitrayler (T3S1 / T3S2) | | | | | | | | | | | | | | | | | | | | | |

GOBIERNO REGIONAL SAN MARTIN
Dirección Regional de Transportes y Comunicaciones

Ing. David Grandez Silva
CIP. N° 284250
RESPONSABLE DE LA UF DRTC SM

| | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Semitrayler (T3S3) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C2R2) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C2R3) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C3R2) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C3R3) | | | | | | | | | | | | | | | | | | | | | |
| IMD TOTAL | 298 | 304 | 311 | 318 | 321 | 330 | 337 | 345 | 351 | 359 | 364 | 371 | 381 | 388 | 398 | 405 | 415 | 423 | 431 | 438 | 452 |



TRÁFICO TOTAL

| Tipo de Vehículo | Año 0 | Año 1 | Año 2 | Año 3 | Año 4 | Año 5 | Año 6 | Año 7 | Año 8 | Año 9 | Año 10 | Año 11 | Año 12 | Año 13 | Año 14 | Año 15 | Año 16 | Año 17 | Año 18 | Año 19 | Año 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Automovil | 71 | 72 | 74 | 75 | 76 | 79 | 80 | 82 | 83 | 86 | 87 | 89 | 90 | 92 | 95 | 96 | 98 | 101 | 103 | 105 | 107 |
| Station Wagon | 15 | 15 | 17 | 17 | 17 | 17 | 18 | 18 | 18 | 19 | 19 | 19 | 20 | 20 | 20 | 21 | 21 | 22 | 22 | 22 | 23 |
| Camioneta (Pickup/Panel) | 67 | 68 | 69 | 72 | 73 | 74 | 76 | 78 | 80 | 81 | 82 | 84 | 87 | 88 | 90 | 91 | 94 | 96 | 98 | 99 | 102 |
| C.Rural (Combi) | 46 | 48 | 49 | 50 | 50 | 51 | 52 | 53 | 55 | 56 | 57 | 58 | 59 | 60 | 63 | 64 | 65 | 66 | 67 | 69 | 71 |
| Micro | | | | | | | | | | | | | | | | | | | | | |
| Bus 2E | 6 | 6 | 6 | 6 | 6 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 10 |
| Bus 3E | | | | | | | | | | | | | | | | | | | | | |
| Bus 4E | | | | | | | | | | | | | | | | | | | | | |
| Camión 2E | 56 | 57 | 58 | 59 | 60 | 61 | 63 | 65 | 66 | 67 | 68 | 69 | 72 | 73 | 74 | 76 | 78 | 79 | 81 | 82 | 84 |
| Camión 3E | 28 | 29 | 29 | 30 | 30 | 32 | 32 | 33 | 33 | 34 | 35 | 35 | 36 | 36 | 37 | 38 | 40 | 40 | 41 | 42 | 42 |
| Camión 4E | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 |
| Semitrayler (T2S1 / T2S2) | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| Semitrayler (T2S3) | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 |
| Semitrayler (T3S1 / T3S2) | | | | | | | | | | | | | | | | | | | | | |
| Semitrayler (T3S3) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C2R2) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C2R3) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C3R2) | | | | | | | | | | | | | | | | | | | | | |
| Trayler (C3R3) | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 |

## 2.4 Proyección de la población del área de infiluencia o beneficiarios directos

Proyección de la población

| Población | Años | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Proyección Población | 298 | 304 | 311 | 317 | 324 | 331 | 338 | 345 | 352 | 360 | 367 | 375 | 383 | 391 | 399 | 408 | 416 | 425 | 434 | 443 | 453 |

| Promedio | 374 |
|---|---|

**NOTA: USAR ESTA HOJA POR CADA ESTACIÓN**

GOBIERNO REGIONAL SAN MARTIN
Dirección Regional de Transportes y Comunicaciones
Ing. David Grandez Silva
CIP. N° 284250
RESPONSABLE DE LA UF DRTC SM

# DETERMINACIÓN DE EJES EQUIVALENTES (EE)

HOJA P1. EJES EQUIVALENTES

| Número de Calzadas | 1 | Nro de Sentidos | 2 | Nro de Carriles - Sentido | 2 |
|---|---|---|---|---|---|
| FD | 0.5 | FC | 0.8 | Tipo de pavimento | Flexible |

TRÁFICO TOTAL

| Tipo de Vehículo | Año 0 | Año 1 | Año 2 | Año 3 | Año 4 | Año 5 | Año 6 | Año 7 | Año 8 | Año 9 | Año 10 | Año 11 | Año 12 | Año 13 | Año 14 | Año 15 | Año 16 | Año 17 | Año 18 | Año 19 | Año 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Automovil | 25,915 | 26,280 | 27,010 | 27,375 | 27,740 | 28,835 | 29,200 | 29,930 | 30,295 | 31,390 | 31,755 | 32,485 | 32,850 | 33,580 | 34,675 | 35,040 | 35,770 | 36,865 | 37,595 | 38,325 | 39,055 |
| Station Wagon | 5,475 | 5,475 | 6,205 | 6,205 | 6,205 | 6,205 | 6,570 | 6,570 | 6,570 | 6,935 | 6,935 | 6,935 | 7,300 | 7,300 | 7,300 | 7,665 | 7,665 | 8,030 | 8,030 | 8,030 | 8,395 |
| Camioneta (Pickup/Panel) | 24,455 | 24,820 | 25,185 | 26,280 | 26,645 | 27,010 | 27,740 | 28,470 | 29,200 | 29,565 | 29,930 | 30,660 | 31,755 | 32,120 | 32,850 | 33,215 | 34,310 | 35,040 | 35,770 | 36,135 | 37,230 |
| C.Rural (Combi) | 16,790 | 17,520 | 17,885 | 18,250 | 18,250 | 18,615 | 18,980 | 19,345 | 20,075 | 20,440 | 20,805 | 21,170 | 21,535 | 21,900 | 22,995 | 23,360 | 23,725 | 24,090 | 24,455 | 25,185 | 25,915 |
| Micro | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Bus 2E | 2,190 | 2,190 | 2,190 | 2,190 | 2,190 | 2,555 | 2,555 | 2,555 | 2,555 | 2,555 | 2,555 | 2,555 | 2,555 | 3,285 | 3,285 | 3,285 | 3,285 | 3,285 | 3,285 | 3,285 | 3,650 |
| Bus 3E | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Bus 4E | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Camión 2E | 20,440 | 20,805 | 21,170 | 21,535 | 21,900 | 22,265 | 22,995 | 23,725 | 24,090 | 24,455 | 24,820 | 25,185 | 26,280 | 26,645 | 27,010 | 27,740 | 28,470 | 28,835 | 29,565 | 29,930 | 30,660 |
| Camión 3E | 10,220 | 10,585 | 10,585 | 10,950 | 10,950 | 11,680 | 11,680 | 12,045 | 12,045 | 12,410 | 12,775 | 12,775 | 13,140 | 13,140 | 13,505 | 13,870 | 14,600 | 14,600 | 14,965 | 15,330 | 15,330 |
| Camión 4E | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 1,095 |
| Semitrayler (T2S1 / T2S2) | 1,095 | 1,095 | 1,095 | 1,095 | 1,095 | 1,095 | 1,095 | 1,095 | 1,095 | 1,095 | 1,095 | 1,460 | 1,460 | 1,460 | 1,460 | 1,460 | 1,460 | 1,460 | 1,460 | 1,460 | 1,460 |
| Semitrayler (T2S3) | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 1,095 |
| Semitrayler (T3S1 / T3S2) | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Semitrayler (T3S3) | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Trayler (C2R2) | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Trayler (C2R3) | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Trayler (C3R2) | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Trayler (C3R3) | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 730 | 1,095 |

2.4 EJES EQUIVALENTES

| Tipo de Vehículo | Factor Equivalencia | FC | FD | Factor de Presión Neumático | Ejes Equivalente |
|---|---|---|---|---|---|
| Automovil | 0.0 | 0.5 | 0.8 | 1.0 | 258 |
| Station Wagon | 0.0 | 0.5 | 0.8 | 1.0 | 56 |
| Camioneta (Pickup/Panel) | 0.0 | 0.5 | 0.8 | 1.0 | 246 |
| C.Rural (Combi) | 0.1 | 0.5 | 0.8 | 1.0 | 16,980 |
| Micro | 4.5 | 0.5 | 0.8 | 1.0 | - |
| Bus 2E | 4.5 | 0.5 | 0.8 | 1.1 | 110,671 |
| Bus 3E | 3.3 | 0.5 | 0.8 | 1.1 | - |
| Bus 4E | 2.8 | 0.5 | 0.8 | 1.1 | - |
| Camión 2E | 4.5 | 0.5 | 0.8 | 1.1 | 1,006,893 |
| Camión 3E | 3.3 | 0.5 | 0.8 | 1.1 | 371,410 |
| Camión 4E | 2.8 | 0.5 | 0.8 | 1.1 | 18,266 |
| Semitrayler (T2S1 / T2S2) | 6.5 | 0.5 | 0.8 | 1.1 | 73,330 |
| Semitrayler (T2S3) | 6.2 | 0.5 | 0.8 | 1.1 | 40,888 |
| Semitrayler (T3S1 / T3S2) | 5.3 | 0.5 | 0.8 | 1.1 | - |
| Semitrayler (T3S3) | 5.0 | 0.5 | 0.8 | 1.1 | - |
| Trayler (C2R2) | 11.0 | 0.5 | 0.8 | 1.1 | - |
| Trayler (C2R3) | 9.8 | 0.5 | 0.8 | 1.1 | - |
| Trayler (C3R2) | 9.8 | 0.5 | 0.8 | 1.1 | - |
| Trayler (C3R3) | 8.5 | 0.5 | 0.8 | 1.1 | 56,246 |
| | | | | E.E totales | 1.70E+06 |

**NOTA: USAR ESTA HOJA POR TRAMO DE VÍA**

**Factores de corrección de vehículos ligeros por unidad de peaje - Promedio (2010-2020)**

**HOJA A4. CF - Liviano**

| Nº | Peaje | Enero | Febrero | Marzo | Abril | Mayo | Junio | Julio | Agosto | Setiembre | Octubre | Noviembre | Diciembre | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Ligeros | Ligeros | Ligeros | Ligeros | Ligeros | Ligeros | Ligeros | Ligeros | Ligeros | Ligeros | Ligeros | Ligeros | Ligeros |
| | | FC | FC | FC | FC | FC | FC | FC | FC | FC | FC | FC | FC | FC |
| 1 | AGUAS CALIENTES | 0.9362 | 0.8626 | 1.0850 | 1.0925 | 1.1626 | 1.1758 | 0.9508 | 0.8846 | 0.9455 | 1.0418 | 1.0414 | 0.9899 | 1.0000 |
| 2 | AGUAS CLARAS | 1.0020 | 1.0567 | 1.1036 | 1.0592 | 1.0095 | 1.0030 | 0.8943 | 0.9075 | 1.0128 | 1.0209 | 1.0749 | 0.9283 | 1.0000 |
| 3 | AMBO | 0.9780 | 1.0546 | 1.1877 | 1.2770 | 1.0179 | 1.0278 | 0.9661 | 0.9541 | 1.0045 | 0.9693 | 0.9846 | 0.8615 | 1.0000 |
| 4 | ATICO | 0.8554 | 0.7148 | 1.0360 | 1.0139 | 1.1484 | 1.1781 | 0.9965 | 0.9877 | 1.1394 | 1.0734 | 1.1453 | 0.8831 | 1.0000 |
| 5 | AYAVIRI | 0.9722 | 0.9244 | 1.0735 | 1.0773 | 1.1098 | 1.1035 | 0.9469 | 0.9272 | 0.9376 | 1.0491 | 1.0215 | 0.9603 | 1.0000 |
| 6 | CAMANA | 0.5958 | 0.4991 | 1.0434 | 1.2433 | 1.3825 | 1.4071 | 1.2562 | 1.2200 | 1.3018 | 1.2559 | 1.2199 | 0.9724 | 1.0000 |
| 7 | CANCAS | 0.8619 | 0.8638 | 1.0755 | 1.1156 | 1.1769 | 1.2257 | 0.9733 | 0.9040 | 1.0565 | 1.0323 | 1.0284 | 0.8897 | 1.0000 |
| 8 | CARACOTO | 1.0517 | 0.9815 | 1.0978 | 1.0548 | 1.0540 | 1.0469 | 0.9914 | 0.8786 | 0.9962 | 0.9920 | 0.9977 | 0.8907 | 1.0000 |
| 9 | CASARACRA | 1.1057 | 1.1747 | 1.2363 | 1.0139 | 0.9312 | 1.0287 | 0.8415 | 0.8825 | 1.0603 | 0.9930 | 1.1250 | 0.9775 | 1.0000 |
| 10 | CATAC | 1.0956 | 1.0575 | 1.3545 | 1.0628 | 1.0762 | 1.0767 | 0.8220 | 0.8684 | 0.9595 | 0.9494 | 1.1134 | 0.9559 | 1.0000 |
| 11 | CCASACANCHA | 1.0047 | 1.0408 | 1.0824 | 1.0548 | 1.0802 | 1.0837 | 0.9767 | 0.9426 | 0.9081 | 0.9961 | 0.9756 | 0.9254 | 1.0000 |
| 12 | CHACAPAMPA | 1.0446 | 0.9661 | 0.9178 | 1.0174 | 1.0405 | 1.1257 | 0.9550 | 0.9888 | 1.0378 | 1.0111 | 1.1305 | 1.0215 | 1.0000 |
| 13 | CHALHUAPUQUIO | 1.1671 | 1.2012 | 1.1909 | 1.0566 | 1.0146 | 1.0442 | 0.7925 | 0.8346 | 1.0198 | 0.9725 | 1.0259 | 0.9319 | 1.0000 |
| 14 | CHICAMA | 0.9671 | 0.9414 | 1.1130 | 1.0394 | 1.0773 | 1.0585 | 0.9554 | 1.0080 | 1.0731 | 1.0420 | 1.0520 | 0.8249 | 1.0000 |
| 15 | CHILCA | 0.6163 | 0.5814 | 0.7890 | 1.0577 | 1.5273 | 1.5796 | 1.2842 | 1.4015 | 1.4833 | 1.2554 | 1.1876 | 0.7400 | 1.0000 |
| 16 | CHULLQUI | 1.0428 | 1.0728 | 1.0509 | 1.0163 | 1.0500 | 0.9407 | 0.9832 | 0.9316 | 0.9915 | 0.9207 | 1.2832 | 0.8829 | 1.0000 |
| 17 | CHULLUCANAS | 1.0135 | 1.0695 | 1.1798 | 1.1501 | 1.0679 | 1.0398 | 0.9751 | 0.9127 | 0.9779 | 0.9474 | 0.9119 | 0.8682 | 1.0000 |
| 18 | CIUDAD DE DIOS | 1.4832 | 0.9071 | 1.1711 | 1.0698 | 1.0027 | 1.0245 | 0.9031 | 0.9123 | 1.0735 | 1.0006 | 1.2021 | 0.8987 | 1.0000 |
| 19 | CORCONA | 1.1024 | 1.1641 | 1.3033 | 1.0335 | 0.9767 | 1.0411 | 0.7767 | 0.8755 | 1.0142 | 1.0035 | 1.1663 | 1.0001 | 1.0000 |
| 20 | CRUCE BAYOVAR | 0.8734 | 0.8855 | 1.1136 | 1.1087 | 1.1470 | 1.1828 | 0.8641 | 0.9783 | 1.1469 | 1.0024 | 1.0777 | 0.7715 | 1.0000 |
| 21 | CUCULI | 0.9948 | 1.0346 | 1.1594 | 1.1180 | 1.1059 | 0.9573 | 0.9652 | 0.9185 | 0.9475 | 0.9719 | 0.9647 | 0.9955 | 1.0000 |
| 22 | DESVIO OLMOS | 0.9497 | 1.0057 | 1.1672 | 1.1939 | 1.1494 | 1.0960 | 0.9411 | 0.8681 | 0.9965 | 0.9618 | 1.0119 | 0.8222 | 1.0000 |
| 23 | DESVIO TALARA | 0.8705 | 0.8863 | 1.1244 | 1.0840 | 1.1438 | 1.1754 | 0.9963 | 0.9874 | 1.1057 | 1.0288 | 1.0309 | 0.8166 | 1.0000 |
| 24 | EL FISCAL | 0.9109 | 0.8549 | 1.0606 | 1.0601 | 1.0691 | 1.1260 | 0.9958 | 0.9853 | 1.0750 | 1.0662 | 1.0647 | 0.9479 | 1.0000 |
| 25 | EL PARAISO | 0.9135 | 0.8984 | 1.0432 | 0.9974 | 1.1183 | 1.1530 | 0.9141 | 0.9811 | 1.0964 | 1.0384 | 1.0935 | 0.8747 | 1.0000 |
| 26 | FORTALEZA | 0.8869 | 0.8317 | 1.0074 | 1.0227 | 1.1568 | 1.1965 | 0.8986 | 1.0119 | 1.1837 | 1.1022 | 1.1702 | 0.7936 | 1.0000 |
| 27 | HUACRAPUQUIO | 1.1193 | 1.1570 | 1.0649 | 1.0486 | 1.1504 | 0.9951 | 0.8705 | 0.9487 | 0.9945 | 0.9710 | 1.1529 | 0.8270 | 1.0000 |
| 28 | HUARMEY | 0.8816 | 0.9056 | 1.1120 | 1.1249 | 1.2557 | 1.2048 | 0.8790 | 0.9703 | 1.1483 | 1.0831 | 1.1634 | 0.7903 | 1.0000 |
| 29 | HUATAYA | | | 2.3609 | 0.9600 | 0.8747 | 0.8705 | 0.7536 | 0.7578 | | 0.7931 | 0.8788 | 0.8353 | 1.0000 |
| 30 | ICA | 0.9310 | 0.8181 | 1.0513 | 1.0461 | 1.1642 | 1.1857 | 0.9741 | 1.0246 | 1.0994 | 1.0312 | 1.1911 | 1.0337 | 1.0000 |
| 31 | ILAVE | 1.0111 | 0.9584 | 1.0360 | 1.0136 | 1.1067 | 1.1613 | 0.9700 | 0.7834 | 1.0335 | 1.0501 | 1.1248 | 0.9861 | 1.0000 |
| 32 | ILO | 0.8275 | 0.8217 | 1.0112 | 1.0814 | 1.0777 | 1.1220 | 1.0989 | 1.0557 | 0.9854 | 1.0498 | 1.0303 | 0.9630 | 1.0000 |
| 33 | JAHUAY - CHINCHA | 0.8291 | 0.8936 | 1.0170 | 0.9967 | 1.1373 | 1.1918 | 0.9486 | 1.0137 | 1.1321 | 1.0340 | 1.0634 | 0.9289 | 1.0000 |
| 34 | LOMA LARGA BAJA | 1.0231 | 1.2771 | 1.3671 | 1.2848 | 1.1408 | 1.0369 | 0.8267 | 0.9064 | 0.9247 | 0.8925 | 0.8826 | 0.8752 | 1.0000 |
| 35 | LUNAHUANA | 1.0119 | 1.0303 | 1.0475 | 0.9587 | 1.0163 | 1.1317 | 0.8233 | 0.9366 | 1.0988 | 0.9949 | 1.0780 | 1.0638 | 1.0000 |
| 36 | MACUSANI | 1.0465 | 0.9990 | 1.0338 | 1.0944 | 1.0979 | 1.1081 | 0.9926 | 0.9493 | 1.0253 | 0.9652 | 1.0158 | 0.8879 | 1.0000 |
| 37 | MARCONA | 0.9525 | 0.9039 | 0.9952 | 1.0106 | 1.0911 | 1.0595 | 1.0346 | 1.0232 | 1.0372 | 1.0290 | 0.9989 | 0.9111 | 1.0000 |
| 38 | MATARANI | 0.4810 | 0.3998 | 0.9717 | 1.5343 | 1.7023 | 1.6718 | 1.6092 | 1.5584 | 1.5827 | 1.4257 | 1.2888 | 0.9119 | 1.0000 |
| 39 | MENOCUCHO | 1.0850 | 0.9812 | 1.0500 | 1.0811 | 1.0266 | 1.0713 | 0.9585 | 0.9192 | 1.0498 | 1.0467 | 1.0452 | 0.8579 | 1.0000 |
| 40 | MOCCE | 1.0115 | 0.9769 | 1.0613 | 1.0650 | 1.0408 | 0.9962 | 0.9898 | 0.9054 | 1.0213 | 1.0118 | 1.0012 | 0.9247 | 1.0000 |
| 41 | MONTALVO | 0.9256 | 0.8896 | 1.0507 | 1.0351 | 1.0336 | 1.1014 | 1.0261 | 0.9905 | 1.0447 | 1.0608 | 1.0346 | 0.9658 | 1.0000 |
| 42 | MORROPE | 0.9291 | 0.9180 | 1.2042 | 1.1140 | 1.1254 | 1.1566 | 0.8933 | 0.9593 | 1.0807 | 0.9811 | 1.0512 | 0.8016 | 1.0000 |
| 43 | MOYOBAMBA | 1.0172 | 0.9993 | 1.0281 | 1.0219 | 0.9910 | 0.9724 | 0.9456 | 0.9603 | 1.0486 | 1.0817 | 1.1373 | 1.0800 | 1.0000 |
| 44 | NAZCA | 0.9690 | 0.9023 | 1.0494 | 1.0595 | 1.0762 | 1.0902 | 0.9243 | 0.9306 | 1.0245 | 1.0160 | 1.0671 | 0.9918 | 1.0000 |
| 45 | PACANGUILLA | 0.9539 | 0.9606 | 1.1189 | 1.1170 | 1.1420 | 1.1808 | 0.9622 | 0.9834 | 1.0937 | 0.9832 | 0.9979 | 0.7654 | 1.0000 |
| 46 | PACRA | 1.0091 | 0.9570 | 1.0209 | 0.9881 | 1.2122 | 1.0758 | 0.8772 | 0.9346 | 1.0435 | 1.0345 | 1.0857 | 0.9795 | 1.0000 |
| 47 | PAITA | 0.8362 | 0.8550 | 0.9600 | 1.1002 | 1.1329 | 1.1228 | 1.0946 | 1.0862 | 1.0113 | 1.0572 | 1.0343 | 0.8543 | 1.0000 |
| 48 | PAMPA CUELLAR | 1.0268 | 0.8142 | 1.0724 | 1.0806 | 1.1616 | 1.1803 | 0.9567 | 0.7957 | 1.0641 | 1.0380 | 1.0557 | 0.9772 | 1.0000 |
| 49 | PAMPA GALERA | 0.9640 | 1.0087 | 1.1311 | 1.1214 | 1.0518 | 1.0972 | 0.8190 | 0.7818 | 1.0505 | 1.0760 | 1.1323 | 0.9586 | 1.0000 |
| 50 | PAMPAMARCA | 0.9412 | 0.9552 | 1.0962 | 1.0681 | 1.1301 | 1.1134 | 0.8653 | 0.8837 | 1.0188 | 1.0975 | 1.1159 | 0.9830 | 1.0000 |
| 51 | PATAHUASI | 1.0505 | 0.9287 | 1.1460 | 1.0872 | 1.1086 | 1.1280 | 0.9169 | 0.8010 | 1.0421 | 0.9869 | 1.0075 | 0.9579 | 1.0000 |
| 52 | PEDRO RUIZ | 0.9661 | 1.0279 | 1.1022 | 1.1233 | 1.1127 | 1.0515 | 0.9396 | 0.9041 | 0.9700 | 0.9832 | 1.0130 | 0.8890 | 1.0000 |
| 53 | PICHIRHUA | 1.0465 | 1.0891 | 1.1333 | 1.0561 | 1.0383 | 1.0173 | 0.9081 | 0.8840 | 0.9812 | 0.9968 | 1.0126 | 0.9151 | 1.0000 |
| 54 | PIURA SULLANA | 1.0607 | 1.0540 | 1.1688 | 1.0898 | 1.0558 | 1.0528 | 0.9697 | 0.9539 | 1.0019 | 0.9640 | 0.9557 | 0.8684 | 1.0000 |
| 55 | PLANCHON | 1.0327 | 1.0756 | 1.0641 | 1.0613 | 1.0628 | 1.0224 | 0.9361 | 0.9131 | 0.9554 | 0.9680 | 1.0417 | 0.9385 | 1.0000 |
| 56 | POMAHUACA | 0.9587 | 0.9933 | 1.1653 | 1.2213 | 1.1511 | 1.0886 | 0.9287 | 0.8481 | 1.0002 | 0.9934 | 1.0133 | 0.8296 | 1.0000 |
| 57 | PONGO | 1.0265 | 1.0591 | 1.0560 | 1.0787 | 1.0546 | 1.0025 | 1.0750 | 0.9103 | 0.9692 | 0.9654 | 0.9778 | 0.9308 | 1.0000 |
| 58 | POZO REDONDO | 0.8942 | 0.8240 | 1.0133 | 1.0783 | 1.1178 | 1.0895 | 1.0494 | 1.0438 | 1.1167 | 1.0579 | 1.0216 | 0.8480 | 1.0000 |
| 59 | PUNTA PERDIDA | 0.9544 | 0.7922 | 1.0980 | 1.1995 | 1.4001 | 1.4069 | 0.8238 | 0.6031 | 1.1767 | 1.0628 | 1.2693 | 1.1014 | 1.0000 |
| 60 | QUIULLA | 1.0974 | 1.1408 | 1.2315 | 1.0452 | 1.0212 | 1.0679 | 0.8140 | 0.8685 | 1.0002 | 1.0063 | 1.1046 | 0.9381 | 1.0000 |
| 61 | RUMICHACA | 1.0577 | 0.9112 | 0.9934 | 0.9745 | 1.2193 | 1.2055 | 0.8748 | 0.8955 | 1.0297 | 1.0766 | 1.1671 | 0.9701 | 1.0000 |
| 62 | SAN ANTON | 1.0691 | 1.0523 | 1.0689 | 1.1318 | 1.0035 | 1.0850 | 1.0541 | 1.0138 | 0.9978 | 0.9206 | 0.9981 | 0.8467 | 1.0000 |
| 63 | SAN GABAN | 1.0594 | 1.0125 | 1.0893 | 1.1043 | 1.1321 | 1.0921 | 0.9727 | 0.8972 | 0.9272 | 0.9528 | 0.9512 | 0.9451 | 1.0000 |
| 64 | SAN LORENZO | 1.0839 | 1.1914 | 1.1227 | 1.2647 | 1.0266 | 1.0376 | 0.8703 | 0.9258 | 0.9436 | 0.9610 | 1.0442 | 0.9270 | 1.0000 |
| 65 | SANTA LUCIA | 1.0087 | 0.8363 | 1.1211 | 1.1106 | 1.1184 | 1.1768 | 0.9560 | 0.7674 | 1.0724 | 1.0214 | 1.0523 | 0.9795 | 1.0000 |
| 66 | SAYLLA | 1.0222 | 0.9785 | 1.1062 | 1.0858 | 1.0613 | 1.0638 | 0.9806 | 0.9142 | 0.9238 | 0.9909 | 0.9947 | 0.9411 | 1.0000 |
| 67 | SERPENTIN DE PASAMA | 1.0535 | 1.0503 | 1.0886 | 1.0814 | 1.0647 | 1.0464 | 0.9694 | 0.8540 | 1.0238 | 1.0450 | 1.0263 | 1.0654 | 1.0000 |
| 68 | SICUYANI | 1.0297 | 0.8278 | 1.0261 | 1.0976 | 1.1347 | 1.1691 | 0.9155 | 0.7740 | 1.0939 | 1.0466 | 1.1455 | 1.0126 | 1.0000 |
| 69 | SOCOS | 1.1903 | 0.9863 | 0.9820 | 0.9188 | 1.0940 | 1.0772 | 0.9340 | 0.9544 | 1.0099 | 1.0057 | 1.0421 | 0.9406 | 1.0000 |
| 70 | TAMBOGRANDE | 0.9297 | 0.9679 | 1.0638 | 1.1030 | 1.2229 | 1.0587 | 1.0396 | 1.0332 | 1.0144 | 0.8993 | 0.9585 | 0.8799 | 1.0000 |
| 71 | TOMASIRI | 0.9972 | 0.9232 | 1.0516 | 1.0804 | 1.1000 | 1.0942 | 1.0293 | 0.9900 | 0.8973 | 1.0404 | 1.0386 | 0.9031 | 1.0000 |
| 72 | TUNAN | 1.0761 | 1.0516 | 1.0703 | 1.0230 | 1.1174 | 1.0413 | 0.8626 | 0.8554 | 0.9819 | 0.9733 | 1.1132 | 0.9816 | 1.0000 |
| 73 | UNION PROGRESO | 1.0256 | 1.0263 | 1.0728 | 1.0366 | 1.0256 | 1.0113 | 0.9544 | 0.9355 | 0.9696 | 1.0061 | 1.1122 | 0.9406 | 1.0000 |
| 74 | UTCUBAMBA | 1.1969 | 1.0343 | 1.0957 | 1.1113 | 1.0756 | 1.0478 | 0.9556 | 0.9124 | 0.9571 | 0.9881 | 0.9318 | 0.8952 | 1.0000 |
| 75 | VARIANTE DE PASAMA | 0.9444 | 0.9203 | 1.0337 | 1.0118 | 1.0802 | 1.1121 | 0.9392 | 0.9927 | 1.0740 | 1.0049 | 1.0472 | 0.9053 | 1.0000 |
| 76 | VARIANTE DE UCHUMA | 0.7411 | 0.6766 | 1.0218 | 1.1452 | 1.1929 | 1.1990 | 1.1217 | 1.0762 | 1.1286 | 1.1417 | 1.1231 | 0.9590 | 1.0000 |
| 77 | VESIQUE | 0.8602 | 0.9031 | 1.0597 | 1.0853 | 1.1302 | 1.1518 | 1.0065 | 1.0684 | 1.1196 | 1.0108 | 1.0305 | 0.8067 | 1.0000 |
| 78 | VIRU | 0.9903 | 0.9640 | 1.0774 | 1.0604 | 1.1153 | 1.1221 | 0.9573 | 1.0148 | 1.2094 | 0.9743 | 0.9961 | 0.7989 | 1.0000 |
| 79 | YAUCA | 0.8794 | 0.7946 | 1.0518 | 1.0285 | 1.1247 | 1.1292 | 0.9653 | 0.9839 | 1.0684 | 1.0963 | 1.1303 | 0.9048 | 1.0000 |

Nota: Los valores presentados, son susceptibles a ser actualizados periódicamente por la OPMI-MTC, sin incurrir en actualización de la Ficha Técnica Estándar.

GOBIERNO REGIONAL SAN MARTIN
Dirección Regional de Transportes y Comunicaciones

Ing. David Grandez Silva
CIP. N° 284250
RESPONSABLE DE LA UF DRTC SM

**Factores de corrección de vehículos pesados por unidad de peaje - Promedio (2010-2020)**

**HOJA A4. CF - Pesado**

| Nº | Peaje | Enero | Febrero | Marzo | Abril | Mayo | Junio | Julio | Agosto | Setiembre | Octubre | Noviembre | Diciembre | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pesados | Pesados | Pesados | Pesados | Pesados | Pesados | Pesados | Pesados | Pesados | Pesados | Pesados | Pesados | Pesados |
| | | FC | FC | FC | FC | FC | FC | FC | FC | FC | FC | FC | FC | FC |
| 1 | AGUAS CALIENTES | 1.0220 | 0.9808 | 1.0324 | 1.0653 | 1.0697 | 1.1028 | 0.9678 | 0.9555 | 0.9682 | 0.9530 | 0.9599 | 1.0098 | 1.0000 |
| 2 | AGUAS CLARAS | 1.0412 | 1.0049 | 1.0248 | 1.0107 | 0.9905 | 0.9823 | 0.9917 | 0.9523 | 0.9768 | 1.0070 | 1.0080 | 1.0344 | 1.0000 |
| 3 | AMBO | 1.0087 | 0.9707 | 1.1288 | 1.2086 | 1.0139 | 1.0238 | 1.0268 | 1.0086 | 0.9899 | 0.9453 | 0.9401 | 0.9497 | 1.0000 |
| 4 | ATICO | 1.0337 | 0.9914 | 1.0298 | 1.0410 | 1.0342 | 1.0302 | 1.0277 | 0.9851 | 1.0265 | 0.9618 | 0.9271 | 0.9440 | 1.0000 |
| 5 | AYAVIRI | 1.0167 | 1.0038 | 1.0708 | 1.0573 | 1.0486 | 1.0366 | 0.9665 | 0.9524 | 0.9501 | 0.9638 | 0.9601 | 1.0007 | 1.0000 |
| 6 | CAMANA | 0.9513 | 0.8973 | 1.0392 | 1.0648 | 1.0702 | 1.0905 | 1.1087 | 0.9914 | 0.9997 | 0.9905 | 0.9708 | 0.9624 | 1.0000 |
| 7 | CANCAS | 1.0323 | 0.9825 | 1.0400 | 1.0412 | 1.0604 | 1.0476 | 1.0117 | 0.9795 | 1.0006 | 0.9550 | 0.9461 | 0.9491 | 1.0000 |
| 8 | CARACOTO | 1.0343 | 1.0013 | 1.0777 | 1.0346 | 1.0568 | 1.0441 | 0.9898 | 0.9297 | 0.9572 | 0.9481 | 1.0071 | 0.9547 | 1.0000 |
| 9 | CASARACRA | 1.0936 | 1.0670 | 1.1094 | 0.9826 | 0.9904 | 0.9900 | 0.9969 | 0.9700 | 0.9760 | 0.9548 | 1.0373 | 1.0652 | 1.0000 |
| 10 | CATAC | 1.0441 | 1.0678 | 1.1999 | 1.0772 | 1.1736 | 0.9553 | 0.9446 | 0.9253 | 0.9566 | 0.9547 | 0.9810 | 0.9634 | 1.0000 |
| 11 | CCASACANCHA | 1.0968 | 1.0852 | 1.0957 | 1.0809 | 1.0239 | 0.9967 | 0.9386 | 0.9525 | 0.9555 | 0.9560 | 0.9332 | 0.9566 | 1.0000 |
| 12 | CHACAPAMPA | 1.1031 | 0.9998 | 0.9655 | 0.9941 | 1.0102 | 1.0186 | 1.0155 | 1.0051 | 0.9767 | 0.9556 | 0.9706 | 1.0601 | 1.0000 |
| 13 | CHALHUAPUQUIO | 1.0680 | 1.0692 | 1.0807 | 1.0683 | 1.0507 | 0.9933 | 0.9468 | 0.9321 | 0.9565 | 0.9463 | 0.9497 | 0.9939 | 1.0000 |
| 14 | CHICAMA | 0.9909 | 0.9874 | 1.0928 | 1.0708 | 1.0543 | 1.0573 | 1.0423 | 0.9933 | 1.0013 | 0.9946 | 0.9510 | 0.8799 | 1.0000 |
| 15 | CHILCA | 0.9571 | 0.9724 | 1.0203 | 1.0381 | 1.0583 | 1.0539 | 1.0372 | 0.9948 | 0.9992 | 0.9880 | 0.9652 | 0.9396 | 1.0000 |
| 16 | CHULLQUI | 0.9571 | 0.9658 | 1.0534 | 1.0776 | 1.0809 | 1.0402 | 1.0171 | 0.9865 | 0.9731 | 0.9169 | 1.2400 | 0.9257 | 1.0000 |
| 17 | CHULUCANAS | 0.9884 | 0.9682 | 1.1388 | 1.1641 | 1.0933 | 1.0508 | 1.0256 | 0.9679 | 0.9628 | 0.9205 | 0.8943 | 0.9220 | 1.0000 |
| 18 | CIUDAD DE DIOS | 1.4397 | 0.9383 | 1.1026 | 0.9966 | 0.9637 | 1.0342 | 1.0428 | 1.0261 | 1.0414 | 0.9841 | 0.9930 | 0.9649 | 1.0000 |
| 19 | CORCONA | 1.0985 | 1.0709 | 1.1124 | 0.9593 | 0.9674 | 0.9840 | 0.9816 | 0.9627 | 0.9804 | 0.9767 | 1.0641 | 1.0861 | 1.0000 |
| 20 | CRUCE BAYOVAR | 0.9845 | 0.9606 | 1.0584 | 1.0655 | 1.0311 | 1.0361 | 1.0157 | 0.9821 | 1.0167 | 0.9774 | 0.9231 | 0.9028 | 1.0000 |
| 21 | CUCULI | 0.9424 | 1.0419 | 1.2043 | 1.1560 | 1.0848 | 0.9783 | 0.9812 | 0.9398 | 0.9207 | 0.9302 | 0.9384 | 1.0749 | 1.0000 |
| 22 | DESVIO OLMOS | 1.0470 | 1.0401 | 1.0986 | 1.0775 | 1.0393 | 1.0176 | 0.9898 | 0.9217 | 0.9405 | 0.9579 | 0.9500 | 0.9817 | 1.0000 |
| 23 | DESVIO TALARA | 1.0063 | 0.9722 | 1.0418 | 1.0405 | 1.0343 | 1.0196 | 1.0721 | 0.9782 | 0.9984 | 0.9792 | 0.9631 | 0.9516 | 1.0000 |
| 24 | EL FISCAL | 0.9889 | 0.9326 | 1.0201 | 1.0355 | 1.0216 | 1.0855 | 1.0665 | 1.0209 | 1.0150 | 0.9799 | 0.9618 | 0.9819 | 1.0000 |
| 25 | EL PARAISO | 1.0084 | 0.9860 | 1.0607 | 1.0471 | 1.0250 | 1.0153 | 1.0105 | 0.9788 | 0.9973 | 0.9801 | 0.9625 | 0.9429 | 1.0000 |
| 26 | FORTALEZA | 0.9938 | 0.9560 | 1.0154 | 1.0346 | 1.0384 | 1.3018 | 1.0249 | 0.9743 | 0.9948 | 0.9789 | 0.9539 | 0.9356 | 1.0000 |
| 27 | HUACRAPUQUIO | 1.0850 | 1.1264 | 1.0529 | 1.0465 | 1.1603 | 1.0254 | 0.9226 | 0.9778 | 0.9218 | 0.9085 | 1.1194 | 0.9334 | 1.0000 |
| 28 | HUARMEY | 1.0456 | 1.0252 | 1.1132 | 1.1416 | 1.1272 | 1.0303 | 0.9991 | 0.9554 | 0.9757 | 0.9617 | 0.9397 | 0.9169 | 1.0000 |
| 29 | HUATAYA | | | 5.1132 | 1.3415 | 1.0567 | 1.1183 | 0.7106 | 0.7730 | | 0.7399 | 0.8727 | 0.9802 | 1.0000 |
| 30 | ICA | 0.9911 | 0.9493 | 1.0440 | 1.0283 | 1.0670 | 1.0747 | 1.0593 | 0.9907 | 0.9632 | 0.9523 | 1.0104 | 1.0032 | 1.0000 |
| 31 | ILAVE | 1.0136 | 0.9470 | 0.9822 | 5.5822 | 1.0245 | 1.0346 | 1.0466 | 0.8667 | 1.0477 | 1.0417 | 1.0885 | 1.0638 | 1.0000 |
| 32 | ILO | 1.0476 | 1.0169 | 1.0590 | 0.9766 | 0.9992 | 1.0420 | 1.0373 | 1.0284 | 0.9669 | 0.9802 | 0.9728 | 0.9625 | 1.0000 |
| 33 | JAHUAY - CHINCHA | 0.9858 | 1.0162 | 1.0441 | 1.0554 | 1.0544 | 1.0480 | 1.0262 | 0.9755 | 0.9777 | 0.9697 | 0.9511 | 0.9478 | 1.0000 |
| 34 | LOMA LARGA BAJA | 0.9981 | 1.1016 | 1.1995 | 1.2276 | 1.1103 | 1.0823 | 0.9596 | 0.9761 | 0.9483 | 0.9331 | 0.8987 | 0.8944 | 1.0000 |
| 35 | LUNAHUANA | 1.1095 | 1.0967 | 1.0541 | 1.0430 | 0.9971 | 1.0739 | 1.0009 | 0.9482 | 0.9827 | 0.9329 | 0.9403 | 0.9907 | 1.0000 |
| 36 | MACUSANI | 1.0895 | 1.0802 | 1.0669 | 1.0219 | 1.0756 | 1.0209 | 1.1741 | 0.9353 | 0.9179 | 0.9386 | 0.9390 | 0.9983 | 1.0000 |
| 37 | MARCONA | 1.0430 | 1.0013 | 0.9583 | 1.0297 | 1.1134 | 1.0444 | 1.0480 | 1.0472 | 1.0260 | 0.9629 | 0.9465 | 0.9320 | 1.0000 |
| 38 | MATARANI | 0.9673 | 0.8698 | 1.0297 | 1.0802 | 1.0657 | 1.0279 | 1.0254 | 1.0446 | 0.9864 | 0.9950 | 0.9822 | 0.9819 | 1.0000 |
| 39 | MENOCUCHO | 1.0933 | 1.0653 | 1.1421 | 1.0364 | 0.9913 | 0.9622 | 0.9505 | 0.9625 | 0.9785 | 0.9841 | 0.9862 | 1.0148 | 1.0000 |
| 40 | MOCCE | 0.9510 | 0.9816 | 1.0770 | 1.1377 | 1.0767 | 0.9655 | 1.0381 | 0.9850 | 0.9950 | 0.9641 | 0.9558 | 0.9435 | 1.0000 |
| 41 | MONTALVO | 0.9754 | 0.9558 | 1.0155 | 1.0254 | 1.0095 | 1.0844 | 1.0785 | 1.0275 | 1.0167 | 0.9888 | 0.9674 | 0.9800 | 1.0000 |
| 42 | MORROPE | 0.9854 | 0.9645 | 1.0770 | 1.0663 | 1.0358 | 1.0336 | 1.0257 | 1.0003 | 1.0155 | 0.9853 | 0.9451 | 0.9179 | 1.0000 |
| 43 | MOYOBAMBA | 1.0123 | 0.9812 | 1.0120 | 1.0299 | 1.0077 | 0.9903 | 1.0012 | 0.9653 | 0.9861 | 1.0086 | 1.0129 | 1.0379 | 1.0000 |
| 44 | NAZCA | 1.0562 | 1.0098 | 1.0329 | 1.0328 | 1.0325 | 1.0269 | 1.0128 | 0.9732 | 0.9609 | 0.9602 | 0.9275 | 1.0491 | 1.0000 |
| 45 | PACANGUILLA | 1.0060 | 0.9822 | 1.0858 | 1.0969 | 1.0766 | 1.0798 | 1.0941 | 1.0166 | 0.9911 | 0.9429 | 0.9142 | 0.8905 | 1.0000 |
| 46 | PACRA | 1.0832 | 1.0100 | 1.0286 | 1.0364 | 1.1083 | 1.0029 | 0.9686 | 0.9412 | 0.9706 | 0.9520 | 0.9859 | 0.9960 | 1.0000 |
| 47 | PAITA | 1.0637 | 0.9986 | 1.0332 | 1.1742 | 1.1086 | 1.0719 | 1.1384 | 1.0669 | 0.9588 | 0.9185 | 0.8559 | 0.9255 | 1.0000 |
| 48 | PAMPA CUELLAR | 1.1081 | 1.0770 | 1.0639 | 1.0083 | 1.1112 | 1.0845 | 0.9918 | 0.9561 | 0.9589 | 0.8916 | 0.9121 | 0.9504 | 1.0000 |
| 49 | PAMPA GALERA | 1.1056 | 1.0973 | 1.0952 | 1.0658 | 1.0378 | 1.0205 | 0.9472 | 0.9311 | 0.9550 | 0.9441 | 0.9360 | 0.9479 | 1.0000 |
| 50 | PAMPAMARCA | 1.0833 | 1.0608 | 1.0800 | 1.0714 | 1.0640 | 1.0321 | 0.9519 | 0.9426 | 0.9317 | 0.9501 | 0.9398 | 0.9688 | 1.0000 |
| 51 | PATAHUASI | 1.0668 | 1.0418 | 1.0805 | 1.0643 | 1.0616 | 1.0569 | 1.0065 | 0.9440 | 0.9475 | 0.9036 | 0.9309 | 0.9274 | 1.0000 |
| 52 | PEDRO RUIZ | 1.0361 | 1.0163 | 1.0483 | 1.0529 | 1.0166 | 0.9901 | 0.9900 | 0.9413 | 0.9610 | 0.9787 | 0.9826 | 1.0212 | 1.0000 |
| 53 | PICHIRHUA | 1.0936 | 1.0887 | 1.1081 | 1.0925 | 1.0545 | 1.0256 | 0.9460 | 0.9321 | 0.9383 | 0.9460 | 0.9298 | 0.9399 | 1.0000 |
| 54 | PIURA SULLANA | 1.0571 | 1.0402 | 1.1277 | 1.0518 | 1.0405 | 1.0119 | 1.0179 | 0.9763 | 0.9950 | 0.9755 | 0.9427 | 0.9189 | 1.0000 |
| 55 | PLANCHON | 1.3722 | 1.2585 | 1.1075 | 1.1700 | 1.0496 | 1.0154 | 0.9275 | 0.8807 | 0.8567 | 0.8959 | 0.9614 | 1.0233 | 1.0000 |
| 56 | POMAHUACA | 1.0674 | 1.0301 | 1.1158 | 1.1109 | 1.0645 | 1.0235 | 0.9882 | 0.9203 | 0.9522 | 0.9370 | 0.9332 | 0.9668 | 1.0000 |
| 57 | PONGO | 1.1145 | 1.0702 | 1.0868 | 1.0354 | 0.9963 | 0.9774 | 1.0443 | 0.9020 | 0.9205 | 0.9680 | 0.9712 | 1.0669 | 1.0000 |
| 58 | POZO REDONDO | 1.0157 | 0.9826 | 1.0195 | 1.0517 | 1.0534 | 1.0459 | 1.0140 | 0.9923 | 1.0345 | 1.0012 | 0.9435 | 0.8997 | 1.0000 |
| 59 | PUNTA PERDIDA | 1.1180 | 1.1218 | 1.0774 | 1.0165 | 1.2427 | 1.1333 | 1.0120 | 0.9370 | 0.9279 | 0.8749 | 0.9204 | 0.9490 | 1.0000 |
| 60 | QUIULLA | 1.1367 | 1.0810 | 1.0874 | 0.9355 | 0.9314 | 0.9736 | 0.9528 | 0.9488 | 0.9841 | 1.0038 | 1.1007 | 1.1109 | 1.0000 |
| 61 | RUMICHACA | 1.0807 | 1.0122 | 1.0302 | 1.0888 | 1.0366 | 1.0099 | 0.9690 | 0.9188 | 0.9654 | 0.9569 | 1.0013 | 0.9972 | 1.0000 |
| 62 | SAN ANTON | 1.1667 | 1.0854 | 1.0365 | 1.0303 | 1.0316 | 1.0426 | 1.0281 | 0.9802 | 0.9333 | 0.9279 | 0.9890 | 0.9485 | 1.0000 |
| 63 | SAN GABAN | 1.1273 | 1.0635 | 1.1206 | 1.0840 | 1.1029 | 1.0740 | 1.1605 | 0.8976 | 0.8572 | 0.9080 | 0.9421 | 1.0128 | 1.0000 |
| 64 | SAN LORENZO | 1.6588 | 1.5639 | 1.3338 | 1.4442 | 1.0532 | 1.0350 | 0.9249 | 0.9112 | 0.8771 | 0.8164 | 0.8616 | 0.9665 | 1.0000 |
| 65 | SANTA LUCIA | 1.0356 | 1.0069 | 1.0716 | 1.0685 | 1.0621 | 1.0856 | 1.0229 | 0.9381 | 0.9411 | 0.9138 | 0.9453 | 0.9392 | 1.0000 |
| 66 | SAYLLA | 1.0693 | 1.0277 | 1.0743 | 1.0610 | 1.0379 | 1.0317 | 0.9662 | 0.9474 | 0.9472 | 0.9579 | 0.9470 | 0.9774 | 1.0000 |
| 67 | SERPENTIN DE PASAM | 0.9940 | 1.0077 | 1.0623 | 1.0455 | 1.0254 | 1.0171 | 1.0086 | 0.9839 | 0.9921 | 0.9743 | 0.9582 | 0.9488 | 1.0000 |
| 68 | SICUYANI | 1.1244 | 1.0164 | 1.0598 | 1.0657 | 1.0985 | 1.1286 | 1.0861 | 0.9971 | 0.9910 | 0.9077 | 0.9308 | 0.9614 | 1.0000 |
| 69 | SOCOS | 1.1347 | 1.0494 | 1.0018 | 1.2878 | 1.0911 | 1.0528 | 1.0231 | 0.9946 | 1.0306 | 1.0022 | 1.0652 | 0.9010 | 1.0000 |
| 70 | TAMBOGRANDE | 0.6024 | 0.7500 | 1.1342 | 1.4458 | 1.5102 | 1.2766 | 1.3328 | 1.3204 | 1.2091 | 1.0369 | 0.9407 | 0.7071 | 1.0000 |
| 71 | TOMASIRI | 0.9789 | 0.9106 | 0.9995 | 1.0539 | 1.0515 | 1.0702 | 1.0580 | 1.0075 | 0.9691 | 1.0064 | 1.0067 | 0.9966 | 1.0000 |
| 72 | TUNAN | 1.0648 | 1.0578 | 1.0849 | 1.0647 | 1.0341 | 0.9359 | 0.9355 | 0.9285 | 0.9763 | 0.9641 | 1.0216 | 1.0031 | 1.0000 |
| 73 | UNION PROGRESO | 1.1522 | 1.1289 | 1.0613 | 1.0342 | 0.9994 | 0.9916 | 0.9577 | 0.9143 | 0.9085 | 0.9675 | 1.0861 | 0.9950 | 1.0000 |
| 74 | UTCUBAMBA | 1.1577 | 1.0364 | 1.0667 | 1.0631 | 1.0240 | 0.9908 | 0.9829 | 0.9364 | 0.9605 | 0.9802 | 0.9216 | 1.0143 | 1.0000 |
| 75 | VARIANTE DE PASAMA | 1.1672 | 0.9188 | 1.0160 | 1.0768 | 0.9922 | 0.9811 | 1.0130 | 0.9895 | 1.0586 | 1.0114 | 0.9803 | 0.9711 | 1.0000 |
| 76 | VARIANTE DE UCHUM | 1.0133 | 0.9729 | 1.0446 | 1.0703 | 1.0573 | 1.0429 | 1.0271 | 0.9797 | 0.9635 | 0.9527 | 0.9549 | 0.9498 | 1.0000 |
| 77 | VESIQUE | 1.0255 | 0.9896 | 1.0600 | 1.0910 | 1.0805 | 1.0745 | 1.0534 | 1.0136 | 1.0300 | 0.9408 | 0.9231 | 0.8962 | 1.0000 |
| 78 | VIRU | 1.0337 | 0.9986 | 1.0637 | 1.0876 | 1.0879 | 1.0847 | 1.0635 | 1.0165 | 1.1260 | 0.9199 | 0.8931 | 0.8927 | 1.0000 |
| 79 | YAUCA | 1.0331 | 0.9871 | 1.0294 | 1.0337 | 1.0300 | 1.0338 | 1.0805 | 0.9767 | 0.9747 | 1.0010 | 0.9281 | 0.9422 | 1.0000 |

Nota: Los valores presentados, son susceptibles a ser actualizados periódicamente por la OPMI-MTC, sin incurrir en actualización de la Ficha Técnica Estándar.

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## Tasas para la Proyección de la Demanda

**FORMATO A5.TC**

| Tasa de Crecimiento Anual de Vehículos Ligeros | |
|---|---|
| Amazonas. | **1.12%** |
| Ancash. | **2.33%** |
| Apurímac. | **1.13%** |
| Arequipa. | **2.90%** |
| Ayacucho. | **1.83%** |
| Cajamarca. | **3.05%** |
| Cusco. | **2.77%** |
| Huancavelica. | **1.17%** |
| Huánuco. | **1.94%** |
| Ica. | **1.10%** |
| Junín. | **2.04%** |
| La Libertad. | **2.61%** |
| Lambayeque. | **2.40%** |
| Lima Provincias. | **2.61%** |
| Loreto. | **0.12%** |
| Madre de Dios. | **2.22%** |
| Moquegua. | **1.97%** |
| Pasco. | **1.72%** |
| Piura. | **2.38%** |
| Puno. | **2.47%** |
| San Martin. | **2.11%** |
| Tacna. | **2.13%** |
| Tumbes. | **0.81%** |
| Ucayali. | **1.73%** |

| Tasa de Crecimiento Anual de Vehículos Pesados | |
|---|---|
| Amazonas. | **2.69%** |
| Ancash. | **1.49%** |
| Apurímac. | **4.50%** |
| Arequipa. | **2.97%** |
| Ayacucho. | **2.90%** |
| Cajamarca. | **1.45%** |
| Cusco. | **3.07%** |
| Huancavelica. | **2.00%** |
| Huánuco. | **3.03%** |
| Ica. | **2.62%** |
| Junín. | **2.84%** |
| La Libertad. | **2.21%** |
| Lambayeque. | **2.54%** |
| Lima Provincias. | **2.34%** |
| Loreto. | **1.48%** |
| Madre de Dios. | **1.38%** |
| Moquegua. | **0.58%** |
| Pasco. | **0.39%** |
| Piura. | **2.37%** |
| Puno. | **2.58%** |
| San Martin. | **2.88%** |
| Tacna. | **2.36%** |
| Tumbes. | **2.40%** |
| Ucayali. | **2.25%** |

Informacion al 2022, sin contar las distorciones estadísticas provocadas por el impacto económico de la COVID 19

Nota: Los valores presentados, son susceptibles a ser actualizados periódicamente por la OPMI-MTC, sin incurrir en actualización de la Ficha Técnica Estándar.

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HOJA B. Situación de la UP

# ESTADO SITUACIONAL DE LA UNIDAD PRODUCTORA

## SITUACION ACTUAL - RESULTADO DE LA VISITA DE CAMPO

| Condiciones Iniciales del Proyecto | |
|---|---|
| **Estado Actual** | El proyecto ubicado en la Provincia de Picota y une las localidades de Tingo de Ponaza – Aypena – Nueva América - Leoncio Prado – San Antonio – Huañipo – Paucar – Alfonso Ugarte – Shamboyacu, la vía comprende una longitud Total de 24+747 Km, el que actualmente presenta deficiencias en su superficie de rodadura, como encalaminado, baches y deficiente drenaje longitudinal y transversal, actualmente para recorrer el tramo de 24+747 Km se necesita de un promedio de 2.00 horas aproximadamente por los continuos baches existentes y tramos donde perdió en su totalidad la carpeta asfáltica. |
| **Velocidad de Diseño** | 40 Km/h |

| Características | Situación SIN PROYECTO | Situación CON PROYECTO |
|---|---|---|
| SINAC | DE LA CARRETERA SM-107, TRAMO: TINGO DE PONASA - SHAMBOYACU | DE LA CARRETERA SM-107, TRAMO: TINGO DE PONASA - SHAMBOYACU |
| Longitud | 24.7 | 24.7 |
| IMDa | 254 | 298 |
| Orografía | Plana | Plana |
| Velocidad Directriz | 30 Km/h | 30 Km/h |
| **ALINEAMIENTO HORIZONTAL** | | |
| Radio Mínimo | 95 m | 95 m |
| Radio Excepcional | 30 m | 30 m |
| Nro de Curvas Por Kilòmetro | | |
| Grados Acumulados/kilometro | | |
| **ALINEAMIENTO VERTICAL** | | |
| Pendiente Mínima | 0.5% | 0.5% |
| Pendiente Máxima | 8.0% | 8.0% |
| Pendiente Máxima Excepcional | 8.0% | 8.0% |
| **SECCIONES TRANSVERSALES** | | |
| Número de Carriles | 2 | 2 |
| Ancho de Plataforma (expresado en metros) | 7 | 7 |
| Ancho de Calzada (expresado en metros) | 6 | 6 |
| Ancho de Berma (expresado en metros) | 1 | 1 |
| Tipo de pavimento | Pavimento flexible (asfáltico) | Solución básica (estabilizado+TSB) |
| Cunetas | Triangular y rectangular | |
| Bombeo | 2.50% | 2.50% |
| Sobreanchos | 1.7 | 1.7 |
| Peralte Máximo | 8.00% | 8.00% |
| Plazoletas de Cruce | | |
| Talud en relleno | Variable según el Estudio Geotécnico | |

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# COSTO DE LA INVERSIÓN

HOJA C.Costos

**ALTERNATIVA N°1**

Longitud Total proyecto

| MEJORAMIENTO DEL SERVICIO DE TRANSITABILIDAD VIAL DEL/LA DE LA CARRETERA SM-107, TRAMO: TINGO DE PONASA - SHAMBOYACU DEL DISTRITO DE TINGO DE PONASA , DE LA PROVINCIA DE PICOTA, DEL DEPARTAMENTO DE SAN MARTIN. | | | | | | | Región Geográfica | SELVA |
|---|---|---|---|---|---|---|---|---|
| Naturaleza de acción 1/ | Activos | Tipo de factor producción | Unidad física | | Dimensión física | | Precio unitario (soles/UM) | Costo total (soles) |
| | | | Unidad de medida | Cantidad | Unidad de medida | Cantidad | | |
| **Tramo 1** | | **Longitud (km)** | 24.7 | | | | | 88,262,797.61 |
| Adquisición | Derecho de vía 2/ | Terrenos | Espacios Físicos | | m2 | | | |
| Construcción | Plataforma | Infraestructura | Espacios Físicos | 1.00 | Km | 24.70 | 1,687,212.24 | 41,674,142.22 |
| Construcción | Intersección | Infraestructura | Espacios Físicos | | m2 | | | |
| Construcción | Protección ribereña vial | Infraestructura | Espacios Físicos | | m | | | |
| Construcción | Talud/ muro de contención | Infraestructura | Espacios Físicos | 1 | m | 540 | 19,991.62 | 10,795,473.70 |
| Construcción | Drenaje Transversales | Infraestructura | Espacios Físicos | 1 | m | 794.35 | 13,409.73 | 10,652,022.55 |
| Construcción | Dispositivos de Control de Tránsito | Equipo | Espacios Físicos | | Km | | | |
| Construcción | Puente Peatonal | Infraestructura | Espacios Físicos | | Und | | | |
| Construcción | Estación de Peaje/Pesaje | Infraestructura | Espacios Físicos | | Und | | | |
| Construcción | Puente | Infraestructura | Espacios Físicos | 1 | m | 127 | 197,961.88 | 25,141,159.14 |
| **Tramo 2** | | **Longitud (km)** | | | | | | |
| | Derecho de vía | Terrenos | Und | | m2 | | | |
| | Plataforma | Infraestructura | Und | | Km | | | |
| | Intersección | Infraestructura | Und | | m2 | | | |
| | Protección ribereña vial | Infraestructura | Und | | m | | | |
| | Talud/ muro de contención | Infraestructura | Und | | m | | | |
| | Drenaje Transversales | Infraestructura | Und | | m | | | |
| | Dispositivos de Control de Tránsito | Equipo | Und | | Km | | | |
| | Puente Peatonal | Infraestructura | Und | | Und | | | |
| | Estación de Peaje/Pesaje | Infraestructura | Und | | Und | | | |
| | Puente | Infraestructura | Und | | m | | | |
| **Tramo 3** | | **Longitud (km)** | | | | | | |
| | Derecho de vía | Terrenos | Und | | m2 | | | |
| | Plataforma | Infraestructura | Und | | Km | | | |
| | Intersección | Infraestructura | Und | | m2 | | | |
| | Protección ribereña vial | Infraestructura | Und | | m | | | |
| | Talud/ muro de contención | Infraestructura | Und | | m | | | |
| | Drenaje Transversales | Infraestructura | Und | | m | | | |
| | Dispositivos de Control de Tránsito | Equipo | Und | | Km | | | |
| | Puente Peatonal | Infraestructura | Und | | Und | | | |
| | Estación de Peaje/Pesaje | Infraestructura | Und | | Und | | | |
| | Puente | Infraestructura | Und | | m | | | |

| | |
|---|---|
| **SUB TOTAL DEL COSTO DIRECTO DE LA INVERSIÓN (A)** | 88,262,797.61 |

| | |
|---|---|
| **GASTOS GENERALES** | 9,267,593.75 |
| **UTILIDAD** | 6,178,395.83 |
| **IMPUESTOS (IGV)** | 18,667,581.69 |
| **SUB TOTAL COSTOS INDIRECTO (B)** | 34,113,571.28 |

| | |
|---|---|
| **SUPERVISIÓN** | 4,833,866.57 |
| **EXPEDIENTE TÉCNICO** | 2,500,000.00 |
| **GESTIÓN DEL PROYECTO** | 2,338,964.14 |
| **SUBTOTAL OTROS COSTOS (C)** | 9,672,830.71 |

| | |
|---|---|
| **COSTO TOTAL DE LA INVERSIÓN (A + B + C)** | 132,049,199.60 |
| **CONTROL CONCURRENTE 3/** | 647,746.00 |

1/ La naturaleza de acción adecuadas deben ser identificados en la Guía Genera Identificación, Formulación y Evaluación de Proyectos de Inversión de la DGPMI.

2/ Activo referido al valor de adquisición del predio.

3/ En el marco de la Ley 31358; solo para proyectos con mayor a cinco millones de

**COSTO DE OPERACIÓN Y MANTENIMIENTO REFERENCIALES (*)**

| Descripción | Precios de Mercado S. /Km | Políticas de Operación |
|---|---|---|
| Mant. Periódico Sin Proyecto | 1,500,218.90 | **1 años** |
| Mant. Rutinario Con Proyecto | 432,250.00 | **1 años *** |
| Mant. Periódico Con Proyecto | 605,200.00 | **3 años *** |

*** Variable en base a la alternativa tecnológica**
**Valor resultado de la estructura del Costo de Mantenimiento**

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# EVALUACION ECONOMICA

HOJA D1.Evaluación

Para IMDA menor o igual a 200 emplear esta metodológia

## ALTERNATIVA Nº1

### A) FACTORES DE CONVERSIÓN A PRECIOS SOCIALES

| Obras | Factor |
|---|---|
| Inversión | 0.79 |
| Mantenimiento y Operación | 0.75 |

| | |
|---|---|
| Tasa Social de Descuento | 8% |
| Valor Residual | 10% |

### B) COSTOS DE INVERSIÓN Y MANTENIMIENTO A PRECIOS DEL MERCADO

| Año | Sin Proyecto | Con Proyecto | |
|---|---|---|---|
| | | Alternativa Solucion | |
| | | Inversión | Mantenimiento* |
| 0 | | 132,049,199.6 | |
| 1 | 1,500,218.9 | | 432,250.0 |
| 2 | 1,500,218.9 | | 432,250.0 |
| 3 | 1,500,218.9 | | 605,200.0 |
| 4 | 1,500,218.9 | | 432,250.0 |
| 5 | 1,500,218.9 | | 432,250.0 |
| 6 | 1,500,218.9 | | 605,200.0 |
| 7 | 1,500,218.9 | | 432,250.0 |
| 8 | 1,500,218.9 | | 432,250.0 |
| 9 | 1,500,218.9 | | 605,200.0 |
| 10 | 1,500,218.9 | | 432,250.0 |
| 11 | 1,500,218.9 | | 432,250.0 |
| 12 | 1,500,218.9 | | 605,200.0 |
| 13 | 1,500,218.9 | | 432,250.0 |
| 14 | 1,500,218.9 | | 432,250.0 |
| 15 | 1,500,218.9 | | 605,200.0 |
| 16 | 1,500,218.9 | | 432,250.0 |
| 17 | 1,500,218.9 | | 432,250.0 |
| 18 | 1,500,218.9 | | 605,200.0 |
| 19 | 1,500,218.9 | | 432,250.0 |
| 20 | 1,500,218.9 | -13,204,920.0 | 432,250.0 |

*** Incluye costo de operación, el cual considera un 10% del costo de mantenimiento rutinario.**

### C) COSTOS DE INVERSIÓN Y MANTENIMIENTO A PRECIOS SOCIALES

| Año | Sin Proyecto | Con Proyecto | |
|---|---|---|---|
| | | Alternativa Solucion | |
| | | Inversión | Mantenimiento* |
| 0 | | 104,318,867.68 | |
| 1 | 1,125,164.2 | | 324,187.5 |
| 2 | 1,125,164.2 | | 324,187.5 |
| 3 | 1,125,164.2 | | 453,900.0 |
| 4 | 1,125,164.2 | | 324,187.5 |
| 5 | 1,125,164.2 | | 324,187.5 |
| 6 | 1,125,164.2 | | 453,900.0 |
| 7 | 1,125,164.2 | | 324,187.5 |
| 8 | 1,125,164.2 | | 324,187.5 |
| 9 | 1,125,164.2 | | 453,900.0 |
| 10 | 1,125,164.2 | | 324,187.5 |
| 11 | 1,125,164.2 | | 324,187.5 |
| 12 | 1,125,164.2 | | 453,900.0 |
| 13 | 1,125,164.2 | | 324,187.5 |
| 14 | 1,125,164.2 | | 324,187.5 |
| 15 | 1,125,164.2 | | 453,900.0 |
| 16 | 1,125,164.2 | | 324,187.5 |
| 17 | 1,125,164.2 | | 324,187.5 |
| 18 | 1,125,164.2 | | 453,900.0 |
| 19 | 1,125,164.2 | | 324,187.5 |
| 20 | 1,125,164.2 | -9,903,690.0 | 324,187.5 |

**D) COSTOS INCREMENTALES**
**En Soles a Precios Sociales**

| Año | Alternativa Soluciòn | |
|---|---|---|
| | Inversión | Mantenimiento* |
| 0 | 104,318,867.7 | |
| 1 | | -800,976.7 |
| 2 | | -800,976.7 |
| 3 | | -671,264.2 |
| 4 | | -800,976.7 |
| 5 | | -800,976.7 |
| 6 | | -671,264.2 |
| 7 | | -800,976.7 |
| 8 | | -800,976.7 |
| 9 | | -671,264.2 |
| 10 | | -800,976.7 |
| 11 | | -800,976.7 |
| 12 | | -671,264.2 |
| 13 | | -800,976.7 |
| 14 | | -800,976.7 |
| 15 | | -671,264.2 |
| 16 | | -800,976.7 |
| 17 | | -800,976.7 |
| 18 | | -671,264.2 |
| 19 | | -800,976.7 |
| 20 | | -800,976.7 |
| | Valor Actual de los Costos (VAC) | 96,829,221.84 |
| | Ratio C-E | 259,040.19 |
| | Costo Anual Equivalente (CAE) | 14,430,409.42 |
| | VAC/Km | 3,920,211.41 |

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## RATIO COSTO EFICIENCIA POR KILÓMETRO A PRECIOS SOCIALES SEGÚN ALTERNATIVA TECNOLÓGICA POR REGIÓN

| IMDA Proyectado (Veh/día) | Solución Técnica Recomendada | Horizonte (Años) 4/ | Costa | | Sierra | | Selva | |
|---|---|---|---|---|---|---|---|---|
| | | | † Plataforma Referencial 5/ (Metros) | Costo/Km VAC (Soles) | † Plataforma Referencial 5/ (Metros) | Costo/Km VAC (Soles) | † Plataforma Referencial 5/ (Metros) | Costo/Km (Soles) |
| 101 - 200 | Solución Básica (Estabilizado + Micropavimento) 1/ | 10 | 6 | 1,072,025 | 6 | 1,312,146 | 6 | 1,232,235 |
| | Solución Básica (Estabilizado + TSB) 1/ | 10 | 6 | 1,467,938 | 6 | 1,627,296 | 6 | 1,674,154 |
| | Solución Básica (Estabilizado + Slurry Seal) 1/ | 10 | 6 | 1,073,481 | 6 | 1,319,321 | 6 | 1,237,504 |
| | Solución Básica (Solo Estabilizado) 2/ | 10 | 6 | 629,454 | 6 | 851,847 | 6 | 1,227,603 |
| 000 - 100 | Solución Básica (Estabilizado + TSB) 1/ | 10 | 5 | 1,132,338 | 5 | 1,213,293 | 5 | 1,253,112 |
| | Solución Básica (Estabilizado + Slurry Seal) 1/ | 10 | 5 | 736,991 | 5 | 900,934 | 5 | 813,243 |
| | Solución Básica (Solo Estabilizado) 2/ | 10 | 5 | 428,639 | 5 | 576,299 | 5 | 806,367 |
| | Afirmado | 10 | 5 | 336,115 | 5 | 386,089 | 5 | 526,603 |

**Donde:**

***1/*** Las Alternativas Tecnológicas descritas en esta tabla, representan soluciones convencionales. No obstante, el formulador podrá proponer otra solución tecnológica, tomando en cuenta factores de localización y disponibilidad de materiales, sin que ello impliqu

***2/*** Para un Número de Ejes Equivalentes Igual o Mayor a: 500,000 EE o 35 (veh. pesados)

***3/*** Para un Número de Ejes Equivalentes Igual o Mayor a: 75,000 EE o 4 (veh. pesados)

***4/*** El horizonte de evaluación, considera la vida útil de las soluciones técnicas señaladas, siempre y cuando el mantenimiento rutinario y periódico se realice de forma oportuna bajo responsabilidad de la institución responsable de ello.

***5/*** Plataforma Referencial = Calzada + berma; expresada en metros. Excepcionalmente podrán utilizarse calzadas de hasta 5 m, con el correspondiente sustento técnico y económico

Para identificar la región natural (Costa, Sierra o Selva) a la que pertenece la UP (carretera interurbana) revisar el siguiente enlace:

https://www.inei.gob.pe/biblioteca-virtual/publicaciones-digitales/



# EVALUACION ECONOMICA BENEFICIO - COSTO

HOJA E1.Evaluación

## ALTERNATIVA 1 Para IMDA mayores a 200 emplear esta metodológia

**Situación Sin Proyecto**

| Región | Topografía | Superficie | Estado |
|---|---|---|---|
| Selva | Ondulado | Afirmado | Malo |

Longitud Total proyecto: 24.7 Km — T.C.: 3.77
Horizonte de evaluación: 20.0 años

**COV Sin Proyecto**

| Auto | Station Wagon | Camioneta | Camioneta Rural | Microbús | Bus 2 Ejes | Bus 3 Ejes | Bus 4 Ejes | Camión 2 Ejes | Camión 3 Ejes | Camión 4 Ejes | Semitrailer 2 Ejes | Semitrailer 3 Ejes | Articulado |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| AU | SW | CA | CR | MB | B2 | B3 | B4 | C2 | C3 | C4 | S2 | S3 | TR |
| $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km |
| 0.449202 | 0.451949 | 0.903932 | 1.006189 | 1.813263 | 3.044367 | 3.096653 | 3.497737 | 0.959158 | 1.441426 | 2.375473 | 2.345004 | 2.426971 | 2.642772 |

**Situación Con Proyecto**

| Región | Topografía | Superficie | Estado |
|---|---|---|---|
| Selva | Ondulado | **Asfaltado** | Bueno |

Recorrido reducido por el tráfico desviado: 0.0 (De corresponder, sino considerar "cero")

**COV Con proyecto**

| Auto | Station Wagon | Camioneta | Camioneta Rural | Microbús | Bus 2 Ejes | Bus 3 Ejes | Bus 4 Ejes | Camión 2 Ejes | Camión 3 Ejes | Camión 4 Ejes | Semitrailer 2 Ejes | Semitrailer 3 Ejes | Articulado |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| AU | SW | CA | CR | MB | B2 | B3 | B4 | C2 | C3 | C4 | S2 | S3 | TR |
| $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km |
| 0.393628 | 0.395443 | 0.781752 | 0.897600 | 1.632481 | 2.797527 | 2.840529 | 3.161157 | 0.664663 | 1.007297 | 1.542155 | 1.687721 | 1.738156 | 1.865717 |
| 0.055574 | 0.056506 | 0.122181 | 0.108589 | 0.180782 | 0.246840 | 0.256124 | 0.336580 | 0.294494 | 0.434130 | 0.833318 | 0.657284 | 0.688815 | 0.777054 |

**ΔCOV**

## BENEFICIOS

**Tráfico Normal**

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Año 1** | 135,998 | 28,808 | 282,385 | 177,157 | - | 50,338 | - | - | 570,536 | 427,906 | 56,646 | 67,020 | 46,823 | 52,822 | **1,896,441.16** |
| **Año 2** | 139,776 | 32,650 | 286,538 | 180,848 | - | 50,338 | - | - | 580,545 | 427,906 | 56,646 | 67,020 | 46,823 | 52,822 | **1,921,912.92** |
| **Año 3** | 141,665 | 32,650 | 298,996 | 184,538 | - | 50,338 | - | - | 590,555 | 442,662 | 56,646 | 67,020 | 46,823 | 52,822 | **1,964,715.52** |
| **Año 4** | 143,554 | 32,650 | 303,149 | 184,538 | - | 50,338 | - | - | 600,564 | 442,662 | 56,646 | 67,020 | 46,823 | 52,822 | **1,980,766.52** |
| **Año 5** | 149,220 | 32,650 | 307,302 | 188,229 | - | 58,728 | - | - | 610,573 | 472,173 | 56,646 | 67,020 | 46,823 | 52,822 | **2,042,186.50** |
| **Año 6** | 151,109 | 34,570 | 315,607 | 191,920 | - | 58,728 | - | - | 630,592 | 472,173 | 56,646 | 67,020 | 46,823 | 52,822 | **2,078,010.96** |
| **Año 7** | 154,887 | 34,570 | 323,913 | 195,611 | - | 58,728 | - | - | 650,611 | 486,928 | 56,646 | 67,020 | 46,823 | 52,822 | **2,128,559.10** |
| **Año 8** | 156,776 | 34,570 | 332,218 | 202,992 | - | 58,728 | - | - | 660,620 | 486,928 | 56,646 | 67,020 | 46,823 | 52,822 | **2,156,144.35** |
| **Año 9** | 162,442 | 36,491 | 336,371 | 206,683 | - | 58,728 | - | - | 670,630 | 501,683 | 56,646 | 67,020 | 46,823 | 52,822 | **2,196,339.80** |
| **Año 10** | 164,331 | 36,491 | 340,524 | 210,374 | - | 58,728 | - | - | 680,639 | 516,439 | 56,646 | 67,020 | 46,823 | 52,822 | **2,230,836.95** |
| **Año 11** | 168,109 | 36,491 | 348,829 | 214,064 | - | 58,728 | - | - | 690,649 | 516,439 | 56,646 | 89,360 | 46,823 | 52,822 | **2,278,960.35** |
| **Año 12** | 169,998 | 38,411 | 361,287 | 217,755 | - | 58,728 | - | - | 720,677 | 531,194 | 56,646 | 89,360 | 46,823 | 52,822 | **2,343,702.32** |
| **Año 13** | 173,776 | 38,411 | 365,440 | 221,446 | - | 75,507 | - | - | 730,686 | 531,194 | 56,646 | 89,360 | 46,823 | 52,822 | **2,382,112.37** |
| **Año 14** | 179,442 | 38,411 | 373,745 | 232,518 | - | 75,507 | - | - | 740,696 | 545,949 | 56,646 | 89,360 | 46,823 | 52,822 | **2,431,921.51** |
| **Año 15** | 181,331 | 40,332 | 377,898 | 236,209 | - | 75,507 | - | - | 760,714 | 560,705 | 56,646 | 89,360 | 46,823 | 52,822 | **2,478,348.62** |
| **Año 16** | 185,109 | 40,332 | 390,356 | 239,900 | - | 75,507 | - | - | 780,733 | 590,216 | 56,646 | 89,360 | 46,823 | 52,822 | **2,547,804.89** |
| **Año 17** | 190,775 | 42,252 | 398,662 | 243,591 | - | 75,507 | - | - | 790,743 | 590,216 | 56,646 | 89,360 | 46,823 | 52,822 | **2,577,397.67** |
| **Año 18** | 194,553 | 42,252 | 406,967 | 247,281 | - | 75,507 | - | - | 810,761 | 604,971 | 56,646 | 89,360 | 46,823 | 52,822 | **2,627,945.82** |
| **Año 19** | 198,331 | 42,252 | 411,120 | 254,663 | - | 75,507 | - | - | 820,771 | 619,726 | 56,646 | 89,360 | 46,823 | 52,822 | **2,668,022.60** |
| **Año 20** | 202,109 | 44,173 | 423,578 | 262,044 | - | 83,897 | - | - | 840,790 | 619,726 | 84,970 | 89,360 | 70,235 | 79,233 | **2,800,114.90** |
| **Beneficios** | 3,343,293 | 739,418 | 6,984,887 | 4,292,361 | 0 | 1,283,626 | 0 | 0 | 13,933,086 | 10,387,795 | 1,161,250 | 1,563,803 | 959,881 | 1,082,846 | **45,732,244.82** |

GOBIERNO REGIONAL SAN MARTIN
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Ing. David Grandez Silva
CIP. N° 284250
RESPONSABLE DE LA UF DRTC SM

| | Auto | Station Wagon | Camioneta | Camioneta Rural | Microbús | Bus 2 Ejes | Bus 3 Ejes | Bus 4 Ejes | Camión 2 Ejes | Camión 3 Ejes | Camión 4 Ejes | Semitrailer 2 Ejes | Semitrailer 3 Ejes | Articulado | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | AU | SW | CA | CR | MB | B2 | B3 | B4 | C2 | C3 | C4 | S2 | S3 | TR | |
| **Tráfico Generado** | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | |
| **Año 1** | 9,444 | 1,921 | 18,687 | 12,918 | - | 4,195 | - | - | 40,038 | 29,511 | 14,162 | 11,170 | 11,706 | 13,205 | **166,956** |
| **Año 2** | 9,444 | 2,881 | 18,687 | 12,918 | - | 4,195 | - | - | 40,038 | 29,511 | 14,162 | 11,170 | 11,706 | 13,205 | **167,916** |
| **Año 3** | 9,444 | 2,881 | 20,764 | 12,918 | - | 4,195 | - | - | 40,038 | 29,511 | 14,162 | 11,170 | 11,706 | 13,205 | **169,993** |
| **Año 4** | 9,444 | 2,881 | 20,764 | 12,918 | - | 4,195 | - | - | 40,038 | 29,511 | 14,162 | 11,170 | 11,706 | 13,205 | **169,993** |
| **Año 5** | 10,389 | 2,881 | 20,764 | 12,918 | - | 4,195 | - | - | 40,038 | 36,888 | 14,162 | 11,170 | 11,706 | 13,205 | **178,315** |
| **Año 6** | 10,389 | 2,881 | 20,764 | 12,918 | - | 4,195 | - | - | 45,042 | 36,888 | 14,162 | 11,170 | 11,706 | 13,205 | **183,319** |
| **Año 7** | 10,389 | 2,881 | 22,840 | 12,918 | - | 4,195 | - | - | 45,042 | 36,888 | 14,162 | 11,170 | 11,706 | 13,205 | **185,396** |
| **Año 8** | 10,389 | 2,881 | 22,840 | 14,763 | - | 4,195 | - | - | 45,042 | 36,888 | 14,162 | 11,170 | 11,706 | 13,205 | **187,241** |
| **Año 9** | 11,333 | 2,881 | 22,840 | 14,763 | - | 4,195 | - | - | 45,042 | 36,888 | 14,162 | 11,170 | 11,706 | 13,205 | **188,186** |
| **Año 10** | 11,333 | 2,881 | 22,840 | 14,763 | - | 4,195 | - | - | 45,042 | 36,888 | 14,162 | 11,170 | 11,706 | 13,205 | **188,186** |
| **Año 11** | 11,333 | 2,881 | 22,840 | 14,763 | - | 4,195 | - | - | 45,042 | 36,888 | 14,162 | 11,170 | 11,706 | 13,205 | **188,186** |
| **Año 12** | 11,333 | 2,881 | 24,916 | 14,763 | - | 4,195 | - | - | 50,047 | 36,888 | 14,162 | 11,170 | 11,706 | 13,205 | **195,267** |
| **Año 13** | 11,333 | 2,881 | 24,916 | 14,763 | - | 8,390 | - | - | 50,047 | 36,888 | 14,162 | 11,170 | 11,706 | 13,205 | **199,462** |
| **Año 14** | 12,278 | 2,881 | 24,916 | 16,608 | - | 8,390 | - | - | 50,047 | 36,888 | 14,162 | 11,170 | 11,706 | 13,205 | **202,251** |
| **Año 15** | 12,278 | 2,881 | 24,916 | 16,608 | - | 8,390 | - | - | 50,047 | 36,888 | 14,162 | 11,170 | 11,706 | 13,205 | **202,251** |
| **Año 16** | 12,278 | 2,881 | 26,993 | 16,608 | - | 8,390 | - | - | 55,052 | 44,266 | 14,162 | 11,170 | 11,706 | 13,205 | **216,710** |
| **Año 17** | 13,222 | 2,881 | 26,993 | 16,608 | - | 8,390 | - | - | 55,052 | 44,266 | 14,162 | 11,170 | 11,706 | 13,205 | **217,655** |
| **Año 18** | 13,222 | 2,881 | 26,993 | 16,608 | - | 8,390 | - | - | 55,052 | 44,266 | 14,162 | 11,170 | 11,706 | 13,205 | **217,655** |
| **Año 19** | 13,222 | 2,881 | 26,993 | 16,608 | - | 8,390 | - | - | 55,052 | 44,266 | 14,162 | 11,170 | 11,706 | 13,205 | **217,655** |
| **Año 20** | 13,222 | 2,881 | 29,069 | 18,454 | - | 8,390 | - | - | 55,052 | 44,266 | 14,162 | 11,170 | 11,706 | 13,205 | **221,576** |
| **Beneficios** | 225,720 | 56,657 | 471,335 | 297,107 | 0 | 117,456 | 0 | 0 | 945,888 | 745,147 | 283,232 | 223,400 | 234,117 | 264,109 | **3,864,167** |
| **Tráfico Desviado** | | | | | | | | | | | | | | | |
| **Año 1** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 2** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 3** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 4** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 5** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 6** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 7** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 8** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 9** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 10** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 11** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 12** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 13** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 14** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 15** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 16** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 17** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 18** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 19** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Año 20** | - | - | - | - | - | - | - | - | - | - | - | - | - | - | **-** |
| **Beneficios** | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | **-** |



## ALTERNATIVA N°1

**A) FACTORES DE CONVERSIÓN A PRECIOS SOCIALES**

| Obras | Factor |
|---|---|
| Inversión | 0.79 |
| Mantenimiento y Operación | 0.75 |

Tasa Social de Descu 8%
Valor Residual 10%

**B) COSTOS DE INVERSIÓN Y MANTENIMIENTO A PRECIOS DEL MERCADO**

| Año | Sin Proyecto | Con Proyecto | |
|---|---|---|---|
| | | Alternativa Solucion | |
| | | Inversión | Mantenimiento* |
| 0 | | 132,049,199.6 | |
| 1 | 1,500,218.9 | | 432,250.0 |
| 2 | 1,500,218.9 | | 432,250.0 |
| 3 | 1,500,218.9 | | 605,200.0 |
| 4 | 1,500,218.9 | | 432,250.0 |
| 5 | 1,500,218.9 | | 432,250.0 |
| 6 | 1,500,218.9 | | 605,200.0 |
| 7 | 1,500,218.9 | | 432,250.0 |
| 8 | 1,500,218.9 | | 432,250.0 |
| 9 | 1,500,218.9 | | 605,200.0 |
| 10 | 1,500,218.9 | | 432,250.0 |
| 11 | 1,500,218.9 | | 432,250.0 |
| 12 | 1,500,218.9 | | 605,200.0 |
| 13 | 1,500,218.9 | | 432,250.0 |
| 14 | 1,500,218.9 | | 432,250.0 |
| 15 | 1,500,218.9 | | 605,200.0 |
| 16 | 1,500,218.9 | | 432,250.0 |
| 17 | 1,500,218.9 | | 432,250.0 |
| 18 | 1,500,218.9 | | 605,200.0 |
| 19 | 1,500,218.9 | | 432,250.0 |
| 20 | 1,500,218.9 | -10,563,936.0 | 432,250.0 |

*** Incluye costo de operación, el cual considera un 10% del costo de mantenimiento rutinario.**

**C) COSTOS DE INVERSIÓN Y MANTENIMIENTO A PRECIOS SOCIALES**

| Año | Sin Proyecto | Con Proyecto | |
|---|---|---|---|
| | | Alternativa Solucion | |
| | | Inversión | Mantenimiento* |
| 0 | | 104,318,867.7 | |
| 1 | 1,125,164.2 | | 324,187.5 |
| 2 | 1,125,164.2 | | 324,187.5 |
| 3 | 1,125,164.2 | | 453,900.0 |
| 4 | 1,125,164.2 | | 324,187.5 |
| 5 | 1,125,164.2 | | 324,187.5 |
| 6 | 1,125,164.2 | | 453,900.0 |
| 7 | 1,125,164.2 | | 324,187.5 |
| 8 | 1,125,164.2 | | 324,187.5 |
| 9 | 1,125,164.2 | | 453,900.0 |
| 10 | 1,125,164.2 | 0.0 | 324,187.5 |
| 11 | 1,125,164.2 | | 324,187.5 |
| 12 | 1,125,164.2 | | 453,900.0 |
| 13 | 1,125,164.2 | | 324,187.5 |
| 14 | 1,125,164.2 | | 324,187.5 |
| 15 | 1,125,164.2 | | 453,900.0 |
| 16 | 1,125,164.2 | | 324,187.5 |
| 17 | 1,125,164.2 | | 324,187.5 |
| 18 | 1,125,164.2 | | 453,900.0 |
| 19 | 1,125,164.2 | | 324,187.5 |
| 20 | 1,125,164.2 | | 324,187.5 |

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**D) COSTOS INCREMENTALES**
**En Soles a Precios Sociales**

| | Año | Alternativa Soluciòn | | Beneficios | FLUJO |
|---|---|---|---|---|---|
| | | Inversión | Mantenimiento* | | |
| | 0 | 104,318,867.7 | | | -104,318,867.7 |
| | 1 | | -800,976.7 | 2,063,397.2 | 2,864,373.8 |
| | 2 | | -800,976.7 | 2,089,829.2 | 2,890,805.9 |
| | 3 | | -671,264.2 | 2,134,708.2 | 2,805,972.3 |
| | 4 | | -800,976.7 | 2,150,759.2 | 2,951,735.8 |
| | 5 | | -800,976.7 | 2,220,501.3 | 3,021,478.0 |
| | 6 | | -671,264.2 | 2,261,330.4 | 2,932,594.6 |
| | 7 | | -800,976.7 | 2,313,954.9 | 3,114,931.6 |
| | 8 | | -800,976.7 | 2,343,385.6 | 3,144,362.2 |
| | 9 | | -671,264.2 | 2,384,525.5 | 3,055,789.6 |
| | 10 | 0.0 | -800,976.7 | 2,419,022.6 | 3,219,999.3 |
| | 11 | | -800,976.7 | 2,467,146.0 | 3,268,122.7 |
| | 12 | | -671,264.2 | 2,538,969.0 | 3,210,233.2 |
| | 13 | | -800,976.7 | 2,581,573.9 | 3,382,550.6 |
| | 14 | | -800,976.7 | 2,634,172.9 | 3,435,149.6 |
| | 15 | | -671,264.2 | 2,680,600.0 | 3,351,864.2 |
| | 16 | | -800,976.7 | 2,764,515.0 | 3,565,491.7 |
| | 17 | | -800,976.7 | 2,795,052.3 | 3,596,028.9 |
| | 18 | | -671,264.2 | 2,845,600.4 | 3,516,864.6 |
| | 19 | | -800,976.7 | 2,885,677.2 | 3,686,653.9 |
| | 20 | | -800,976.7 | 3,021,691.2 | 3,822,667.9 |
| Actualizado | 8% | 104,318,867.7 | -7,489,645.8 | 23,192,425.8 | |
| | | | | VAN | 30,682,071.67 |
| | | | | B/C | 2.24 |
| | | | | TIR | 8.10% |

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HOJA E2. COV Interurbano

| Región | Topografía | Superficie | Estado | COV para Carreteras Interurbanas | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Auto | Station Wagon | Camioneta | Camioneta Rural | Microbús | Bus 2 Ejes | Bus 3 Ejes | Bus 4 Ejes | Camión 2 Ejes | Camión 3 Ejes | Camión 4 Ejes | Semitrailer 2 Ejes | Semitrailer 3 Ejes | Articulado |
| | | | | AU | SW | CA | CR | MB | B2 | B3 | B4 | C2 | C3 | C4 | S2 | S3 | TR |
| | | | | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km |
| Costa | Plano | Concreto Hidráulico | Bueno | 0.291423 | 0.297343 | 0.558305 | 0.700603 | 1.233408 | 1.994261 | 2.120180 | 2.388761 | 0.483342 | 0.654240 | 1.011659 | 1.125892 | 1.338216 | 1.446336 |
| Costa | Plano | Concreto Hidráulico | Regular | 0.304609 | 0.313114 | 0.585488 | 0.722383 | 1.259343 | 2.029704 | 2.157772 | 2.439933 | 0.546557 | 0.752200 | 1.142510 | 1.236702 | 1.475983 | 1.601717 |
| Costa | Plano | Concreto Hidráulico | Malo | 0.321931 | 0.333970 | 0.623431 | 0.754777 | 1.296422 | 2.085118 | 2.213397 | 2.510525 | 0.606926 | 0.847710 | 1.272431 | 1.350239 | 1.616936 | 1.760647 |
| Costa | Plano | Asfaltado | Bueno | 0.295456 | 0.302149 | 0.566404 | 0.706942 | 1.241171 | 2.004539 | 2.131315 | 2.404290 | 0.504937 | 0.687381 | 1.055479 | 1.162782 | 1.384060 | 1.498025 |
| Costa | Plano | Asfaltado | Regular | 0.315594 | 0.326332 | 0.609355 | 0.742548 | 1.282455 | 2.063728 | 2.192224 | 2.484187 | 0.586961 | 0.815980 | 1.229141 | 1.311837 | 1.569324 | 1.706988 |
| Costa | Plano | Asfaltado | Malo | 0.336622 | 0.351690 | 0.656535 | 0.784185 | 1.330253 | 2.138610 | 2.265454 | 2.573501 | 0.646681 | 0.911172 | 1.359164 | 1.429296 | 1.714698 | 1.870692 |
| Costa | Plano | Tratamiento Superficial | Bueno | 0.299833 | 0.307384 | 0.575419 | 0.714150 | 1.249749 | 2.016223 | 2.143730 | 2.421228 | 0.525954 | 0.719978 | 1.099073 | 1.199669 | 1.429929 | 1.549762 |
| Costa | Plano | Tratamiento Superficial | Regular | 0.321931 | 0.333970 | 0.623431 | 0.754777 | 1.296422 | 2.085118 | 2.213397 | 2.510525 | 0.606926 | 0.847710 | 1.272431 | 1.350239 | 1.616936 | 1.760647 |
| Costa | Plano | Tratamiento Superficial | Malo | 0.364665 | 0.385530 | 0.720343 | 0.842190 | 1.398323 | 2.249933 | 2.372104 | 2.698159 | 0.706540 | 1.007083 | 1.490153 | 1.554228 | 1.868540 | 2.043436 |
| Costa | Plano | Afirmado | Bueno | 0.336622 | 0.351690 | 0.656535 | 0.784185 | 1.330253 | 2.138610 | 2.265454 | 2.573501 | 0.646681 | 0.911172 | 1.359164 | 1.429296 | 1.714698 | 1.870692 |
| Costa | Plano | Afirmado | Regular | 0.375854 | 0.399040 | 0.745807 | 0.865515 | 1.426087 | 2.295955 | 2.415999 | 2.748521 | 0.726690 | 1.039390 | 1.534167 | 1.597524 | 1.921703 | 2.103017 |
| Costa | Plano | Afirmado | Malo | 0.372197 | 0.387900 | 0.741955 | 0.898850 | 1.562992 | 2.523656 | 2.633716 | 2.994502 | 0.808887 | 1.170946 | 1.712553 | 1.777488 | 2.142040 | 2.349470 |
| Costa | Plano | Sin Afirmar | Regular | 0.411888 | 0.427915 | 0.820088 | 1.009462 | 1.722377 | 2.817560 | 2.910664 | 3.279779 | 0.794311 | 1.157929 | 1.701276 | 1.800889 | 2.167332 | 2.375927 |
| Costa | Plano | Sin Afirmar | Malo | 0.439024 | 0.457289 | 0.883007 | 1.082945 | 1.866362 | 3.045499 | 3.135227 | 3.534808 | 0.881358 | 1.295241 | 1.884996 | 1.981149 | 2.388227 | 2.623021 |
| Costa | Plano | Trocha | Regular | 0.424831 | 0.441986 | 0.850271 | 1.044389 | 1.789823 | 2.924219 | 3.015731 | 3.399604 | 0.837391 | 1.226009 | 1.792533 | 1.889947 | 2.276592 | 2.498219 |
| Costa | Plano | Trocha | Malo | 0.506400 | 0.528923 | 1.034498 | 1.266559 | 2.213828 | 3.608563 | 3.691651 | 4.112703 | 1.065244 | 1.582419 | 2.266386 | 2.359898 | 2.850281 | 3.138616 |
| Costa | Ondulado | Concreto Hidráulico | Bueno | 0.317078 | 0.323226 | 0.628177 | 0.801046 | 1.405361 | 2.324584 | 2.529262 | 2.827265 | 0.566068 | 0.809568 | 1.283247 | 1.364524 | 1.594627 | 1.715170 |
| Costa | Ondulado | Concreto Hidráulico | Regular | 0.330033 | 0.338744 | 0.654323 | 0.821327 | 1.429704 | 2.357277 | 2.563955 | 2.875169 | 0.629478 | 0.907309 | 1.414662 | 1.473972 | 1.731108 | 1.868650 |
| Costa | Ondulado | Concreto Hidráulico | Malo | 0.346707 | 0.358890 | 0.689825 | 0.850217 | 1.462912 | 2.405225 | 2.612370 | 2.937933 | 0.689747 | 1.002156 | 1.544793 | 1.583697 | 1.868259 | 2.022751 |
| Costa | Ondulado | Asfaltado | Bueno | 0.321065 | 0.327982 | 0.636049 | 0.807055 | 1.412762 | 2.334230 | 2.539747 | 2.842039 | 0.587751 | 0.842671 | 1.327280 | 1.401128 | 1.640211 | 1.766424 |
| Costa | Ondulado | Asfaltado | Regular | 0.340648 | 0.351556 | 0.676749 | 0.839423 | 1.450571 | 2.387029 | 2.594236 | 2.914877 | 0.669859 | 0.970711 | 1.501485 | 1.546934 | 1.822297 | 1.971145 |
| Costa | Ondulado | Asfaltado | Malo | 0.360625 | 0.375764 | 0.720331 | 0.875907 | 1.492297 | 2.449883 | 2.656078 | 2.991948 | 0.729193 | 1.064824 | 1.631372 | 1.658232 | 1.961406 | 2.127192 |
| Costa | Ondulado | Tratamiento Superficial | Bueno | 0.325368 | 0.333137 | 0.644726 | 0.813775 | 1.420826 | 2.345070 | 2.551229 | 2.857922 | 0.608835 | 0.875198 | 1.371065 | 1.437581 | 1.685674 | 1.817553 |
| Costa | Ondulado | Tratamiento Superficial | Regular | 0.346707 | 0.358890 | 0.689825 | 0.850217 | 1.462912 | 2.405225 | 2.612370 | 2.937933 | 0.689747 | 1.002156 | 1.544793 | 1.583697 | 1.868259 | 2.022751 |
| Costa | Ondulado | Tratamiento Superficial | Malo | 0.386920 | 0.407713 | 0.778858 | 0.926362 | 1.550544 | 2.541788 | 2.744159 | 3.096777 | 0.788136 | 1.158897 | 1.761573 | 1.773513 | 2.105244 | 2.288135 |
| Costa | Ondulado | Afirmado | Bueno | 0.360625 | 0.375764 | 0.720331 | 0.875907 | 1.492297 | 2.449883 | 2.656078 | 2.991948 | 0.729193 | 1.064824 | 1.631372 | 1.658232 | 1.961406 | 2.127192 |
| Costa | Ondulado | Afirmado | Regular | 0.397380 | 0.420441 | 0.802268 | 0.946750 | 1.574259 | 2.579936 | 2.780356 | 3.138935 | 0.807850 | 1.190408 | 1.805159 | 1.813000 | 2.154437 | 2.343122 |
| Costa | Ondulado | Afirmado | Malo | 0.390327 | 0.405719 | 0.789474 | 0.967308 | 1.692447 | 2.772475 | 2.962633 | 3.346600 | 0.887663 | 1.317887 | 1.980969 | 1.976414 | 2.357336 | 2.569878 |
| Costa | Ondulado | Sin Afirmar | Regular | 0.440490 | 0.456064 | 0.882625 | 1.098914 | 1.893125 | 3.142629 | 3.301445 | 3.692888 | 0.839869 | 1.280723 | 1.952313 | 1.998097 | 2.380511 | 2.593327 |
| Costa | Ondulado | Sin Afirmar | Malo | 0.464248 | 0.482025 | 0.937812 | 1.161388 | 2.017779 | 3.336765 | 3.491425 | 3.911565 | 0.926152 | 1.413949 | 2.132688 | 2.165217 | 2.587360 | 2.825239 |
| Costa | Ondulado | Trocha | Regular | 0.451819 | 0.468500 | 0.909024 | 1.128455 | 1.951217 | 3.232841 | 3.389822 | 3.795176 | 0.882577 | 1.346808 | 2.042004 | 2.080519 | 2.482621 | 2.707874 |
| Costa | Ondulado | Trocha | Malo | 0.524313 | 0.546359 | 1.074077 | 1.323295 | 2.326257 | 3.834197 | 3.977428 | 4.415215 | 1.108791 | 1.693189 | 2.506186 | 2.523552 | 3.028723 | 3.318641 |
| Costa | Accidentado | Concreto Hidráulico | Bueno | 0.368532 | 0.374483 | 0.754959 | 0.981341 | 1.804804 | 3.065968 | 3.433157 | 3.799023 | 0.658585 | 1.055831 | 1.720972 | 1.803195 | 2.064255 | 2.215285 |
| Costa | Accidentado | Concreto Hidráulico | Regular | 0.381119 | 0.389612 | 0.779313 | 0.998737 | 1.826061 | 3.094862 | 3.464647 | 3.843265 | 0.723185 | 1.155020 | 1.856875 | 1.916520 | 2.205097 | 2.373086 |
| Costa | Accidentado | Concreto Hidráulico | Malo | 0.396968 | 0.408885 | 0.811715 | 1.023179 | 1.852946 | 3.132786 | 3.504071 | 3.888303 | 0.784559 | 1.251008 | 1.991332 | 2.028535 | 2.344888 | 2.529781 |
| Costa | Accidentado | Asfaltado | Bueno | 0.372430 | 0.379144 | 0.762341 | 0.986542 | 1.811410 | 3.074852 | 3.442900 | 3.822000 | 0.666066 | 1.089438 | 1.766507 | 1.841183 | 2.111392 | 2.268077 |
| Costa | Accidentado | Asfaltado | Regular | 0.391253 | 0.401915 | 0.799853 | 1.014096 | 1.843201 | 3.118844 | 3.489811 | 3.877409 | 0.764317 | 1.219229 | 1.946612 | 1.991277 | 2.298335 | 2.477604 |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Auto | Station Wagon | Camioneta | Camioneta Rural | Microbús | Bus 2 Ejes | Bus 3 Ejes | Bus 4 Ejes | Camión 2 Ejes | Camión 3 Ejes | Camión 4 Ejes | Semitrailer 2 Ejes | Semitrailer 3 Ejes | Articulado |
| | | | | AU | SW | CA | CR | MB | B2 | B3 | B4 | C2 | C3 | C4 | S2 | S3 | TR |
| | | | | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km |
| Costa | Accidentado | Asfaltado | Malo | 0.409935 | 0.424757 | 0.839174 | 1.044567 | 1.875323 | 3.165493 | 3.536750 | 3.938547 | 0.824654 | 1.314168 | 2.080609 | 2.103009 | 2.438080 | 2.634174 |
| Costa | Accidentado | Tratamiento Superficial | Bueno | 0.376614 | 0.384173 | 0.770429 | 0.992312 | 1.818469 | 3.084445 | 3.453447 | 3.827719 | 0.702151 | 1.122450 | 1.811791 | 1.878944 | 2.158325 | 2.320663 |
| Costa | Accidentado | Tratamiento Superficial | Regular | 0.396968 | 0.408885 | 0.811715 | 1.023179 | 1.852946 | 3.132786 | 3.504071 | 3.896303 | 0.784559 | 1.251008 | 1.991332 | 2.028535 | 2.344888 | 2.529781 |
| Costa | Accidentado | Tratamiento Superficial | Malo | 0.434042 | 0.454403 | 0.891502 | 1.086258 | 1.917802 | 3.229686 | 3.598749 | 4.015661 | 0.884389 | 1.408442 | 2.214441 | 2.215210 | 2.578768 | 2.791493 |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
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| | | | | AU | SW | CA | CR | MB | B2 | B3 | B4 | C2 | C3 | C4 | S2 | S3 | TR |
| | | | | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km |
| Costa | Accidentado | Afirmado | Bueno | 0.409935 | 0.424757 | 0.839174 | 1.044567 | 1.875323 | 3.165493 | 3.536750 | 3.938547 | 0.824654 | 1.314168 | 2.080609 | 2.103009 | 2.438080 | 2.634174 |
| Costa | Accidentado | Afirmado | Regular | 0.443556 | 0.466139 | 0.912428 | 1.103132 | 1.934793 | 3.255914 | 3.623575 | 4.045742 | 0.904309 | 1.439856 | 2.259103 | 2.252903 | 2.626091 | 2.844300 |
| Costa | Accidentado | Afirmado | Malo | 0.431735 | 0.446454 | 0.888071 | 1.107313 | 2.019695 | 3.389612 | 3.748015 | 4.192443 | 0.984643 | 1.566046 | 2.438428 | 2.406204 | 2.818715 | 3.058542 |
| Costa | Accidentado | Sin Afirmar | Regular | 0.483310 | 0.497895 | 0.974239 | 1.228638 | 2.193930 | 3.715672 | 4.024297 | 4.468036 | 0.934471 | 1.542137 | 2.425616 | 2.414149 | 2.827643 | 3.065996 |
| Costa | Accidentado | Sin Afirmar | Malo | 0.503605 | 0.520382 | 1.021247 | 1.279540 | 2.295019 | 3.870875 | 4.173792 | 4.643308 | 1.019550 | 1.669998 | 2.606207 | 2.572436 | 3.026534 | 3.286473 |
| Costa | Accidentado | Trocha | Regular | 0.493022 | 0.508707 | 0.996704 | 1.252617 | 2.240902 | 3.787510 | 4.093694 | 4.549974 | 0.976622 | 1.605762 | 2.515636 | 2.492509 | 2.926121 | 3.175281 |
| Costa | Accidentado | Trocha | Malo | 0.554990 | 0.576077 | 1.139454 | 1.415842 | 2.548141 | 4.278180 | 4.562104 | 5.041964 | 1.199770 | 1.935265 | 2.976164 | 2.909919 | 3.449423 | 3.753457 |
| Sierra | Plano | Concreto Hidráulico | Bueno | 0.390510 | 0.379865 | 0.719820 | 0.717827 | 1.171021 | 1.919634 | 1.958646 | 2.250337 | 0.574479 | 0.772659 | 1.089664 | 1.406302 | 1.478210 | 1.594454 |
| Sierra | Plano | Concreto Hidráulico | Regular | 0.404461 | 0.395082 | 0.745192 | 0.736306 | 1.191135 | 1.947040 | 1.987572 | 2.292530 | 0.645262 | 0.863228 | 1.219381 | 1.535460 | 1.613213 | 1.746925 |
| Sierra | Plano | Concreto Hidráulico | Malo | 0.422093 | 0.414411 | 0.778822 | 0.760910 | 1.216023 | 1.980848 | 2.022981 | 2.342318 | 0.712738 | 0.950833 | 1.347841 | 1.664069 | 1.747809 | 1.899098 |
| Sierra | Plano | Asfaltado | Bueno | 0.394822 | 0.384558 | 0.727525 | 0.723439 | 1.177312 | 1.928228 | 1.967739 | 2.263740 | 0.598634 | 0.803378 | 1.133133 | 1.449536 | 1.523379 | 1.645439 |
| Sierra | Plano | Asfaltado | Regular | 0.415729 | 0.407427 | 0.766520 | 0.751889 | 1.207071 | 1.968685 | 2.010273 | 2.324682 | 0.690470 | 0.921810 | 1.305082 | 1.621135 | 1.702860 | 1.848273 |
| Sierra | Plano | Asfaltado | Malo | 0.436549 | 0.430295 | 0.807268 | 0.781848 | 1.236314 | 2.008465 | 2.051726 | 2.381369 | 0.756870 | 1.008625 | 1.433356 | 1.750528 | 1.838368 | 2.001477 |
| Sierra | Plano | Tratamiento Superficial | Bueno | 0.399459 | 0.389616 | 0.735951 | 0.729573 | 1.183995 | 1.937332 | 1.977349 | 2.277770 | 0.622177 | 0.833514 | 1.176351 | 1.492565 | 1.568357 | 1.696238 |
| Sierra | Plano | Tratamiento Superficial | Regular | 0.422093 | 0.414411 | 0.778822 | 0.760910 | 1.216023 | 1.980848 | 2.022981 | 2.342318 | 0.712738 | 0.950833 | 1.347841 | 1.664069 | 1.747809 | 1.899098 |
| Sierra | Plano | Tratamiento Superficial | Malo | 0.463463 | 0.459909 | 0.861375 | 0.821859 | 1.274029 | 2.060023 | 2.105091 | 2.451430 | 0.822626 | 1.095291 | 1.562075 | 1.882889 | 1.977095 | 2.158159 |
| Sierra | Plano | Afirmado | Bueno | 0.436549 | 0.430295 | 0.807268 | 0.781848 | 1.236314 | 2.008465 | 2.051726 | 2.381369 | 0.756870 | 1.008625 | 1.433356 | 1.750528 | 1.838368 | 2.001477 |
| Sierra | Plano | Afirmado | Regular | 0.474096 | 0.471619 | 0.882976 | 0.837859 | 1.288888 | 2.080413 | 2.126117 | 2.478385 | 0.844548 | 1.124303 | 1.605203 | 1.928015 | 2.024407 | 2.211530 |
| Sierra | Plano | Afirmado | Malo | 0.458709 | 0.451762 | 0.861673 | 0.836689 | 1.361822 | 2.180968 | 2.229402 | 2.607401 | 0.932856 | 1.241617 | 1.779371 | 2.114948 | 2.220402 | 2.432190 |
| Sierra | Plano | Sin Afirmar | Regular | 0.545796 | 0.535196 | 1.026057 | 1.006776 | 1.606519 | 2.561964 | 2.611997 | 3.009095 | 1.011804 | 1.321404 | 1.862704 | 2.238490 | 2.344699 | 2.562834 |
| Sierra | Plano | Sin Afirmar | Malo | 0.565655 | 0.555848 | 1.070159 | 1.045363 | 1.683942 | 2.666643 | 2.719312 | 3.144120 | 1.098566 | 1.437394 | 2.035952 | 2.419342 | 2.534400 | 2.776968 |
| Sierra | Plano | Trocha | Regular | 0.555358 | 0.545172 | 1.047185 | 1.025111 | 1.642702 | 2.610806 | 2.662140 | 3.072685 | 1.055036 | 1.379120 | 1.948983 | 2.327906 | 2.438519 | 2.668805 |
| Sierra | Plano | Trocha | Malo | 0.614962 | 0.606276 | 1.180648 | 1.145675 | 1.865202 | 2.919831 | 2.978196 | 3.418189 | 1.277135 | 1.677536 | 2.392103 | 2.807442 | 2.940603 | 3.233656 |
| Sierra | Ondulado | Concreto Hidráulico | Bueno | 0.402700 | 0.391360 | 0.758197 | 0.758197 | 1.208400 | 2.104508 | 2.153794 | 2.461350 | 0.632811 | 0.963089 | 1.327376 | 1.626503 | 1.672402 | 1.793420 |
| Sierra | Ondulado | Concreto Hidráulico | Regular | 0.416552 | 0.406491 | 0.782996 | 0.776031 | 1.228311 | 2.133278 | 2.183465 | 2.504300 | 0.703947 | 1.053541 | 1.458697 | 1.756930 | 1.808505 | 1.946809 |
| Sierra | Ondulado | Concreto Hidráulico | Malo | 0.434059 | 0.425677 | 0.815749 | 0.799679 | 1.252833 | 2.167959 | 2.219208 | 2.554399 | 0.771709 | 1.140875 | 1.588619 | 1.886343 | 1.943776 | 2.099445 |
| Sierra | Ondulado | Asfaltado | Bueno | 0.406984 | 0.396028 | 0.765735 | 0.763619 | 1.214634 | 2.113587 | 2.163160 | 2.475032 | 0.657088 | 0.993778 | 1.371388 | 1.670182 | 1.717958 | 1.844732 |
| Sierra | Ondulado | Asfaltado | Regular | 0.427738 | 0.418745 | 0.803783 | 0.791022 | 1.244029 | 2.155585 | 2.206457 | 2.536734 | 0.749356 | 1.111968 | 1.545397 | 1.843226 | 1.898681 | 2.048549 |
| Sierra | Ondulado | Asfaltado | Malo | 0.448389 | 0.441436 | 0.843374 | 0.819737 | 1.272730 | 2.195676 | 2.247762 | 2.593207 | 0.815967 | 1.198336 | 1.674965 | 1.972808 | 2.034290 | 2.201605 |
| Sierra | Ondulado | Tratamiento Superficial | Bueno | 0.411589 | 0.401059 | 0.773972 | 0.769540 | 1.221252 | 2.123151 | 2.173023 | 2.489320 | 0.680749 | 1.023877 | 1.415142 | 1.713639 | 1.763307 | 1.895842 |
| Sierra | Ondulado | Tratamiento Superficial | Regular | 0.434059 | 0.425677 | 0.815749 | 0.799679 | 1.252833 | 2.167959 | 2.219208 | 2.554399 | 0.771709 | 1.140875 | 1.588619 | 1.886343 | 1.943776 | 2.099445 |
| Sierra | Ondulado | Tratamiento Superficial | Malo | 0.474986 | 0.470740 | 0.895956 | 0.858177 | 1.309543 | 2.246348 | 2.300504 | 2.661929 | 0.881780 | 1.284204 | 1.804633 | 2.103984 | 2.171845 | 2.356813 |
| Sierra | Ondulado | Afirmado | Bueno | 0.448389 | 0.441436 | 0.843374 | 0.819737 | 1.272730 | 2.195676 | 2.247762 | 2.593207 | 0.815967 | 1.198336 | 1.674965 | 1.972808 | 2.034290 | 2.201605 |
| Sierra | Ondulado | Afirmado | Regular | 0.485475 | 0.482310 | 0.916987 | 0.873590 | 1.324007 | 2.266115 | 2.321060 | 2.688218 | 0.903680 | 1.312859 | 1.847986 | 2.148323 | 2.218441 | 2.409347 |
| Sierra | Ondulado | Afirmado | Malo | 0.469300 | 0.461692 | 0.892855 | 0.869499 | 1.394862 | 2.362351 | 2.421016 | 2.813454 | 0.991664 | 1.428246 | 2.022542 | 2.330057 | 2.409969 | 2.624969 |
| Sierra | Ondulado | Sin Afirmar | Regular | 0.559382 | 0.548034 | 1.060261 | 1.042599 | 1.651443 | 2.734317 | 2.802562 | 3.212308 | 1.044063 | 1.471680 | 2.067526 | 2.419788 | 2.509091 | 2.728884 |
| Sierra | Ondulado | Sin Afirmar | Malo | 0.578649 | 0.568119 | 1.102460 | 1.079213 | 1.726885 | 2.836545 | 2.908205 | 3.341565 | 1.131565 | 1.587840 | 2.243126 | 2.600229 | 2.699310 | 2.943354 |
| Sierra | Ondulado | Trocha | Regular | 0.568675 | 0.557752 | 1.080502 | 1.060015 | 1.702495 | 2.795295 | 2.865204 | 3.275146 | 1.087661 | 1.529504 | 2.155010 | 2.509201 | 2.603279 | 2.835138 |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Auto | Station Wagon | Camioneta | Camioneta Rural | Microbús | Bus 2 Ejes | Bus 3 Ejes | Bus 4 Ejes | Camión 2 Ejes | Camión 3 Ejes | Camión 4 Ejes | Semitrailer 2 Ejes | Semitrailer 3 Ejes | Articulado |
| | | | | AU | SW | CA | CR | MB | B2 | B3 | B4 | C2 | C3 | C4 | S2 | S3 | TR |
| | | | | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km |
| Sierra | Ondulado | Trocha | Malo | 0.626157 | 0.616830 | 1.207935 | 1.174313 | 1.902473 | 3.080094 | 3.160028 | 3.611691 | 1.311719 | 1.827908 | 2.603421 | 2.983589 | 3.104857 | 3.398966 |
| Sierra | Accidentado | Concreto Hidráulico | Bueno | 0.435119 | 0.421850 | 0.845740 | 0.851536 | 1.319566 | 2.558866 | 2.626642 | 2.976070 | 0.705939 | 1.331487 | 1.736538 | 2.073012 | 2.088965 | 2.237233 |
| Sierra | Accidentado | Concreto Hidráulico | Regular | 0.448874 | 0.436872 | 0.869834 | 0.868426 | 1.338694 | 2.588700 | 2.658330 | 3.020823 | 0.778205 | 1.424165 | 1.872548 | 2.207356 | 2.230045 | 2.395224 |
| Sierra | Accidentado | Concreto Hidráulico | Malo | 0.466109 | 0.455810 | 0.901393 | 0.890833 | 1.361984 | 2.622992 | 2.695250 | 3.071795 | 0.847097 | 1.513557 | 2.007136 | 2.340299 | 2.369995 | 2.552065 |
| Sierra | Accidentado | Asfaltado | Bueno | 0.439380 | 0.426491 | 0.853082 | 0.856629 | 1.325570 | 2.568681 | 2.636731 | 2.990411 | 0.730591 | 1.362927 | 1.782107 | 2.118011 | 2.136186 | 2.290091 |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Auto | Station Wagon | Camioneta | Camioneta Rural | Microbús | Bus 2 Ejes | Bus 3 Ejes | Bus 4 Ejes | Camión 2 Ejes | Camión 3 Ejes | Camión 4 Ejes | Semitrailer 2 Ejes | Semitrailer 3 Ejes | Articulado |
| | | | | AU | SW | CA | CR | MB | B2 | B3 | B4 | C2 | C3 | C4 | S2 | S3 | TR |
| | | | | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km |
| Sierra | Accidentado | Asfaltado | Regular | 0.459909 | 0.448987 | 0.889894 | 0.882661 | 1.353660 | 2.610903 | 2.682244 | 3.053989 | 0.824369 | 1.483991 | 1.962369 | 2.296081 | 2.323404 | 2.499848 |
| Sierra | Accidentado | Asfaltado | Malo | 0.480116 | 0.471256 | 0.927833 | 0.909661 | 1.380649 | 2.649519 | 2.723761 | 3.110272 | 0.892092 | 1.572223 | 2.096518 | 2.428619 | 2.463197 | 2.656499 |
| Sierra | Accidentado | Tratamiento Superficial | Bueno | 0.443954 | 0.431486 | 0.861087 | 0.862224 | 1.331930 | 2.578449 | 2.647274 | 3.005308 | 0.754629 | 1.393770 | 1.827426 | 2.162780 | 2.183198 | 2.342741 |
| Sierra | Accidentado | Tratamiento Superficial | Regular | 0.466109 | 0.455810 | 0.901393 | 0.890833 | 1.361984 | 2.622992 | 2.695250 | 3.071795 | 0.847097 | 1.513557 | 2.007136 | 2.340299 | 2.369995 | 2.552065 |
| Sierra | Accidentado | Tratamiento Superficial | Malo | 0.505976 | 0.499845 | 0.977797 | 0.945337 | 1.414762 | 2.696421 | 2.774092 | 3.176538 | 0.958962 | 1.659538 | 2.230535 | 2.561325 | 2.603657 | 2.813728 |
| Sierra | Accidentado | Afirmado | Bueno | 0.480116 | 0.471256 | 0.927833 | 0.909661 | 1.380649 | 2.649519 | 2.723761 | 3.110272 | 0.892092 | 1.572223 | 2.096518 | 2.428619 | 2.463197 | 2.656499 |
| Sierra | Accidentado | Afirmado | Regular | 0.516138 | 0.511099 | 0.997696 | 0.959560 | 1.428060 | 2.714298 | 2.793253 | 3.201293 | 0.981196 | 1.688558 | 2.275266 | 2.605761 | 2.650815 | 2.866450 |
| Sierra | Accidentado | Afirmado | Malo | 0.498216 | 0.488792 | 0.968072 | 0.949711 | 1.492828 | 2.799372 | 2.884303 | 3.316279 | 1.070415 | 1.804700 | 2.454898 | 2.785435 | 2.842231 | 3.079971 |
| Sierra | Accidentado | Sin Afirmar | Regular | 0.584805 | 0.571782 | 1.121275 | 1.107992 | 1.735614 | 3.098258 | 3.202239 | 3.643629 | 1.108521 | 1.820764 | 2.473742 | 2.830351 | 2.898883 | 3.134178 |
| Sierra | Accidentado | Sin Afirmar | Malo | 0.603081 | 0.590930 | 1.160332 | 1.141258 | 1.807695 | 3.195577 | 3.305400 | 3.773588 | 1.197847 | 1.937777 | 2.655180 | 3.012328 | 3.092838 | 3.350915 |
| Sierra | Accidentado | Trocha | Regular | 0.593646 | 0.581073 | 1.140030 | 1.123823 | 1.769350 | 3.144025 | 3.250806 | 3.705233 | 1.153027 | 1.879121 | 2.564196 | 2.920804 | 2.995165 | 3.241834 |
| Sierra | Accidentado | Trocha | Malo | 0.647618 | 0.636830 | 1.257826 | 1.227889 | 1.973608 | 3.420552 | 3.545114 | 4.025658 | 1.381798 | 2.177067 | 3.026108 | 3.392518 | 3.500994 | 3.805433 |
| Selva | Plano | Concreto Hidráulico | Bueno | 0.379904 | 0.381676 | 0.741213 | 0.851040 | 1.563269 | 2.610863 | 2.647160 | 2.949910 | 0.585797 | 0.794162 | 1.254490 | 1.436511 | 1.509222 | 1.626979 |
| Selva | Plano | Concreto Hidráulico | Regular | 0.392477 | 0.394754 | 0.766356 | 0.869563 | 1.584514 | 2.638021 | 2.675794 | 2.991769 | 0.648533 | 0.884630 | 1.420417 | 1.565399 | 1.644027 | 1.779245 |
| Selva | Plano | Concreto Hidráulico | Malo | 0.408174 | 0.411214 | 0.799403 | 0.894256 | 1.610139 | 2.671214 | 2.710537 | 3.040796 | 0.707911 | 0.972051 | 1.585082 | 1.693463 | 1.778123 | 1.930893 |
| Selva | Plano | Asfaltado | Bueno | 0.383807 | 0.385723 | 0.748870 | 0.856663 | 1.569963 | 2.619401 | 2.656183 | 2.963233 | 0.607274 | 0.824854 | 1.310030 | 1.479672 | 1.554345 | 1.677916 |
| Selva | Plano | Asfaltado | Regular | 0.402530 | 0.405283 | 0.787346 | 0.885199 | 1.601003 | 2.659311 | 2.698107 | 3.023477 | 0.688349 | 0.943101 | 1.530255 | 1.650756 | 1.733391 | 1.880295 |
| Selva | Plano | Asfaltado | Malo | 0.420919 | 0.424646 | 0.827185 | 0.915302 | 1.630556 | 2.698108 | 2.738518 | 3.078970 | 0.746600 | 1.029654 | 1.694733 | 1.779286 | 1.868050 | 2.032608 |
| Selva | Plano | Tratamiento Superficial | Bueno | 0.387987 | 0.390071 | 0.757222 | 0.862812 | 1.577027 | 2.628425 | 2.665699 | 2.977156 | 0.628134 | 0.854957 | 1.365318 | 1.522614 | 1.599259 | 1.728649 |
| Selva | Plano | Tratamiento Superficial | Regular | 0.408174 | 0.411214 | 0.799403 | 0.894256 | 1.610139 | 2.671214 | 2.710537 | 3.040796 | 0.707911 | 0.972051 | 1.585082 | 1.693463 | 1.778123 | 1.930893 |
| Selva | Plano | Tratamiento Superficial | Malo | 0.444444 | 0.449543 | 0.879800 | 0.955606 | 1.667724 | 2.748009 | 2.790154 | 3.147035 | 0.804101 | 1.115914 | 1.859660 | 1.910124 | 2.005200 | 2.187634 |
| Selva | Plano | Afirmado | Bueno | 0.420919 | 0.424646 | 0.827185 | 0.915302 | 1.630556 | 2.698108 | 2.738518 | 3.078970 | 0.746600 | 1.029654 | 1.694733 | 1.779286 | 1.868050 | 2.032608 |
| Selva | Plano | Afirmado | Regular | 0.453688 | 0.459352 | 0.900764 | 0.971755 | 1.682182 | 2.767697 | 2.810452 | 3.173141 | 0.823243 | 1.144755 | 1.914861 | 1.954574 | 2.051801 | 2.240257 |
| Selva | Plano | Afirmado | Malo | 0.440379 | 0.443218 | 0.876265 | 0.971601 | 1.752475 | 2.864994 | 2.910369 | 3.298139 | 0.900276 | 1.261191 | 2.137359 | 2.138037 | 2.244115 | 2.457046 |
| Selva | Plano | Sin Afirmar | Regular | 0.528377 | 0.530147 | 1.050049 | 1.184963 | 2.093691 | 3.441424 | 3.489230 | 3.899888 | 0.991295 | 1.354436 | 2.245247 | 2.278107 | 2.384061 | 2.604160 |
| Selva | Plano | Sin Afirmar | Malo | 0.545715 | 0.547742 | 1.091358 | 1.224265 | 2.168914 | 3.542467 | 3.592874 | 4.030628 | 1.066639 | 1.469259 | 2.466435 | 2.455660 | 2.570460 | 2.814824 |
| Selva | Plano | Trocha | Regular | 0.536745 | 0.538647 | 1.069880 | 1.203598 | 2.128921 | 3.488535 | 3.537622 | 3.961448 | 1.028850 | 1.411610 | 2.355503 | 2.366012 | 2.476372 | 2.708546 |
| Selva | Plano | Trocha | Malo | 0.588349 | 0.590953 | 1.194261 | 1.327967 | 2.343497 | 3.789139 | 3.845230 | 4.296338 | 1.221611 | 1.706136 | 2.918476 | 2.834833 | 2.967701 | 3.262089 |
| Selva | Ondulado | Concreto Hidráulico | Bueno | 0.389743 | 0.391421 | 0.774248 | 0.892160 | 1.625718 | 2.788430 | 2.831037 | 3.147428 | 0.643024 | 0.976591 | 1.486036 | 1.644016 | 1.692624 | 1.814421 |
| Selva | Ondulado | Concreto Hidráulico | Regular | 0.402285 | 0.404455 | 0.798878 | 0.910068 | 1.647148 | 2.817219 | 2.861056 | 3.190633 | 0.706243 | 1.067086 | 1.653688 | 1.774498 | 1.828627 | 1.967740 |
| Selva | Ondulado | Concreto Hidráulico | Malo | 0.417922 | 0.420855 | 0.831194 | 0.933878 | 1.672846 | 2.851766 | 2.897005 | 3.240887 | 0.766066 | 1.154432 | 1.819969 | 1.903843 | 1.963688 | 2.120207 |
| Selva | Ondulado | Asfaltado | Bueno | 0.393628 | 0.395443 | 0.781752 | 0.897600 | 1.632481 | 2.797527 | 2.840529 | 3.161157 | 0.664663 | 1.007297 | 1.542155 | 1.687721 | 1.738156 | 1.865717 |
| Selva | Ondulado | Asfaltado | Regular | 0.412306 | 0.414951 | 0.819411 | 0.925153 | 1.663705 | 2.839459 | 2.884207 | 3.223196 | 0.746363 | 1.125525 | 1.764623 | 1.860769 | 1.918678 | 2.069384 |
| Selva | Ondulado | Asfaltado | Malo | 0.430585 | 0.434207 | 0.858416 | 0.954273 | 1.693199 | 2.879268 | 2.925580 | 3.279649 | 0.805010 | 1.211877 | 1.930577 | 1.990135 | 2.053963 | 2.222150 |
| Selva | Ondulado | Tratamiento Superficial | Bueno | 0.397804 | 0.399780 | 0.789934 | 0.903545 | 1.639607 | 2.807097 | 2.850509 | 3.175556 | 0.685685 | 1.037410 | 1.598020 | 1.731198 | 1.783473 | 1.916804 |
| Selva | Ondulado | Tratamiento Superficial | Regular | 0.417922 | 0.420855 | 0.831194 | 0.933878 | 1.672846 | 2.851766 | 2.897005 | 3.240887 | 0.766066 | 1.154432 | 1.819969 | 1.903843 | 1.963688 | 2.120207 |
| Selva | Ondulado | Tratamiento Superficial | Malo | 0.453902 | 0.458899 | 0.910038 | 0.993360 | 1.730014 | 2.929414 | 2.977608 | 3.348041 | 0.862894 | 1.[illegible] | [illegible] | 2.096682 | 2.120747 | 2.190903 | 2.376735 |
| Selva | Ondulado | Afirmado | Bueno | 0.430585 | 0.434207 | 0.858416 | 0.954273 | 1.693199 | 2.879268 | 2.925580 | 3.279649 | 0.805010 | 1.211877 | 1.930577 | 1.990135 | 2.053963 | 2.222150 |

GOBIERNO REGIONAL SAN MARTIN
Dirección Regional de Transportes y Comunicaciones
Ing. David Grandez Silva
CIP. N° 284250
RESPONSABLE DE LA UF DRTC SM

| Región | Topografía | Superficie | Estado | COV para Carreteras Interurbanas | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Auto | Station Wagon | Camioneta | Camioneta Rural | Microbús | Bus 2 Ejes | Bus 3 Ejes | Bus 4 Ejes | Camión 2 Ejes | Camión 3 Ejes | Camión 4 Ejes | Semitrailer 2 Ejes | Semitrailer 3 Ejes | Articulado |
| | | | | AU | SW | CA | CR | MB | B2 | B3 | B4 | C2 | C3 | C4 | S2 | S3 | TR |
| | | | | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km |
| Selva | Ondulado | Afirmado | Regular | 0.463048 | 0.468612 | 0.930578 | 1.008988 | 1.744270 | 2.948962 | 2.997871 | 3.374092 | 0.882017 | 1.326287 | 2.152194 | 2.164813 | 2.237193 | 2.428956 |
| Selva | Ondulado | Afirmado | Malo | 0.449202 | 0.451949 | 0.903932 | 1.006189 | 1.813263 | 3.044367 | 3.096653 | 3.497737 | 0.959158 | 1.441426 | 2.375473 | 2.345004 | 2.426971 | 2.642772 |
| Selva | Ondulado | Sin Afirmar | Regular | 0.539486 | 0.541135 | 1.079906 | 1.222046 | 2.157159 | 3.606271 | 3.670836 | 4.092695 | 1.025926 | 1.498550 | 2.441665 | 2.449465 | 2.541424 | 2.762922 |
| Selva | Ondulado | Sin Afirmar | Malo | 0.556424 | 0.558336 | 1.119791 | 1.259595 | 2.231492 | 3.706804 | 3.774596 | 4.223493 | 1.102259 | 1.614062 | 2.665649 | 2.627954 | 2.729427 | 2.975057 |
| Selva | Ondulado | Trocha | Regular | 0.547673 | 0.549457 | 1.099077 | 1.239870 | 2.192008 | 3.653276 | 3.719400 | 4.154407 | 1.063969 | 1.556082 | 2.553336 | 2.537994 | 2.634614 | 2.868124 |

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| Región | Topografía | Superficie | Estado | COV para Carreteras Interurbanas | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Auto | Station Wagon | Camioneta | Camioneta Rural | Microbús | Bus 2 Ejes | Bus 3 Ejes | Bus 4 Ejes | Camión 2 Ejes | Camión 3 Ejes | Camión 4 Ejes | Semitrailer 2 Ejes | Semitrailer 3 Ejes | Articulado |
| | | | | AU | SW | CA | CR | MB | B2 | B3 | B4 | C2 | C3 | C4 | S2 | S3 | TR |
| | | | | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km | $/veh-km |
| Selva | Ondulado | Trocha | Malo | 0.597794 | 0.600299 | 1.218801 | 1.358494 | 2.402745 | 3.948801 | 4.024433 | 4.486352 | 1.259388 | 1.852092 | 3.122952 | 3.005648 | 3.128483 | 3.423822 |
| Selva | Accidentado | Concreto Hidráulico | Bueno | 0.416909 | 0.418475 | 0.853129 | 0.990121 | 1.800386 | 3.249306 | 3.301369 | 3.658663 | 0.719656 | 1.337565 | 1.895030 | 2.079721 | 2.098318 | 2.245953 |
| Selva | Accidentado | Concreto Hidráulico | Regular | 0.429336 | 0.431396 | 0.877161 | 1.007287 | 1.821615 | 3.279066 | 3.333177 | 3.703532 | 0.784238 | 1.430263 | 2.067443 | 2.214135 | 2.239448 | 2.403992 |
| Selva | Accidentado | Concreto Hidráulico | Malo | 0.444748 | 0.447562 | 0.908501 | 1.029890 | 1.846776 | 3.313902 | 3.370358 | 3.754756 | 0.845449 | 1.519677 | 2.238456 | 2.347135 | 2.379424 | 2.560865 |
| Selva | Accidentado | Asfaltado | Bueno | 0.420773 | 0.422479 | 0.860463 | 0.995350 | 1.807103 | 3.258769 | 3.311488 | 3.673032 | 0.741745 | 1.369012 | 1.952732 | 2.124744 | 2.145557 | 2.298827 |
| Selva | Accidentado | Asfaltado | Regular | 0.439221 | 0.441750 | 0.897098 | 1.021634 | 1.837868 | 3.301610 | 3.357244 | 3.736846 | 0.825282 | 1.490104 | 2.181545 | 2.302900 | 2.332829 | 2.508640 |
| Selva | Accidentado | Asfaltado | Malo | 0.457176 | 0.460674 | 0.934670 | 1.048955 | 1.866442 | 3.340921 | 3.399169 | 3.793523 | 0.885317 | 1.578357 | 2.352135 | 2.435475 | 2.472616 | 2.665302 |
| Selva | Accidentado | Tratamiento Superficial | Bueno | 0.424906 | 0.426775 | 0.868448 | 1.001051 | 1.814166 | 3.268670 | 3.322070 | 3.687967 | 0.763222 | 1.399862 | 2.010185 | 2.169536 | 2.192586 | 2.351493 |
| Selva | Accidentado | Tratamiento Superficial | Regular | 0.444748 | 0.447562 | 0.908501 | 1.029890 | 1.846776 | 3.313902 | 3.370358 | 3.754756 | 0.845449 | 1.519677 | 2.238456 | 2.347135 | 2.379424 | 2.560865 |
| Selva | Accidentado | Tratamiento Superficial | Malo | 0.479958 | 0.484826 | 0.983987 | 1.085205 | 1.901511 | 3.388848 | 3.450238 | 3.860499 | 0.944477 | 1.665693 | 2.522668 | 2.568168 | 2.612992 | 2.822485 |
| Selva | Accidentado | Afirmado | Bueno | 0.457176 | 0.460674 | 0.934670 | 1.048955 | 1.866442 | 3.340921 | 3.399169 | 3.793523 | 0.885317 | 1.578357 | 2.352135 | 2.435475 | 2.472616 | 2.665302 |
| Selva | Accidentado | Afirmado | Regular | 0.488868 | 0.494302 | 1.003599 | 1.099691 | 1.914953 | 3.407166 | 3.469748 | 3.885589 | 0.964133 | 1.694720 | 2.579563 | 2.612583 | 2.660098 | 2.875173 |
| Selva | Accidentado | Afirmado | Malo | 0.473747 | 0.476366 | 0.972445 | 1.091340 | 1.979323 | 3.494725 | 3.562934 | 4.002614 | 1.042992 | 1.810885 | 2.807899 | 2.792068 | 2.851144 | 3.088441 |
| Selva | Accidentado | Sin Afirmar | Regular | 0.560749 | 0.562304 | 1.134810 | 1.290176 | 2.290260 | 3.965995 | 4.061286 | 4.512833 | 1.095433 | 1.834898 | 2.838674 | 2.847317 | 2.920163 | 3.155388 |
| Selva | Accidentado | Sin Afirmar | Malo | 0.576971 | 0.578786 | 1.172132 | 1.324570 | 2.362796 | 4.064344 | 4.165080 | 4.643239 | 1.174100 | 1.951904 | 3.068936 | 3.028629 | 3.113105 | 3.371258 |
| Selva | Accidentado | Trocha | Regular | 0.568610 | 0.570298 | 1.152769 | 1.306508 | 2.324335 | 4.012199 | 4.110097 | 4.574599 | 1.134632 | 1.893256 | 2.953524 | 2.937482 | 3.015998 | 3.262667 |
| Selva | Accidentado | Trocha | Malo | 0.616148 | 0.618564 | 1.264565 | 1.415351 | 2.527447 | 4.294692 | 4.409331 | 4.899160 | 1.336190 | 2.191030 | 3.537938 | 3.406448 | 3.517911 | 3.822717 |

Nota 1: Los valores presentados incluyen el valor social del tiempo y ahorro por reducción de emisiones de CO2 equivalente.

Nota2 : Los valores presentados, son susceptibles a ser actualizados periódicamente por la OPMI-MTC, sin incurrir en actualización de la Ficha Técnica Estándar.

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